

Universal Specificity: Mechanistic Unification of Time Dilation, Gravitation, and Inertia

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Abstract

Relativistic phenomena admit two complementary perspectives: describing what happens through spacetime geometry and explaining why it happens through physical mechanisms. This paper develops a mechanistic framework, *Universal Specificity*, that seeks to explain why those phenomena occur while recovering the observable regularities described geometrically by relativity. The framework is grounded in the *axiom of universal presentism*: the recognition, induced from universal observation, that only the present physically exists, all existence and interaction are confined to it, and time is the interval over which physical change occurs. Because observed physical change is mediated by interactions between spatially separated entities, the rate at which those interactions are completed, termed the *speed of causality*, determines how much physical change occurs during a given interval. Applying Mill's Methods of Agreement and Difference, the speed of causality is identified as the metric-independent causal condition governing both kinetic and gravitational time dilation. This condition is equivalently represented by specific internal energy, providing a common physical basis for unifying time dilation, gravitation, and inertia and yielding an *equivalence law* that mechanistically explains the empirical success of the equivalence principle. The framework reproduces the observations described geometrically by relativity while opening revealing questions concerning deeper physical mechanisms thus opening new avenues toward a quantum theory of gravity, a unified field theory, a theory of everything, and ultimately a constitutional theory of everything.

Keywords: relativistic physics, time dilation, gravitation, inertia, equivalence principle, presentism

1 Introduction: *Descriptio an Explicatio*

Scientific inquiry advances through two complementary pursuits. It seeks both to describe *what* regularities are observed in nature through increasingly precise mathematical relationships and to explain *why* those regularities occur through physical mechanism. These complementary objectives answer fundamentally different scientific questions: description characterizes relationships among observable quantities, whereas explanation seeks the physical processes responsible for those relationships. A mathematical framework may therefore reproduce observations with remarkable precision with or without capturing the underlying physical mechanisms that give rise to them. Likewise, different explanatory frameworks may account for the same observed phenomena yet differ fundamentally in their ontology and physical interpretation. The distinction between successful description and successful mechanistic explanation has consequently played a central role throughout the historical development of physics [1].

History provides numerous examples in which increasingly precise mathematical descriptions of observed regularities eventually reached a point where deeper physical explanation became necessary for further scientific progress [1]. Ptolemy’s epicyclic model of planetary motion, for example, accurately reproduced astronomical observations through an intricate geometric construction that made no appeal to underlying physical mechanisms. Kepler’s laws replaced independent geometric constructions with a single governing influence centered on the Sun, and Newton later unified terrestrial and celestial motion under a single physical law governed by mass. This progression illustrates that predictive success alone does not uniquely determine whether the underlying physical mechanisms responsible for observed phenomena were properly identified [2, 3].

The relationship between geometric description and mechanistic explanation is not unidirectional. Just as successful geometric descriptions may later be complemented by deeper mechanistic explanations, successful mechanistic theories may also be reformulated geometrically without altering their empirical predictions. A canonical example is *geometrized Newtonian gravitation* (Newton–Cartan theory), in which Newtonian gravitation is represented through spacetime structure rather than as a force acting within Euclidean space. Force-based and geometric formulations therefore constitute alternative descriptions of the same observable dynamics rather than distinct empirical theories [4–6]. This demonstrates that successful geometry need not uniquely determine physical interpretation and naturally raises the converse question: might the geometric description of gravitation provided by relativity likewise be complemented by a coherent mechanistic explanation?

Einstein himself repeatedly emphasized the distinction between mathematical formalism and physical content. In *Geometry and Experience*, he argued that geometry acquires empirical significance only through its connection with physical laws rather than through mathematics alone [7]. Throughout his career, he pursued increasingly unified physical descriptions of gravitation and electromagnetism, reflecting his enduring pursuit of deeper physical explanation beyond geometric description [8, 9]. His early investigations of variable-speed-of-light (VSL) models prior to general relativity [10–16], together with his later reconsideration of related ideas [17, 18], illustrate

that the search for deeper physical interpretation remained an active component of his scientific program.

Consistent mechanistic interpretations of the empirical predictions of special relativity have long been recognized within Lorentzian formulations [19–22]. A comparably unified mechanistic account spanning both special and general relativistic phenomena, however, remains comparatively underdeveloped. In particular, no generally accepted physical mechanism presently explains kinetic time dilation, gravitational time dilation, gravitation, and inertia as consequences of a single underlying physical mechanism.

Motivated by this explanatory question, the present paper develops Universal Specificity, a mechanistic framework that seeks to explain the relativistic phenomena described geometrically by relativity while reproducing the same observational regularities. It is grounded in the *axiom of universal presentism*: the recognition, induced from universal observation, that only the present physically exists and that all existence and interaction are confined to it. Within this framework, time is understood as the interval over which physical change occurs, while the rate at which interactions are completed determines how much physical change occurs during that interval. These foundational concepts are developed in Section 2.

Section 3 identifies this interaction rate, termed the *speed of causality*, as the physical mechanism responsible for relativistic time dilation. Section 4 then shows how a reduction in the speed of causality corresponds to diminished internal motion, quantifiable by an object’s *specific internal energy*, whose variation provides the common physical basis underlying kinetic and gravitational time dilation, gravitation, and inertia, culminating in the derivation of an *equivalence law*. Finally, Section 5 explores how the resulting framework naturally reveals deeper mechanistic questions, thereby opening new avenues toward a quantum theory of gravity, a unified field theory, and ultimately a theory of everything.

We begin by establishing the conceptual and methodological foundations upon which the remainder of the paper is built.

2 Laying the Groundwork for Mechanistic Discovery

Relativistic phenomena—including time dilation, gravitation, and inertial resistance—are described with extraordinary empirical success by the relativistic formalism. This section establishes the conceptual and methodological groundwork required for the inductive discovery of the physical mechanisms underlying those phenomena, developed in the sections that follow.

To establish that foundation, we first contrast geometric description and mechanistic explanation, clarifying their complementary roles in scientific progress through deeper understanding, broader unification, continued scientific discovery, and ultimately greater scientific control. We then distinguish the epistemic methods appropriate for validating these complementary scientific objectives: the hypothetico–deductive construction of descriptive frameworks and the inductive discovery of physical mechanisms through axiomatic and causal induction.

With the epistemic foundations established, we next develop the foundational concepts upon which the mechanistic paradigm rests—simultaneity, time, time dilation, and measurement—and contrast each with its counterpart in the geometric paradigm. These conceptual foundations reveal the points at which the mechanistic and geometric paradigms naturally diverge, establishing the geometric–mechanistic fork in the road developed in the following subsection. With both the epistemic and conceptual foundations established, Mill’s Methods can then be applied in the next section to identify the specific physical condition responsible for time dilation, initiating the sequence of causal discoveries developed throughout the remainder of the paper.

2.1 On Geometric Description and Mechanistic Explanation

Throughout the history of science, two complementary approaches have repeatedly advanced our understanding of nature: *geometric description* and *mechanistic explanation*. Both can achieve predictive success, yet they answer fundamentally different questions. Mechanistic explanation is fundamentally grounded in entities acting under specific physical conditions. It seeks to explain phenomena by identifying the entities involved, how they act, and the physical conditions under which those actions change.

The defining hallmark of mechanistic explanation is therefore *entities acting*. By *action*, we mean a change of state of an entity—either internally or in its relations to other entities. Mechanistic explanation therefore rests upon the law of causality, understood as the law of identity applied to action. The law of identity states that each thing has a specific nature that determines what it is; the law of causality states that each thing must act in accordance with that nature. An entity cannot act in contradiction to what it is. Causality, in this sense, is not merely temporal succession, but the necessity of a particular change of state following from a thing’s identity under given conditions [23–25].¹

The law of causality is therefore the fundamental reason predictive success is possible at all. Given identical essential conditions, identical entities must act identically. This is what makes successful geometric descriptions and mechanistic explanations possible in the first place. Mechanistic explanations therefore do not replace the regularities captured by successful geometric descriptions; rather, they seek to explain why those regularities exist by identifying the entities, interactions, and physical conditions from which they arise.

This distinction is crucial. A geometric description can achieve remarkable empirical success while remaining agnostic about the underlying physical mechanisms responsible for those regularities. It answers the question *how do phenomena behave?* without necessarily answering *what produces that behavior?* Mechanistic explanation, by contrast, is explicitly concerned with the latter. It seeks to identify causes, not merely correlations or patterns.

Importantly, the distinction is not one of correctness versus error. History contains many cases in which geometric descriptions and mechanistic explanations are empirically equivalent while differing in explanatory content. A geometric description may organize phenomena with extraordinary precision, yet still leave open the possibility

¹Throughout this paper, “causality” refers to the law of identity applied to action, not merely to temporal ordering.

of a deeper causal account. The central question, then, is not whether a geometric description works, but what physical mechanisms give rise to the regularities it successfully describes.

This distinction sets the stage for the analysis that follows. Relativity provides one of the most successful geometric descriptions ever developed. The task of this paper is not to replace relativity's geometric description, but to discover the physical mechanisms responsible for the relativistic phenomena it successfully describes.

2.2 Scientific Progress Through Description and Discovery

Successful geometric descriptions advance scientific understanding by capturing stable observational regularities in precise mathematical relationships, thereby enabling reliable prediction. By organizing observations into precise mathematical relationships, they also expand the range and precision of empirical evidence available for causal discovery. Discovering the physical mechanisms responsible for those regularities extends this scientific progression by providing deeper understanding of physical entities, broader unification through common causes, and new avenues for further scientific discovery. This progression forms a continuing cycle of scientific progress, as each mechanistic discovery reveals new opportunities for the next. Together, these advances ultimately yield greater scientific control through prediction with intervention, the principal practical benefit of scientific progress. The history of science repeatedly illustrates this progression.

The first scientific benefit of discovering physical mechanisms is a deeper understanding of physical entities. Entities are known only through how they act and interact. An entity that never interacted with anything would be undetectable and therefore of no possible scientific significance. It is through interactions that we discover what entities exist, infer their identities, and determine the physical conditions under which their actions change. Mechanistic explanations therefore deepen understanding by identifying the entities involved, the interactions they undergo, and the physical conditions responsible for those interactions.

The history of astronomy illustrates this progression. The Ptolemaic system provided genuine scientific understanding by accurately characterizing how planets appeared to move across the night sky. Through increasingly sophisticated geometric constructions, it revealed stable observational regularities and enabled increasingly reliable prediction of celestial events and seasonal cycles useful for navigation, agriculture, and calendar keeping. This descriptive understanding proved invaluable in practice. What it did not explain, however, was why those regularities existed. Planetary motion remained a geometric pattern rather than the consequence of interactions among physical entities. Consequently, the entities responsible for planetary motion and the physical conditions governing their behavior remained unknown.

Kepler derived mathematical laws describing planetary motion and interpreted them mechanistically by proposing that the Sun exerted a governing influence on the planets. Although the physical nature of that influence remained unknown, planetary motion was no longer understood merely as a geometric regularity but as the consequence of a physical interaction. Newton later identified universal gravitation

and mass as the underlying physical mechanism. Together, these developments progressively deepened our understanding of both the entities responsible for planetary motion and the reasons for their behavior [3, 26].

The second scientific benefit of discovering physical mechanisms is broader unification through common causes. Kepler replaced independent geometric constructions for each planet with a common set of mathematical laws applicable to all planetary motion. Newton subsequently unified phenomena that had previously appeared unrelated—falling bodies on Earth, planetary orbits, and ocean tides—by identifying a single causal law governing gravitational attraction. These unifications were not achieved by geometric abstraction, but by recognizing that the same kinds of entities were acting according to the same causal law. Unification through common causes is scientifically valuable because it reduces the number of independent principles required to explain nature. Phenomena that once appeared to require separate rules become understood as different manifestations of the same underlying cause, yielding a simpler and more unified understanding of the physical world [3, 26].

The third scientific benefit of discovering physical mechanisms is that each discovery opens new avenues for further scientific discovery. A purely geometric description may continue to improve predictive accuracy while simultaneously masking the physical questions whose answers lead to deeper discovery. The mature Ptolemaic system illustrates this limitation. Although increasingly refined geometric constructions improved prediction, planetary motion remained a geometric fact rather than the consequence of physical interactions. As a result, further progress largely consisted of refining the geometric description itself rather than investigating underlying physical causes. Kepler’s mechanistic interpretation transformed that situation by proposing that the Sun exerted a governing influence on planetary motion. The central scientific question was no longer *How can planetary motion be described more accurately?*, but rather *What physical mechanism produces the Sun’s influence?* Newton’s discovery of universal gravitation answered that question while simultaneously leaving open the next: by what physical mechanism does gravitation itself act at a distance? Mechanistic understanding therefore does not conclude scientific inquiry; it continually expands the frontier of scientific discovery by revealing new entities, interactions, and physical conditions that themselves become subjects of investigation.

The fourth scientific benefit of discovering physical mechanisms is greater scientific control. The preceding benefits describe how mechanistic discovery advances scientific knowledge through deeper understanding, broader unification, and continued discovery. Their ultimate practical significance, however, lies in the greater scientific control that such knowledge makes possible. The ability to predict, intervene, and intentionally shape future outcomes is the principal practical benefit of scientific progress. Geometric descriptions enable reliable prediction by revealing observational regularities that can be exploited in practice. The Ptolemaic system, for example, supported increasingly accurate calendars and seasonal planning despite providing no causal account of planetary motion. Mechanistic explanations extend that capability by revealing the physical conditions that produce those regularities, making it possible to understand what must be changed to achieve new outcomes. Newtonian gravitation did

not merely describe planetary motion; it identified the physical mechanism governing it, enabling prediction with intervention. Orbital mechanics, trajectory correction, and ultimately spaceflight became possible not merely through the description of trajectories, but through laws relating controlled changes in physical conditions to the subsequent motion of physical entities. Control over the future required knowing what physical conditions to change in order to produce a desired outcome.

The same scientific progression can be pursued in relativistic physics. Relativity provides one of the most successful geometric descriptions ever developed, accurately relating observations across an extraordinary range of physical conditions. By organizing observational regularities into an exceptionally precise mathematical framework, relativity likewise expands the range and precision of empirical evidence available for causal discovery. Because stable observational regularities arise only from entities acting according to their identities, the implication is that these regularities likewise arise from underlying physical mechanisms. The next opportunity for scientific progress therefore lies not solely in the further refinement of relativity's geometric description, but in the inductive discovery of the physical mechanisms responsible for the relativistic phenomena it successfully describes.

Discovering those mechanisms promises the same scientific progression: progressively deeper understanding of the entities involved and the reasons for their actions, broader unification through common causes, and new avenues for further scientific discovery. Together, these advances ultimately expand our capacity for scientific control through prediction with intervention. Because the validation of mechanistic explanations differs fundamentally from the validation of descriptive frameworks, the pursuit of physical mechanisms requires a different epistemic foundation. We therefore turn next to the epistemic foundations upon which the remainder of this work is built.

2.3 Epistemic Foundations: Deduction and Induction

The epistemic method appropriate for scientific inquiry depends upon the kind of claim one seeks to validate. If the objective is to validate a mathematical description of observational regularities, the hypothetico-deductive method provides a powerful means of constructing and refining descriptive frameworks. If the objective is to establish that a proposed physical mechanism is the true cause of those regularities, however, induction becomes indispensable. Because geometric descriptions and mechanistic explanations pursue different scientific objectives, they consequently rely upon different epistemic methods for validating their central claims. Geometric descriptions can be validated against observation through hypothetico-deductive reasoning, whereas mechanistic explanations ultimately require induction to establish that their proposed mechanisms correspond to reality. Mechanistic hypotheses may certainly be proposed through hypothetico-deductive reasoning, but deduction alone cannot establish that the proposed mechanism is the true cause of the observed regularities. Recognizing this methodological distinction is essential for faithfully developing a mechanistic framework that identifies physical causes while remaining mathematically consistent with observational regularities.

2.3.1 Geometric Validation: Hypothetico–Deductive Method

Geometric frameworks are characteristically developed using the hypothetico–deductive (HD) method. In this approach, candidate principles are proposed, their mathematical consequences are deduced, and those consequences are compared with observation. Frameworks whose predictions agree with observation are retained and refined, whereas those that fail are revised or discarded [27, 28].

This method has proven extraordinarily effective because it systematically evolves descriptive frameworks toward increasingly accurate mathematical representations of observational regularities. Through repeated comparison with observation, mathematical descriptions progressively improve while preserving only those formulations that successfully account for the empirical evidence. In this sense, the method functions as an iterative process of empirical refinement, progressively retaining the descriptive principles that best correspond with observation while discarding those that do not. Its strength therefore lies in optimizing descriptive adequacy rather than establishing that its adopted principles correspond to the underlying physical causes [24, 25, 29].

However, its limitation follows directly from the nature of deduction itself. Deduction can preserve the truth of its premises but cannot establish them. At least one premise in every deductive framework must be a generalization by definition, and generalizations cannot themselves be validated deductively. Consequently, hypothetico–deductive frameworks may achieve extraordinary predictive success while leaving open whether their foundational assumptions correctly identify the mechanisms responsible for the observed regularities.

Within such frameworks, explanatory depth is therefore often constrained by the initial assumptions. Once a postulate is adopted, the explanatory burden shifts to the mathematical structure it generates. Questions concerning why those postulates hold, whether the resulting ontology corresponds to reality, or what physical entities and interactions give rise to the observed regularities typically remain outside the scope of the framework itself. Geometry and mathematical description consequently become the principal focus of the theory [7].

2.3.2 Mechanistic Validation: Axiomatic and Causal Induction

Because mechanistic frameworks seek to identify the entities, interactions, and physical conditions responsible for observed regularities, they require a different epistemic foundation for validating their explanatory claims. Rather than beginning from assumed postulates, they can begin from facts that are directly and universally accessible and that cannot be coherently denied. These foundational recognitions are established through *axiomatic induction*: the identification of truths that are implicit in all observation and cognition.

Chief among these are the axioms of existence and identity: that something exists, and that whatever exists is something specific. From these follows the law of causality, here understood as identity applied to action—that entities act in accordance with what they are. These principles are not hypotheses adopted for convenience, nor conclusions inferred from theory; they are recognitions grounded in the conditions

of awareness itself. Without them, the very possibility of explanation, prediction, or controlled experiment would collapse [23, 24, 30].

Causal induction builds directly upon this axiomatic foundation. Because entities act according to their identities, and because identical essential conditions produce identical actions, causal relationships can be discovered by systematically varying conditions and observing corresponding changes in behavior. This logic is formalized in Mill’s Methods of Induction [31, 32], which provide explicit procedures for isolating causal relationships.

Mill articulated two complementary methods that play a central role throughout the analysis that follows [31, 32]:

- **Method of Agreement (MoA):** If an effect persists while conditions vary, the cause must lie in what remains invariant.
- **Method of Difference (MoD):** If a change in conditions is accompanied by a corresponding change in effect, the cause must lie among the conditions that changed.

These methods, illustrated in Figure 1, introduce no new assumptions. Rather, they operationalize the axioms of identity and causality by providing explicit procedures for identifying causal relationships through controlled comparison. Whereas the hypothetico–deductive method excels at constructing descriptive frameworks, Mill’s methods provide a principled means of discovering physical mechanisms directly from observed reality.

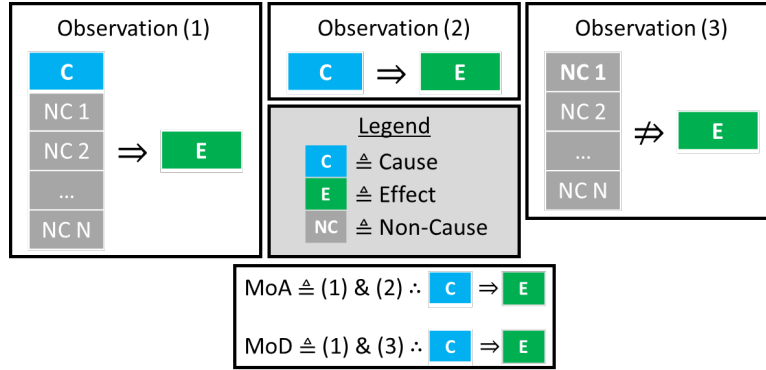


Fig. 1: Mill’s method of agreement and method of difference.

The distinction between geometric description and mechanistic explanation is therefore reflected in the epistemic methods by which they are validated. Geometric frameworks excel at constructing increasingly accurate descriptions of observational regularities, whereas mechanistic frameworks require axiomatic and causal induction to establish that their proposed mechanisms correspond to reality. Having established

the epistemic method appropriate for discovering physical mechanisms, we next lay the conceptual foundation upon which a mechanistic paradigm is constructed.

2.4 The Conceptual Foundation of this Mechanistic Paradigm

Having established the epistemic foundations appropriate for discovering physical mechanisms, we next establish the foundational concepts upon which this mechanistic paradigm rests. Unlike the geometric paradigm, they are not consequences of relativistic postulates, but foundational concepts of the mechanistic paradigm itself.

The discussion proceeds from existence (simultaneity) to the evolution of existence (time), to the alteration of that evolution (time dilation), and finally to the means by which those changes are observed (measurement). Each concept is first established within the mechanistic paradigm and then contrasted with its counterpart in the geometric paradigm. Together, these concepts provide the conceptual foundation required for the causal discoveries developed in the sections that follow.

2.4.1 Simultaneity

Mechanistic Paradigm: Universal Presentism

The mechanistic paradigm developed in this paper begins from a foundational recognition: all observation, interaction, and measurement occur in the present. Every act of awareness, every experiment, and every physical interaction takes place now. References to the past exist only as present records or memories, and projections of the future are formed only from present facts.

From this universally accessible observation follows an axiomatic recognition: *whatever exists, exists now*. This principle—*universal presentism*—is not a hypothesis or assumed postulate, but an axiom discovered through axiomatic induction. It is implied in all cognition and cannot be coherently denied without presupposing it [23, 24, 29, 30].

Universal presentism is a metaphysical claim, not an epistemological one. It does not assert that distant simultaneity can be operationally determined; rather, it asserts that all existence is confined to the present regardless of our ability to determine distant simultaneity operationally. If two spatially separated entities both exist, their existence is confined to the same present. Disagreement between observers about simultaneity therefore reflects limitations of measurement and knowledge, not indeterminacy in existence.

Geometric Paradigm: Relativity of Simultaneity

The geometric paradigm adopts a different definition of simultaneity. Within relativity, simultaneity is not treated as a universal fact but as a frame-dependent relation. It is deduced from the postulate that the speed of light in vacuum is invariant for all inertial observers. Because distant clock synchronization depends on light signals, observers in relative motion assign different simultaneity relations to the same events. In this view, planes of simultaneity rotate through spacetime according to an observer's state of motion or position within a gravitational potential [33–36].

This conception is internally consistent within the geometric paradigm, but simultaneity is not treated as a fact about what exists. Rather, it is a coordinate-dependent relation that emerges from the mathematical structure required to preserve light-speed invariance.

The distinction therefore lies not merely in how simultaneity is described, but in what simultaneity is taken to be. Within the mechanistic paradigm, simultaneity is a fact of existence; within the geometric paradigm, it is a coordinate-dependent relation emerging from the theory's mathematical structure.

2.4.2 Time

Mechanistic Paradigm: Time as the Interval Over Which Change Occurs

If universal presentism confines all existence to the present, then a crucial task remains: to account for change. The present is not static. What exists now is continually transforming—what we observe, experience, and interact with at one moment differs from what exists moments later. Yet if only the present exists, then this transformation must occur *within* the present itself.

To remain consistent with the law of identity, change cannot be instantaneous—an entity cannot simultaneously be in a state of existence and not be in that same state. From universally accessible observation, processes unfold through intermediate states, and these transformations occur over measurable intervals. Time is therefore induced as *the interval over which change occurs*.

This conception of time is grounded in causality. Entities act according to their nature, and when identical essential conditions recur, identical changes unfold over identical durations. Time is not an entity or dimension that acts on matter; it is a relational measure that compares one process of change to another.

Clocks do not measure an independent substance called time. They compare the progression of one physical process to another. If all physical change were to cease, time would cease with it.

This conception integrates naturally with universal presentism: if only the present exists, and change occurs within the present, then time is nothing more—and nothing less—than the measured duration of such change.

Geometric Paradigm: Time as a Spacetime Dimension

The geometric paradigm adopts a different definition of time. Within relativity, time is treated as a coordinate within a four-dimensional spacetime manifold. Events are located by temporal and spatial coordinates, and physical processes are represented as worldlines. Time is not defined by change, but by its role within the metric structure that relates events [33–36].

Within this conception, physical processes do not unfold *in* time so much as trace paths through spacetime. Time is a geometric parameter, interchangeable with space under Lorentz transformations [34–36].

The distinction therefore lies not merely in how time is described, but in what time is taken to be. Within the mechanistic paradigm, time is the interval over which physical change occurs. Within the geometric paradigm, time is a coordinate within spacetime that relates events.

2.4.3 Time Dilation

Mechanistic Paradigm: Slowing of Physical Processes

The mechanistic paradigm developed here defines time dilation as a physical effect that alters the rate at which an entity's internal processes operate, rather than a change in time as an independent entity. Because time is not taken to exist independently, but rather as a relational measure of physical change, it cannot itself be altered directly. When we say time dilates, we therefore mean that the rates of physical processes are reduced such that the same physical processes require longer intervals to complete.

Consider the following example. Identical clocks that remain co-located under the same physical conditions operate at the same rate. When identical clocks are subjected to different physical conditions, the rates of their internal processes can differ, so less physical change occurs during the same interval. Once reunited under common conditions, the clocks again operate at the same rate, but they generally indicate different accumulated intervals resulting from the period of separation. It is not time itself that has changed; rather, one clock underwent less physical change during the same interval because its internal processes operated more slowly.

What distinguishes time dilation from other changes affecting measuring instruments is that it does not selectively alter some processes while leaving others unchanged. Rather, the rates of all physical processes are altered uniformly under the same physical conditions. Under this conception, kinematic and gravitational time dilation are naturally unified: what differs is the source of the altered physical conditions, not the nature of time itself.

Because time is the interval over which change occurs, time dilation corresponds to a real lengthening of the interval required for the same physical change to occur, caused by altered physical conditions acting upon physical entities.

Time dilation is thus defined as:

Within the mechanistic paradigm, time dilation refers to a uniform reduction in the rates of physical processes caused by altered physical conditions, such that the same physical processes require longer intervals to complete.

Geometric Paradigm: Differences in Spacetime Path Lengths

The geometric paradigm adopts a different definition of time dilation. Within relativity, time dilation appears as a difference in proper time along different worldlines. Within the geometric paradigm, nothing physically happens to clocks or processes. Different trajectories simply possess different invariant lengths in spacetime [34–36].

For example, one observer may traverse two events in one year while another traverses the same events in six months. Within the geometric paradigm, this difference is attributed to one worldline possessing a shorter proper-time interval through spacetime. Time dilation is therefore not treated as a physical effect acting upon clocks or physical processes, but as a geometric consequence of spacetime itself. Within the geometric paradigm, the observed differences in elapsed time are fully accounted for by the geometry of spacetime, without introducing an additional physical mechanism acting upon clocks or other physical processes [34–36].

The distinction therefore lies not merely in how time dilation is described, but in what is taken to change and what ultimately explains the observed effect. Within the mechanistic paradigm, altered physical conditions slow the rates of physical processes, causing the same physical processes to require longer intervals to complete. Within the geometric paradigm, differing spacetime intervals fully account for the observed differences in elapsed time, without introducing an additional physical mechanism acting upon clocks or other physical processes.

2.4.4 Measurement

Mechanistic Paradigm: Measuring Devices as Physical Systems

Within the mechanistic paradigm, measuring devices are themselves physical systems and are therefore subject to the same causal influences as every other physical entity. If altered physical conditions slow physical processes or change the distances over which they occur, then clocks, rods, and every other instrument used to perform measurements must be altered as well.

Measurements therefore reflect not only the external phenomena being observed, but also the physical state of the instruments performing the observation. Apparent invariances must consequently be interpreted with care, since they may arise either from genuine invariance in the external phenomenon or from corresponding changes in the measuring systems themselves.

Every measurement is fundamentally a comparison between physical systems. Consequently, an observed difference alone cannot establish whether the measured system changed, the measuring system changed, or both changed together. Conversely, an unchanged measurement does not establish that neither system changed, since both may have changed by the same amount.

This recognition is essential for causal analysis. Because measurement alone cannot determine which physical system changed—or whether both changed together—causal identification requires accounting for possible changes in both the measuring system and the system being measured. Only then can genuine physical invariance be distinguished from apparent invariance produced by compensating changes within the measuring instruments or the measured system.

Geometric Paradigm: Measuring Devices as Local Standards

The geometric paradigm adopts a different role for measurement. Within relativity, rods and clocks serve as local standards by which spacetime intervals are operationally defined. The measured constancy and isotropy of the speed of light are incorporated directly into this operational framework through the light postulate and the associated synchronization procedures. Consequently, observed differences in measurements are attributed to the geometric relationships between observers rather than to additional physical mechanisms acting upon the measuring devices themselves [33–36].

The distinction therefore lies not merely in how measurements are interpreted, but in the causal role assigned to measuring devices. Within the mechanistic paradigm, measuring devices are ordinary physical systems whose behavior must itself be explained by the same physical mechanisms affecting everything else. Within the geometric paradigm, measuring devices provide the operational basis for defining

spacetime relationships, and the observed measurement differences are accounted for through the geometry of spacetime rather than additional physical changes acting upon the instruments.

The order in which these concepts were developed is itself a consequence of the mechanistic paradigm. Beginning from existence, simultaneity provides the foundation for understanding time; time provides the foundation for understanding time dilation; and each, in turn, informs how measurement must be understood. Within the geometric paradigm, the logical progression proceeds in essentially the opposite direction: operational measurement provides the starting point for the relativistic postulates, from which the mathematical relationships governing time and time dilation are deduced, while relativity of simultaneity emerges alongside them as a consequence of the same relativistic structure. With these contrasting conceptual foundations established, the mechanistic paradigm is now equipped to proceed from the observed relativistic regularities toward the discovery of their physical causes.

2.5 Conclusion: *Via ad Inventionem ab Inductione Incipit*

This section has established the methodological and conceptual groundwork required for mechanistic discovery. We first distinguished geometric description from mechanistic explanation and showed that each scientific objective requires an epistemic method appropriate to the claims it seeks to validate. Whereas the hypothetico-deductive method provides a powerful means of constructing and validating mathematical descriptions against observation, validating physical causes ultimately requires induction.

On this epistemic foundation, we established the foundational concepts of the mechanistic paradigm: simultaneity as a fact of existence through universal presentism, time as the interval over which change occurs, time dilation as the slowing of physical processes, and measurement as a physical interaction involving measuring systems that are themselves subject to causal influence. Contrasting each with its geometric counterpart revealed that these conceptual differences reflect a deeper distinction in scientific objective. The geometric paradigm seeks to describe observed regularities through spacetime geometry, whereas the mechanistic paradigm treats those regularities as effects whose physical causes remain to be discovered.

The groundwork for causal discovery is therefore complete. Relativity already provides extraordinarily successful quantitative predictions of the observational regularities under investigation; those predictions can consequently serve as empirical constraints upon the mechanistic explanations developed here. The task ahead is not to replace those successful predictions, but to determine what physical conditions produce the effects they describe. The following section begins this process by applying Mill's Methods of Agreement and Difference to isolate the physical condition responsible for time dilation, establishing the first causal link in the mechanistic progression toward the unification of time dilation, gravitation, and inertia.

3 Inducing the Cause of Time Dilation

In the preceding section, time dilation was defined within the mechanistic paradigm as a real physical effect: the uniform slowing of physical processes such that the same processes require longer intervals to complete. The task now is to discover the physical condition responsible for that effect in both kinetic and gravitational contexts.

The investigation proceeds inductively using Mill's Methods of Agreement and Difference. Rather than assuming the relevant causal parameter at the outset, competing candidates are progressively isolated and tested against the conditions under which time dilation occurs. The analysis begins with kinetic time dilation, where the physical conditions associated with time dilation are simplest, then extends to include gravitational time dilation. Through this progression, a single causal relationship is sought that accounts for time dilation across all cases.

Before this causal analysis can proceed, however, the observationally successful predictions of relativity must first be placed within the mechanistic paradigm developed in the preceding section. This requires establishing why the geometric paradigm cannot itself isolate the physical cause of time dilation, and how the same empirical predictions can instead provide the quantitative constraints required for causal discovery.

3.1 Opening the Door to Causal Discovery

Relativity provides extraordinarily successful quantitative predictions of time dilation. Within the geometric paradigm, these predictions are expressed through relationships between spacetime intervals and require no additional physical mechanism acting upon the systems involved. Once the observed regularities are accounted for by the metric, the descriptive objective of the geometric paradigm has been satisfied. Within the mechanistic paradigm, however, the same predictions acquire a different role. If time dilation represents a real slowing of physical processes, then the predicted change becomes an effect for which a physical cause remains to be discovered. Relativistic predictions may therefore be retained as empirical constraints without requiring their geometric interpretation to be retained with them.

The difficulty in discovering the physical cause of time dilation directly within the relativistic framework arises from the symmetry built into its description. Mill's Methods of Agreement and Difference require causal factors to correspond systematically with the effect across the cases being compared so that those factors can be isolated. In special relativity, however, the reciprocity of inertial frames prevents the effect from being universally assigned to either system as a physical change. For two systems in uniform relative motion, either may regard itself as stationary and the other as moving, and each assigns the corresponding time dilation effect to the other. More generally, for any given system, a continuum of inertial frames exists with different relative velocities, each assigning a different magnitude of time dilation to that system. No inertial frame is privileged within the geometric paradigm as identifying the system's physical condition. Consequently, the reciprocal relationships alone cannot universally identify which system is physically slowed or assign a unique magnitude to

that slowing as a physical property of the system. Without such agreement, the conditions producing the effect cannot be varied and isolated in the manner required by Mill’s Methods. An analogous obstacle to causal isolation arises in the gravitational context and will be addressed when the analysis is extended to gravitational time dilation. Thus, while the geometric paradigm successfully predicts time dilation, its symmetries leave the causal conditions producing that effect underdetermined: within the geometric paradigm, the door to causal discovery remains closed.

The key to opening that door lies within the mechanistic paradigm established in the preceding section: the additional independent physical constraint supplied by universal presentism removes this causal ambiguity. Universal presentism establishes that all existence and interaction are confined to a single physically real present. Special relativity, by contrast, derives relativity of simultaneity from its postulates, with different inertial frames assigning different and mutually incompatible simultaneity relations [33–35, 37]. There is only one physically real present, while the continuum of inertial frames defines a corresponding continuum of different planes of simultaneity. These simultaneity relations cannot all correspond to the single present required by universal presentism. Consequently, only one of these simultaneity relations can correspond to physical reality.

Absent gravitational effects, the inertial frame possessing this simultaneity relation is defined here as the *preferred frame*: the inertial frame whose plane of simultaneity corresponds to the physically real present. In gravitational settings, the same requirement generalizes to a preferred foliation whose simultaneity plane corresponds to the universal present, together with the local preferred frame associated with that foliation at each location. Where gravitational effects vanish, these local preferred frames reduce to a single global preferred inertial frame.

Introducing such a frame immediately raises an empirical question: if one inertial frame is physically preferred, why do measurements performed in uniformly moving frames continue to exhibit the observational equivalence predicted by special relativity?

The answer follows from the mechanistic conception of measurement established in the preceding section. In the absence of gravitational effects, the physical one-way speed of light is isotropic in the preferred frame and anisotropic relative to frames moving with respect to it. This physical anisotropy does not imply that observers in those frames will measure an anisotropic speed of light. Measurements of the one-way speed employing Einstein synchronization continue to return c in each direction, while two-way measurements independently return c . Because rods and clocks are themselves physical systems, systematic changes in the measuring systems can compensate for differences in the underlying physical quantities and thereby preserve these observed invariances.

The null result of the Michelson–Morley experiment provides the classic experimental example of this measured two-way isotropy and therefore does not by itself exclude a preferred frame. Lorentzian dynamical interpretations historically demonstrated how physical changes to moving measuring systems can render a preferred frame empirically indistinguishable through such measurements while reproducing the same observable relationships described by special relativity [21, 22, 37, 38]. Appendix A develops the

corresponding mechanistic derivation used here, showing how the Lorentz transformations and the null Michelson–Morley result are recovered when the measuring systems themselves are permitted to change.

The preferred frame is therefore an ontological reference frame without necessarily being an operationally identifiable one. Directly identifying motion relative to it would require access to a calibrated one-way speed of light without first assuming a synchronization convention, which remains operationally inaccessible [39–41]. Its operational inaccessibility does not prevent it from supplying the physical reference required for causal analysis, just as the inability to identify that frame experimentally does not prevent the Lorentz transformations from supplying empirically successful quantitative predictions.

The result is precisely the combination required for causal discovery. The preferred frame does not remove the observational reciprocity captured by the Lorentz transformations; it removes the *causal ambiguity* that prevents the physical conditions associated with time dilation from being isolated. Motion relative to the preferred frame supplies a nonreciprocal physical condition between otherwise observationally equivalent systems, while the Lorentz transformations provide the quantitative relationships that any successful mechanistic explanation must reproduce. Mill’s Methods can therefore be applied without abandoning the empirical predictions of special relativity.

With these conditions established, causal discovery can now begin. We first consider kinetic time dilation, where controlled comparisons between systems at rest and in motion relative to the preferred frame allow competing causal candidates to be progressively isolated and eliminated.

3.2 Evaluating Competing Hypotheses via Mill’s Methods

The causal analysis now proceeds by testing competing hypotheses against controlled cases using Mill’s Methods of Agreement and Difference. Candidates inconsistent with the presence, absence, or magnitude of time dilation are progressively eliminated, while those remaining viable are refined and carried forward into increasingly general cases.

The plausible causal factors can be organized into three broad families. The first concerns the *motion of the system*: whether time dilation is caused by relative motion generally or by some specific property of motion with respect to the preferred frame. The second concerns *changes in the motion of the system*: whether the effect originates during acceleration or in the physical action required to change the system’s state of motion. The third concerns the *rate of causal interaction within the system*: whether changes in the rate at which interactions between spatially separated entities are completed alter the rate of physical change itself. Each family begins with its simplest plausible candidate and is progressively refined as competing hypotheses are eliminated.

We begin with kinetic time dilation, where these physical conditions can be examined in their simplest form, before introducing gravitational effects to separate candidates that remain observationally coupled in the kinetic case.

3.2.1 Kinetic Cases Only

We begin with uniform motion relative to the preferred frame in the absence of gravitational effects. Before the effect can be compared across physical conditions, however, a distinction must be made between *calibrated* and *uncalibrated* measurements. As established previously, measuring devices are themselves physical systems and are therefore subject to the same physical effects as the systems they measure. A measurement taken directly from an affected measuring system without correcting for those effects is referred to here as *uncalibrated*. A measurement corrected to the physical standard of the preferred frame is referred to as *calibrated*.

Measurement and Representation Notation.

The notation used in this work distinguishes both the physical state of the measuring instrument and the manner in which its measurement is represented. A prime denotes a direct, uncalibrated operational quantity measured by instruments co-located and comoving with the affected system. A subscript- $\star$ denotes a direct, uncalibrated operational quantity measured by instruments co-located and comoving with the affected system, but is calibrated for the preferred frame time coordinate. A subscript-0 denotes a direct, uncalibrated operational quantity measured by instruments co-located with the affected system, but stationary with respect to the preferred frame, but is calibrated for the preferred frame time coordinate. Lastly, an unsubscripted and unprimed quantities denotes an operational measurement made by an instrument calibrated to the preferred frame coordinates, hence retaining the coordinate meaning assigned by the particular representation under consideration. Thus, in general,

$$q' : \text{ uncalibrated comoving measurement, } \quad (1)$$

$$q_\star : \text{ comoving measurement, calibrated time but not spatial measurements, } \quad (2)$$

$$q_0 : \text{ co-located measurement, calibrated time but not spatial measurements, } \quad (3)$$

$$q : \text{ measurement calibrated to preferred frame coordinates. } \quad (4)$$

Operational identification of the preferred frame is not required for the analysis. If the preferred frame were known, measuring systems could in principle be calibrated with respect to it. Moreover, the analysis can be performed for any candidate inertial frame taken to be the preferred frame: if that frame were the physically preferred frame, the corresponding corrected measurements would be the calibrated quantities. We may therefore proceed as though the preferred frame is known, without requiring its experimental identification, since the conclusions derived herein apply to whichever inertial frame ultimately corresponds to the physically real present.

The effect to be explained is the reduction in the rate of physical change within a system moving relative to this frame. From the Lorentz transformations (see Appendix A.5), this reduction is quantified by the ratio of the uncalibrated time interval dt' measured by the affected system to the corresponding calibrated interval dt :

$$\frac{dt'}{dt} = \sqrt{1 - \frac{\frac{1}{2}v_p^2}{\frac{1}{2}c^2}} = \gamma_K^{-1}, \quad (5)$$

where:

- dt' is the uncalibrated time interval measured in the system under consideration,
- dt is the corresponding calibrated time interval,
- v_p is the velocity of the system relative to the preferred frame, and
- c is the invariant numerical value returned by uncalibrated two-way light-speed measurements.

For any nonzero v_p , $dt'/dt < 1$, corresponding within the mechanistic paradigm to less physical change occurring within the moving system during the same calibrated interval. The causal question is therefore precise: *what physical condition corresponds systematically with this reduction and can account for its magnitude?*

Motion of the System: From Relative Velocity to Preferred-Frame Speed

The first family of hypotheses concerns the motion of the system itself. A natural starting point is the candidate inherited from special relativity, *frame-wise relative velocity*: perhaps time dilation is caused simply by motion relative to another physical system. This hypothesis fails under the Method of Agreement. Consider two systems in relative motion where one remains at rest in the preferred frame. As the relative velocity between them changes, the system at rest in the preferred frame remains undilated. The effect therefore remains unchanged despite variation in frame-wise relative velocity, ruling out relative velocity itself as the causal factor (**MoA**).

The hypothesis can be refined by replacing arbitrary relative velocity with velocity v_p measured specifically with respect to the preferred frame. This introduces the required nonreciprocal physical reference, but the signed velocity still contains more information than the effect depends upon. Systems moving at $v_p = +v$ and $v_p = -v$ experience identical time dilation because Equation (5) depends upon v_p^2 . Reversing the direction of motion therefore changes v_p while leaving the effect unchanged. Signed velocity is consequently ruled out (**MoA**).

What survives is the *speed* of the system relative to the preferred frame, $|v_p|$. Unlike relative velocity or signed preferred-frame velocity, $|v_p|$ varies monotonically with the magnitude of kinetic time dilation and vanishes when kinetic time dilation is absent. The motion family therefore narrows to speed with respect to the preferred frame as its remaining viable candidate.

Changes in Motion: From Acceleration to Specific Work

The second family concerns the physical process by which the system's motion changes. A natural first hypothesis is *acceleration*. Acceleration is present when a system is brought from rest to motion relative to the preferred frame and is often invoked when discussing the asymmetry of the twin paradox. It is not, however, present while kinetic

time dilation persists during subsequent uniform motion. A system can coast inertially with zero acceleration while continuing to exhibit time dilation. The effect therefore differs from the undilated case even when acceleration is absent, ruling out acceleration as the sustaining cause of kinetic time dilation (**MoD**).

The hypothesis may then be refined from acceleration to the *work done* in changing the system's motion. Work done remains associated with the physical change that produced the new state of motion even after acceleration has ceased. Total work done, however, does not correspond uniquely with the magnitude of time dilation. The same total work done on systems of different mass produces different speeds and therefore different amounts of time dilation (**MoD**). Conversely, different amounts of total work done can produce the same speed, and hence the same time dilation, when the masses differ appropriately (**MoA**). Total work done therefore fails to track the effect uniquely under both the Method of Difference and Method of Agreement.

The natural refinement is *specific work done with respect to the preferred frame*: the work associated specifically with changing the system's motion after the inertial contribution has first been removed. As derived from first principles in Appendix B, the specific quantities are constructed beginning with specific momentum, followed by specific force and then specific work. Removing inertia prior to integration is important because the inertial contribution can itself vary throughout the change in motion and therefore cannot, in general, be cleanly factored from the integrated quantity afterward.

Let w_p denote the specific work done with respect to the preferred frame required to establish the system's kinetic state relative to that frame. The derivation in Appendix B gives

$$w_p = \frac{1}{2}v_p^2. \quad (6)$$

Equation (5) can therefore be written directly in terms of specific work:

$$\frac{dt'}{dt} = \sqrt{1 - \frac{w_p}{\frac{1}{2}c^2}} = \gamma_K^{-1}. \quad (7)$$

Thus, within the kinetic cases considered here, specific work and preferred-frame speed remain inseparable as candidate causal factors: each uniquely determines the other and therefore corresponds identically with the time-dilation effect.

Rate of Causal Interaction: From Light Propagation to Interaction Cycles

The third family concerns the rate at which causal interactions can occur within the system. Because physical change requires interactions between spatially separated entities, a natural hypothesis is that kinetic time dilation results from a reduction in the speed at which causal influence propagates relative to the moving system.

The directional speed of light provides the simplest candidate. In a system moving uniformly relative to the preferred frame, the physical speed of light relative to that

system becomes anisotropic. In the transverse direction, the propagation speed is reduced by exactly the factor associated with kinetic time dilation (see Appendix A), making the hypothesis initially promising [21]. Along the direction of motion, however, light propagates too slowly in the direction of travel and too quickly against the direction of travel to match the observed time-dilation factor. The proposed cause is therefore direction dependent while the observed slowing of physical processes is isotropic. Because the effect remains uniform despite these directional differences, the raw directional speed of light cannot itself be the causal quantity (**MoA**).

This failure suggests that the relevant quantity may instead involve propagation over a complete interaction cycle—a single back-and-forth interaction between spatially separated entities—rather than propagation in either direction considered independently. A natural refinement is therefore the average two-way speed of light relative to the system. To distinguish the physical propagation speeds from the invariant numerical value c returned by uncalibrated measurements, let $\bar{c}'$ denote the uncalibrated average two-way light speed relative to the system and $\bar{c}$ the corresponding calibrated average two-way speed.

Averaging the opposing directions removes the forward–reverse asymmetry, but it does not remove the directional dependence of the resulting two-way propagation speed. As shown in Appendix A, the calibrated two-way propagation speed along the direction of motion decreases more than the calibrated two-way propagation speed in the transverse direction. The proposed cause therefore remains anisotropic while the time-dilation effect remains isotropic. Because the effect is unchanged despite this variation in the candidate causal quantity, the average two-way propagation speed is likewise ruled out (**MoA**).

The remaining discrepancy reveals what propagation speed alone neglects. An interaction between spatially separated entities depends not only upon how rapidly causal influence propagates, but also upon the physical path length over which that propagation must occur. The same physical conditions that alter propagation speed also alter the separation between interacting entities through length contraction. An anisotropic propagation speed therefore need not imply an anisotropic interaction rate if the path length required to complete the interaction cycle changes correspondingly—and it does (see Appendix A).

The relevant causal quantity is consequently the *interaction cycle rate*: the frequency with which causal interactions between spatially separated entities can be completed. Consider a complete interaction cycle between two entities. Let $d\ell'$ denote the uncalibrated interaction-cycle path length and $d\ell$ the corresponding calibrated interaction-cycle path length. Likewise, let f' denote the uncalibrated interaction-cycle frequency measured within the affected system and f the corresponding calibrated interaction-cycle frequency. The interaction-cycle frequencies may therefore be written as

$$f' = \frac{\bar{c}'}{d\ell'}, \quad (8)$$

$$f = \frac{\bar{c}}{d\ell}. \quad (9)$$

Their ratio is

$$\begin{aligned}\frac{f}{f'} &= \frac{\bar{c}/d\ell}{\bar{c}'/d\ell'} \\ &= \frac{\bar{c}}{\bar{c}'} \frac{d\ell'}{d\ell}.\end{aligned}\tag{10}$$

The calibrated average propagation speed over the interaction-cycle path is

$$\bar{c} = \frac{d\ell}{dt}.\tag{11}$$

Substituting this expression for $\bar{c}$ into the numerator of the preceding ratio gives

$$\begin{aligned}\frac{f}{f'} &= \frac{1}{\bar{c}'} \left(\frac{d\ell}{dt} \frac{d\ell'}{d\ell} \right) \\ &= \frac{1}{\bar{c}'} \frac{d\ell'}{dt}.\end{aligned}\tag{12}$$

The quantity $d\ell'/dt$ represents the uncalibrated interaction-cycle path length $d\ell'$ completed per calibrated time interval dt . We define this quantity as the *speed of causality*, $c_\star$:

$$c_\star \equiv \frac{d\ell'}{dt}.\tag{13}$$

The frequency ratio therefore becomes

$$\frac{f}{f'} = \frac{c_\star}{\bar{c}'}.\tag{14}$$

The uncalibrated average two-way light-speed measurement remains invariant (see Appendix A), so

$$\bar{c}' = c = \frac{d\ell'}{dt'}.\tag{15}$$

Consequently,

$$\frac{c_\star}{c} = \frac{d\ell'/dt}{d\ell'/dt'} = \frac{dt'}{dt}.\tag{16}$$

Combining this result with the interaction-cycle frequencies gives

$$\frac{f}{f'} = \frac{c_\star}{c} = \frac{dt'}{dt} = \gamma_K^{-1}. \quad (17)$$

For purely kinetic time dilation,

$$\begin{aligned} c_\star &= c \sqrt{1 - \frac{\frac{1}{2}v_p^2}{\frac{1}{2}c^2}} \\ &= c \sqrt{1 - \frac{w_p}{\frac{1}{2}c^2}} = c\gamma_K^{-1}. \end{aligned} \quad (18)$$

What survives as the cause of time dilation within rate of causal interaction family is the interaction cycle rate, f . In addition, this rate has an equivalent representation as an interaction speed that causal influence propagates, the speed of causality, $c_\star$, which has a direct physical interpretation. It is a speed-valued representation of the rate at which causal interaction cycles can be completed once both altered propagation and altered interaction-cycle path length are taken into account. Unlike the raw directional or calibrated two-way propagation speeds, $c_\star$ is isotropic with respect to the internal processes of the affected system and varies in exactly the same proportion as the observed reduction in physical change.

Surviving Kinematic Candidates

The kinetic cases therefore narrow the original hypotheses to one surviving member from each causal family: *speed with respect to the preferred frame* $|v_p|$, *specific work done with respect to the preferred frame* w_p , and the *speed of causality* $c_\star$. Importantly, these survivors are not independent. Within the kinetic context they form a natural causal sequence: specific work done with respect to the preferred frame changes the system's speed relative to the preferred frame, which changes the speed of causality, which in turn varies in exact proportion to the observed slowing of physical processes.

Their mathematical relationship can be summarized directly as

$$\frac{dt'}{dt} = \frac{c_\star}{c} = \sqrt{1 - \frac{\frac{1}{2}v_p^2}{\frac{1}{2}c^2}} = \sqrt{1 - \frac{w_p}{\frac{1}{2}c^2}}. \quad (19)$$

Thus, the three surviving candidates are mathematically linked in the same order suggested by their physical relationship:

$$w_p \longrightarrow |v_p| \longrightarrow c_\star \longrightarrow \frac{dt'}{dt}. \quad (20)$$

Kinematic cases alone cannot yet determine which member of this chain represents the more general causal condition, because all three remain systematically coupled whenever kinetic time dilation occurs. The speed of causality is nevertheless the surviving candidate most proximal to the observed effect: it directly represents the reduction in the rate at which causal interaction cycles are completed, while specific work done and preferred-frame speed describe physical conditions further upstream in the same sequence. Gravitational time dilation provides the additional cases required to separate these remaining candidates and determine which relationship persists beyond the kinetic context.

3.2.2 Gravitational Cases Only

The kinetic analysis left one viable candidate from each of the three causal families: speed with respect to the preferred frame, $|v_p|$; specific work done with respect to the preferred frame, w_p ; and the speed of causality, c_* . Because these quantities remain systematically coupled in purely kinetic cases, those cases alone cannot determine which represents the more general causal condition. Gravitational time dilation provides the cases required to separate them.

Before these hypotheses can be tested, however, the causal ambiguity identified previously for general relativity must be resolved. As in the kinetic case, the difficulty is not a failure to predict elapsed proper time, but the absence within the geometric paradigm of a uniquely identified amount of time dilation that can be assigned as a physical property of a system in a given state.

The Gravitational Causal Ambiguity

Measurements such as the Shapiro time delay establish that the round-trip propagation of light through a gravitational field requires more time than corresponding propagation outside the field [42]. What such measurements do not establish is how that additional propagation time is distributed between the two one-way directions. The measured round-trip delay is therefore compatible with multiple underlying one-way descriptions.

This freedom is reflected in different coordinate representations of the same gravitational system. In Schwarzschild coordinates, the radial coordinate speed of light decreases symmetrically in the inward and outward directions (see Appendix C). In Painlevé–Gullstrand coordinates, the same gravitational system can instead be represented by an asymmetric inward–outward propagation structure, as illustrated physically by the river model (see Appendix D). The inward-flowing and outward-flowing Painlevé–Gullstrand descriptions provide limiting alternatives, with a continuum of intermediate choices between them.

These differences in one-way light speed are not merely differences in propagation coordinates; they also determine how spatially separated clocks are synchronized and therefore which distant events are assigned to the same simultaneity plane. They also assign different amounts of time dilation to a given system in a given state of motion, just as the continuum of inertial frames does in the special-relativistic case. The physical conditions responsible for time dilation therefore cannot be uniquely isolated within the geometric paradigm.

The same independent constraint used in the mechanistic paradigm for the kinetic case continues to resolve this ambiguity. Universal presentism requires a single physically real present. Accordingly, among the admissible representations of a gravitational system, only the one whose simultaneity plane corresponds to that present can represent physical reality. Which particular representation satisfies this requirement need not be operationally known for the causal analysis to proceed. The admissible coordinate representations or foliations may instead be leveraged to determine which causal relationship survives regardless of which foliation ultimately corresponds to physical reality.

This establishes the requirement for the gravitational analysis: a universal cause of time dilation will be metric-independent. The causal relationship sought here must remain valid whichever admissible representation ultimately corresponds to the physically real present, just as the surviving causal candidates in the kinetic analysis had to remain valid regardless of which inertial frame ultimately proves to be preferred.

Metric-Independent Constraint

Relativistic metrics may therefore be retained in the mechanistic paradigm for the same reason that the Lorentz transformations were retained in the kinetic analysis: their empirically successful predictions provide quantitative constraints without requiring adoption of their conventional geometric interpretation. Within the mechanistic paradigm, relativistic metrics are instead interpreted as a mapping between calibrated and uncalibrated measurements of the same underlying physical process. A general constraint on time dilation can consequently be established before any particular gravitational metric is considered.

For a timelike interval, relativistic metrics satisfy

$$ds = c dt' \tag{21}$$

where ds , c , and dt' are conventionally interpreted as the spacetime interval, the speed of light constant, and the proper-time interval, respectively [34–36].

Dividing Equation (21) by $c dt$ gives

$$\begin{aligned} \frac{ds}{c dt} &= \frac{c dt'}{c dt} \\ &= \frac{dt'}{dt}. \end{aligned} \tag{22}$$

This relationship establishes that, within any given admissible representation, ds/dt varies in exact proportion to the rate of physical change relative to that representation's coordinate time.² Because Equation (22) follows from the general timelike metric interval, this relationship holds for whichever admissible metric representation ultimately corresponds to the physically real present.

²This relationship is general in form; it does not require ds/dt to have the same numerical value under changes of temporal coordinate.

Applying the Schwarzschild Metric

The remaining causal hypotheses can now be tested using any admissible representation capable of separating them. For a stationary spherical gravitating body, the Schwarzschild metric provides a convenient and empirically successful description. We therefore provisionally treat its stationary coordinate frame as though its simultaneity plane corresponds to the physically real present and determine what follows.

The Schwarzschild line element is [35]

$$ds^2 = \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 - \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 - r^2 d\Omega^2. \quad (23)$$

For a system stationary with respect to these coordinates,

$$dr = d\Omega = 0, \quad (24)$$

and therefore

$$\frac{ds}{dt} = c \sqrt{1 - \frac{2GM}{rc^2}}. \quad (25)$$

Using Equation (22),

$$\frac{dt'}{dt} = \sqrt{1 - \frac{2GM}{rc^2}} = \gamma_g^{-1}. \quad (26)$$

The Schwarzschild representation therefore supplies a gravitational case in which the three candidates surviving the kinetic analysis can be distinguished.

Testing Speed with Respect to the Preferred Frame

The first surviving candidate is speed with respect to the preferred frame, $|v_p|$. Consider two identical stationary systems located at different radii within the gravitational field. Under the Schwarzschild candidate presently being considered, both remain stationary relative to the same candidate preferred frame, so that

$$|v_p| = 0 \quad (27)$$

for each. Nevertheless, Equation (26) predicts different rates of physical change at the two radii. The effect therefore changes while the proposed causal factor remains unchanged, ruling out $|v_p|$ as a metric-independent cause of time dilation (**MoD**), even though it remains correlated with the effect under another admissible representation (see Appendix D).

Testing Specific Work Done with Respect to the Preferred Frame

The second surviving candidate is specific work done with respect to the preferred frame, w_p . As defined in the kinetic analysis and developed from first principles in Appendix B, this quantity represents the specific work required to establish a system's state of motion relative to the preferred frame.

For the stationary systems considered under the Schwarzschild candidate, no kinetic state relative to the candidate preferred frame has been established:

$$w_p = 0. \quad (28)$$

Yet the systems experience different amounts of gravitational time dilation depending upon their positions in the gravitational field. The effect therefore again changes while the proposed causal factor remains unchanged, ruling out w_p as a metric-independent cause of time dilation (**MoD**), even though it too remains correlated with the effect under another admissible representation (see Appendix D).

Testing the Speed of Causality

The final surviving candidate is the speed of causality, c_* . Testing this candidate requires returning to the metric interval. From the kinetic analysis, dt' represents the *uncalibrated time interval* over which the corresponding physical interaction cycle is completed. Let $d\ell'$ denote the uncalibrated interaction-cycle path length completed during dt' . The uncalibrated measurement of the interaction-cycle speed remains

$$c = \frac{d\ell'}{dt'}. \quad (29)$$

Substituting Equation (29) into Equation (21) gives

$$ds = dt' \frac{d\ell'}{dt'} = d\ell'. \quad (30)$$

The same quantity that appears geometrically as the invariant spacetime interval therefore admits a direct mechanistic interpretation: ds represents the uncalibrated interaction-cycle path length $d\ell'$.

Dividing by the calibrated time interval dt ,

$$\frac{ds}{dt} = \frac{d\ell'}{dt}. \quad (31)$$

But the kinetic analysis defined the right-hand side as the speed of causality:

$$c_* \equiv \frac{d\ell'}{dt}. \quad (32)$$

Hence,

$$\frac{ds}{dt} = c_\star. \quad (33)$$

Substituting this result into Equation (22) gives

$$\boxed{\frac{c_\star}{c} = \frac{dt'}{dt}}. \quad (34)$$

The final candidate therefore possesses precisely the property that preferred-frame speed and specific work lack: its relationship to time dilation is independent of which admissible representation ultimately corresponds to the physically real present.

For the stationary Schwarzschild case,

$$c_\star = c\sqrt{1 - \frac{2GM}{rc^2}}, \quad (35)$$

so $c_\star$ decreases with decreasing radius in exactly the same proportion as the gravitational time-dilation factor. Under a Painlevé–Gullstrand representation, the assigned one-way propagation speeds, simultaneity plane, and preferred-frame motion differ, yet the same relationship between $c_\star$ and time dilation remains valid (see Appendix D).

The speed of causality therefore satisfies Mill’s Methods across the kinetic and gravitational cases. Wherever the time-dilation effect is the same, $c_\star/c$ has the same value despite changes in the admissible representation and in the upstream physical conditions. Likewise, whenever the rate of physical change differs, $c_\star$ differs in exactly the same proportion. Of the three candidates surviving the kinetic analysis, $c_\star$ alone remains universally coupled to the effect regardless of which admissible metric corresponds to physical reality.

Surviving Gravitational Candidate

The gravitational cases break the coupling among the three candidates that survived the kinetic analysis. Preferred-frame speed and specific work may remain correlated with time dilation under some admissible representations or foliations, such as the Painlevé–Gullstrand river model, but both fail under the Schwarzschild representation. They therefore cannot constitute the metric-independent cause of time dilation.

The speed of causality alone survives every admissible metric representation because its relationship to the effect follows directly from the metric interval:

$$\boxed{\frac{dt'}{dt} = \frac{c_\star}{c}}. \quad (36)$$

The gravitational analysis therefore identifies the speed of causality as the causal condition common to both kinetic and gravitational time dilation, regardless of which admissible representation ultimately corresponds to the physically real present.

Because Equation (34) was established from the general timelike metric interval rather than from either the kinetic or Schwarzschild cases specifically, its applicability to the general case has already been established in principle. Independent first-principles derivations of $c_\star$ for the kinetic, gravitational, and combined cases are provided in Appendix E, where $c_\star$ is constructed directly from calibrated light propagation and uncalibrated interaction-cycle path length. The causal analysis of time dilation is therefore complete.

3.3 Conclusion: *Ex Inductione, Causa Invenitur*

Through the successive application of Mill's Methods of Agreement and Difference, competing hypotheses for the metric-independent cause of time dilation were not merely eliminated but progressively refined, thereby revealing which physical conditions were relevant and which were not. This inductive process isolated a single causal quantity that survived every test: the *speed of causality*, $c_\star$.

The analysis demonstrated that $c_\star$ alone remains universally coupled to the effect regardless of whether time dilation is considered in kinetic or gravitational contexts, or represented through different admissible coordinate representations and foliations. Whenever the time-dilation effect remains unchanged, $c_\star$ remains unchanged (**MoA**); whenever the effect varies, $c_\star$ varies in exactly the same proportion (**MoD**). It therefore satisfies both of Mill's criteria across all cases examined.

This result does more than identify a cause. It unifies kinetic and gravitational time dilation under a single causal condition. Although different admissible metric representations assign different one-way propagation structures, simultaneity planes, preferred-frame motions, and coordinate descriptions to the same observable phenomena, the relationship between $c_\star$ and the time-dilation factor remains unchanged.

Within the mechanistic paradigm, relativistic metrics therefore acquire a different physical interpretation. Rather than treating their geometric formalism as itself the explanation for relativistic effects, they may be understood as quantitative mappings between calibrated and uncalibrated measurements of the same underlying physical processes. The particular metric determines how those quantities are related under a given representation, while the causal relationship between $c_\star$ and the rate of physical change remains invariant across the admissible alternatives. The empirical success of relativistic metrics is therefore retained without requiring that any particular geometric representation be assigned ontological privilege.

The inductive analysis therefore identifies $c_\star$ as the universal speed of causality governing relativistic time dilation. This result completes the causal analysis of time dilation and establishes a unified physical basis for its kinetic and gravitational manifestations.

The analysis that follows considers an equivalent representation of this same causal condition in terms of specific internal energy. Expressing $c_\star$ in this alternative

form extends the preceding unification beyond time dilation alone and provides the foundation for unifying time dilation, gravitation, and inertia.

4 Unifying Time Dilation, Gravitation, and Inertia

The previous section established that time dilation, whether kinetic or gravitational, is governed by the speed of causality, c_* , an equivalent representation of the frequency with which interaction cycles are completed between physically separated entities. Time dilation was therefore shown to arise from a specific physical condition rather than from geometry alone.

One immediate consequence of this result is that the internal motion of entities undergoing time dilation must also slow. Whenever time dilation occurs, internal processes take longer to complete relative to systems experiencing less time dilation, or equivalently, they proceed more slowly. Because clocks are themselves physical systems whose operation depends upon those same internal processes, measurements of time dilation necessarily reflect changes in internal motion.

Around gravitating bodies, these reductions in internal motion vary systematically with position, implying the existence of a measurable field that modulates the local value of c_* and, consequently, the rate of all physical change, including motion. The physical agent responsible for producing this field remains unknown, much as Kepler did not yet know what property of the Sun governed planetary motion [3], or Newton what local mechanism produced gravitation itself [26]. What matters at this stage is that the field's effects are measurable and systematic. As the local value of c_* decreases, time dilation increases and internal processes slow proportionally. The field therefore represents the present boundary between known effect and unknown cause while simultaneously pointing toward the next frontier of discovery.

The purpose of this section is to establish a unified causal account of time dilation, gravitation, and inertia. We proceed in four stages. First, we develop a scalar quantity that characterizes the internal motion inferred from time dilation. This quantity, which serves as an equivalent representation of c_* , is identified as specific internal energy. Second, we demonstrate that this scalar naturally reproduces the correct equations of motion through a Lagrangian formulation, confirming that the induced mechanistic quantities remain consistent with relativistic predictions. Third, guided by this predictive correspondence, we hypothesize that spatial gradients in this scalar constitute the causal origin of both gravitation and inertia. In the fourth stage, we apply Mill's Methods of Agreement and Difference to the phenomena of weightedness and weightlessness to isolate the true causal factor inductively. Finally, having identified the causal origin of gravitation and inertia, we induce the Equivalence Law, explaining the physical basis for the observational regularity captured by the equivalence principle.

4.1 Inducing Specific Internal Energy from Time Dilation

Time dilation corresponds to a reduction in the rate at which internal processes occur. As an object approaches the speed of light or nears an event horizon, its capacity for internal change diminishes. It is therefore natural to ask whether the empirically

established relationship governing time dilation reveals a more fundamental scalar quantity that characterizes internal motion.

Consider the Schwarzschild time-dilation relation expressed as

$$\frac{dt'}{dt} = \sqrt{\frac{\frac{1}{2}c^2 - \frac{GM}{r} - \frac{1}{2}\dot{r}^2\gamma_g^2 - \frac{1}{2}r^2\dot{\Omega}^2}{\frac{1}{2}c^2}}. \quad (37)$$

A remarkable observation follows from this expression: the numerator consists entirely of recognizable specific-energy terms. As demonstrated in Appendix B, each term in the numerator can be expressed as a specific-energy quantity. The first term, $c^2/2$, serves as the reference energy. The second term represents a change in specific gravitational potential energy, while the remaining terms represent changes in specific kinetic energy.

Solving for the numerator yields

$$\frac{1}{2}c^2 \left(\frac{dt'}{dt} \right)^2 = e_{P_0} - e_{K_0}. \quad (38)$$

Where (see Appendix B),

$$e_{P_0} = \frac{1}{2}c^2 - \frac{GM}{r} \quad (39)$$

$$e_{K_0} = \frac{1}{2}\dot{r}_0^2 + \frac{1}{2}r_0^2\dot{\Omega}_0^2 = \frac{1}{2}\dot{r}^2\gamma_g^2 + \frac{1}{2}r^2\dot{\Omega}^2 \quad (40)$$

$$(41)$$

The same quantity can also be expressed in terms of the speed of causality:

$$\frac{1}{2}c_\star^2 = \frac{1}{2}c^2 \left(\frac{dt'}{dt} \right)^2 = e_{P_0} - e_{K_0}. \quad (42)$$

Although first identified through the Schwarzschild metric, this scalar is therefore not restricted to any particular metric construction. Rather, it is an alternative representation of the same metric-independent causal condition previously identified through $c_\star$. Expressed as a specific energy, this quantity provides a scalar measure of the interaction-cycle rate between physically separated entities and therefore quantifies the *internal motion of those interactions*.

We therefore identify this scalar as the system's *specific internal energy*,

$$e_{I_\star} = \frac{1}{2}c_\star^2 = \frac{1}{2} \left(\frac{ds}{dt} \right)^2 = \frac{1}{2}g_{\mu\nu} \frac{x^\mu}{dt} \frac{x^\nu}{dt}. \quad (43)$$

To develop a physical interpretation of this quantity, it is instructive to consider purely kinetic and gravitational cases, summarized in Figure 2.

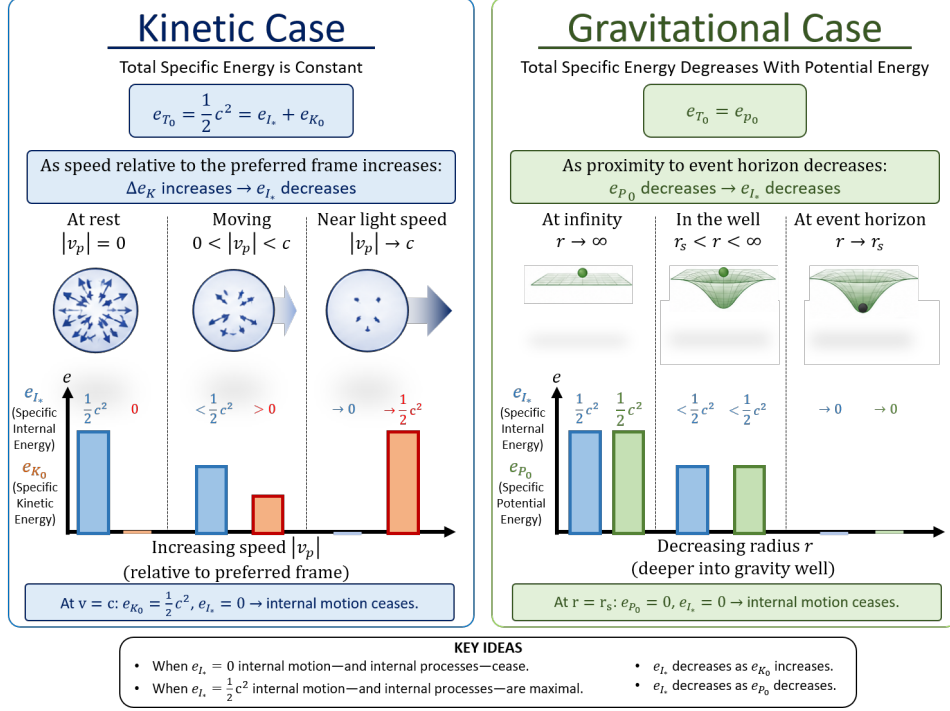


Fig. 2: Specific Internal Energy Cases.

In the absence of gravitation, the total specific energy remains fixed at

$$e_{T_0} = e_{\max} = \frac{1}{2}c^2. \quad (44)$$

The specific internal and kinetic energies therefore satisfy

$$e_{T_0} = e_{I_*} + e_{K_0}. \quad (45)$$

At rest with respect to the preferred frame, the specific internal energy reaches its maximum value,

$$e_{I_*} = e_{T_0} = e_{\max} = \frac{1}{2}c^2. \quad (46)$$

As translational motion increases, specific kinetic energy increases while specific internal energy decreases by an equal amount. In the limiting case as the speed approaches c , the specific internal energy approaches zero and internal motion ceases.

For a system held stationary in the Schwarzschild representation, an analogous relationship exists in purely gravitational cases. As the system is held at spatial infinity, where the gravitational contribution vanishes, the specific internal energy again reaches its maximum value,

$$e_{I_*} = e_{T_0} = e_{P_0} = \frac{1}{2}c^2. \quad (47)$$

As the system is held at progressively smaller radius, its specific internal energy decreases. In the limit $r \rightarrow r_s$, $e_{I_*} \rightarrow 0$, corresponding to the cessation of internal motion.

These kinetic and gravitational cases, reveal that specific internal energy provides a common scalar description of internal motion across both forms of time dilation. The identification of specific internal energy therefore represents the induction of a new scalar quantity from the observed relationships governing time dilation. This quantity provides the conceptual foundation for the analysis that follows.

4.2 Gravitation and Inertia

The previous subsection established specific internal energy as a scalar quantity induced from the observed relationships governing time dilation. Although identified through the Schwarzschild time-dilation relation, the resulting quantity was shown to be an alternative representation of the same metric-independent causal condition previously identified through c_* .

As demonstrated in Appendix B, this scalar naturally decomposes into specific internal, potential, and kinetic energy terms:

$$e_{I_*} = e_{P_0} - e_{K_0}, \quad (48)$$

$$e_{P_0} = \frac{1}{2}c^2 - \frac{GM}{r}, \quad (49)$$

$$e_{K_0} = \frac{1}{2}\dot{r}_0^2 + \frac{1}{2}r^2\dot{\Omega}_0^2 = \frac{1}{2}\dot{r}^2\gamma_g^2 + \frac{1}{2}r^2\dot{\Omega}^2. \quad (50)$$

This decomposition reveals an important correspondence. The specific internal energy induced from time dilation is simply the negative of the corresponding Lagrangian expression. This observation provides a predictive connection between the mechanistic quantities developed here and the mathematical formalism of relativity.

4.2.1 The Emergence of the Specific Lagrangian

Recognizing that

$$e_{I_\star} = e_{P_0} - e_{K_0} = -(e_{K_0} - e_{P_0}), \quad (51)$$

we immediately identify the specific internal energy as the negative of the specific Lagrangian:

$$e_{I_\star} = -\mathcal{L}_0(t, r_0, \dot{r}_0, \Omega_0, \dot{\Omega}_0) = -(e_{K_0} - e_{P_0}), \quad (52)$$

with $\Omega_0 = \Omega$ in this Schwarzschild representation. From this relation, the following radial coordinate acceleration follows from the Euler–Lagrange equations and the chain rule (Appendix F):

$$\ddot{r}_\nabla = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2 + r\dot{\Omega}^2\gamma_g^{-2}. \quad (53)$$

This establishes an important predictive result: once specific internal energy is introduced as a scalar quantity induced from time dilation, the standard Lagrangian machinery naturally generates components of the radial equation of motion.

Predictive agreement, however, does not establish causation. Lagrangian formalisms reliably reproduce equations of motion even when the underlying physical origin of their terms remains unknown. Predictive consistency may therefore motivate a causal hypothesis, but it cannot by itself identify the physical cause underlying the observed motion.

4.2.2 The Causal-Potential Hypothesis

The quantity $e_{I_\star}$ measures a system’s capacity for internal change. When $e_{I_\star} = 0$, the system is frozen internally and incapable of internal change. When $e_{I_\star} = e_{\max}$, its internal motion and capacity for change are maximal. Time dilation is therefore not merely a reduction in $c_\star$; it quantifies a reduction in causal capacity.

Because $e_{I_\star}$ is a scalar that measures causal capacity, and therefore a potential for action, it is natural to interpret it as a form of potential energy. We therefore define

$$e_{U_\star} \equiv e_{I_\star}, \quad (54)$$

and refer to $e_{U_\star}$ as the system’s *causal potential*. This does not introduce a new physical quantity. Rather, it reinterprets the previously induced specific internal energy according to the role it will play in the following hypothesis.

At this stage, we introduce the following hypothesis:

Hypothesis (Causal-Potential Gradient): Spatial gradients in causal potential generate the specific forces responsible for gravitation and inertial resistance, analogously to the way in which gradients in other potentials generate forces throughout mechanics, electromagnetism, and thermodynamics.

This hypothesis is strongly motivated by the general role of potentials across physics [43–45]; however, it remains only a hypothesis at this stage. Before attempting an inductive demonstration, we first determine whether its mathematical expression is consistent with the predictive framework already established through the Lagrangian.

4.2.3 Predictive Validation of the Causal-Potential Hypothesis

If the causal-potential hypothesis is correct, then its gradient-generated specific force contribution is obtained from the spatial partial gradient of the causal potential while holding the generalized velocities fixed (Appendix G):

$$\begin{aligned} f_{g_0} &= -\nabla_0 e_{U_*} = -\nabla_0 e_{I_*} = -\nabla_0 \frac{1}{2} c_*^2 \\ &= -\frac{GM}{r^2} \gamma_g^{-1} + r \dot{\Omega}^2 \gamma_g^{-1}. \end{aligned} \quad (55)$$

Applying the chain rule to convert from uncalibrated path length to calibrated coordinates yields

$$f_g = -\frac{GM}{r^2} \gamma_g^{-2} + \frac{GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^2 + r \dot{\Omega}^2 \gamma_g^{-2} \triangleq -\nabla e_U, \quad (56)$$

which coincides exactly with $\ddot{r}_\nabla$ in Equation (53).

This agreement is significant, but it remains *predictive*: it shows that the causal-potential hypothesis reproduces the same energy-gradient term already present in the Lagrangian dynamics. This predictive success is promising and motivates continued investigation. However, predictive agreement alone cannot establish causation. As discussed in Section 2, causal discovery requires an inductive demonstration, which we defer until the following subsection.

4.2.4 A Mechanistic Decomposition of Relativistic Acceleration

To fully integrate this picture with relativity’s standard prediction, note that relativity’s Schwarzschild radial coordinate acceleration is (Appendix H):

$$\ddot{r} = -\frac{GM}{r^2} \gamma_g^{-2} + \frac{3GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^2 + r \dot{\Omega}^2 \gamma_g^{-2}. \quad (57)$$

This expression differs from $\ddot{r}_\nabla$ by a factor of three in the middle term. The question, then, is: *Why the difference?*

A closer examination reveals that Equation (57) can be decomposed into two contributions that combine linearly, one of which is precisely $\ddot{r}_\nabla$:

$$\ddot{r} = \ddot{r}_\nabla + \ddot{r}_V. \quad (58)$$

The additional term is

$$\ddot{r}_V = \ddot{r} - \ddot{r}_\nabla = \frac{2GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^2. \quad (59)$$

Equation (53) reveals another important property. In this coordinate representation, the acceleration reaches its maximum magnitude for objects without initial radial motion and decreases continuously until it vanishes at the radial coordinate speed of light (including light itself).

However, the Schwarzschild radial coordinate speed of light varies with position (Appendix C):

$$c_r = \frac{dr}{dt} = \pm c \gamma_g^{-2}. \quad (60)$$

Thus, light experiences a coordinate acceleration due to the Velocity Damping Field (VDF) surrounding gravitating objects:

$$\begin{aligned} \ddot{r}_{c_r} &= \frac{d^2 r}{dt^2} = \frac{dc_r}{dt} = \frac{dc_r}{dr} \frac{dr}{dt} = \frac{d(\pm c \gamma_g^{-2})}{dr} \frac{dr}{dt} \\ &= \frac{2GM}{r^2} \frac{c_r}{c^2} \gamma_g^2 \frac{dr}{dt} = \frac{2GM}{r^2} \frac{c_r^2}{c^2} \gamma_g^2. \end{aligned} \quad (61)$$

The same principle applies to all material objects because the ratio between an object's velocity and the local speed of light remains invariant in both calibrated and uncalibrated representations:

$$\frac{\dot{r}}{c_r} = \frac{\dot{r}'}{c'_r} = \frac{\dot{r}'}{c}. \quad (62)$$

An object could therefore achieve its present velocity with an initial velocity $\dot{r}_{r \rightarrow \infty}$ if it applied a counteracting force equal to the local gravitational acceleration induced by Equation (53).

Even though this counteracts the local energy-gradient acceleration, the velocity ratio remains fixed such that

$$\frac{\dot{r}}{c_r} = \text{constant} = \frac{\dot{r}_{r \rightarrow \infty}}{c}, \quad (63)$$

which implies

$$\dot{r} = \frac{\dot{r}_{r \rightarrow \infty}}{c} c_r = \dot{r}_{r \rightarrow \infty} \gamma_g^{-2}. \quad (64)$$

Hence, the same cause that slows the coordinate speed of light near a massive body—the VDF—also slows the speed of every other object according to

$$\begin{aligned}\ddot{r}_V &= \frac{d^2 r}{dt^2} = \frac{d\dot{r}}{dt} = \frac{d\left(\frac{\dot{r}_{r \rightarrow \infty}}{c} c_r\right)}{dr} \frac{dr}{dt} \\ &= \frac{\dot{r}}{c_r} \frac{dc_r}{dr} \frac{dr}{dt} = \frac{2GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^2 \quad \blacksquare\end{aligned}\tag{65}$$

Making the total Schwarzschild radial coordinate acceleration decomposition

$$\ddot{r} = \ddot{r}_\nabla + \ddot{r}_V = -\nabla e_U + \frac{\dot{r}}{c_r} \frac{dc_r}{dt}.\tag{66}$$

Although these contributions arise through different mechanisms, both originate from the same underlying field. The first reflects the VDF’s spatial modulation of causal potential, producing an energy-gradient contribution to motion, whereas the second reflects the VDF’s direct modulation of local velocity. This decomposition provides a bridge between relativity’s successful predictive form and a mechanistic interpretation: relativity supplies the total effect, while the mechanistic picture distinguishes two physically different contributions.

At this stage, the causal-potential hypothesis has demonstrated predictive consistency: the negative gradient of e_{U*} reproduces the energy-gradient contribution to relativistic acceleration, and the full Schwarzschild result can be decomposed into energy-gradient and velocity-damping components. Yet, as emphasized in Section 2, predictive consistency alone cannot establish causal origin.

If gradients in causal potential are truly the cause of force, then their presence and absence must systematically coincide with the presence and absence of weight. This is precisely the criterion required by Mill’s Methods of Agreement and Difference. We therefore turn to the phenomena of weight and weightlessness to determine whether gradients in causal potential cause weight.

4.2.5 Inducing the Cause of Weight and Weightlessness

We now apply Mill’s Methods of Agreement (**MoA**) and Difference (**MoD**) to determine whether the presence and absence of weight systematically coincide with the presence and absence of a sustained net causal-potential gradient. The aim is not to assume a mechanism, but to isolate the physical condition uniquely associated with weight and weightlessness.

Although the following cases are discussed qualitatively, the underlying causal-potential gradients and resulting specific forces have already been derived quantitatively in the preceding subsection and in Appendix I. The qualitative descriptions that follow are therefore interpretations of previously established quantitative results rather than independent assumptions.

Consider four canonical cases spanning the combinations required for the present comparison:

1. **Stationary in a nonuniform VDF (standing on Earth).**
2. **Uniform acceleration in empty space (uniform VDF).**
3. **Free fall in a nonuniform VDF.**
4. **Floating in empty space.**

Case 1: Stationary in a nonuniform VDF.

A spatial variation in the VDF establishes a gradient in causal potential across the body. The corresponding specific force follows directly from the energy-gradient relations derived previously. Because motion is constrained, this produces a sustained *net causal-potential gradient*. Internal stress manifests as the sensation of weight and is measurable as a contact force on the ground.

Case 2: Uniform acceleration in empty space.

Acceleration itself produces a front–rear gradient in causal potential across the body (Appendix I). Again, a sustained *net causal-potential gradient* is present, and internal stress manifests as the sensation of weight and is measurable as a contact reaction force opposing the direction of acceleration.

Case 3: Free fall in a nonuniform VDF.

The spatial variation in the VDF still establishes an external gradient in causal potential across the body. For a sufficiently small freely falling body over which the gravitational field may be treated as locally uniform, and neglecting tidal variation across the body, the acceleration-induced kinematic gradient locally opposes the gravitationally induced internal gradient. To leading order under these assumptions, the two gradients cancel (Appendices I and J).

The result is not the absence of a field gradient, but the cancellation of the *net causal-potential gradient* within the body.

The internal stress associated with the sensation of weight vanishes, and the contact force registered by a comoving scale becomes zero. The body is therefore weightless.

Free fall therefore represents a state of *dynamic internal equilibrium* under the locally uniform-field approximation.³ Although the external field remains present, the acceleration-induced gradient exactly cancels the gravitationally induced internal gradient. Any deviation from this state immediately reintroduces a nonzero net causal-potential gradient, causing internal stress—and thus the sensation of weight—to reappear.

Weightlessness therefore signifies not the absence of a field or of acceleration, but the restoration of *net causal-potential equilibrium*.

Case 4: Floating in empty space.

No field-induced gradient exists, and no acceleration-induced gradient exists. The causal potential is spatially uniform. No net causal-potential gradient is present, and the internal stress associated with the sensation of weight is absent. The body is weightless.

³Tidal effects, which arise from variation of the gravitational gradient across a finite body, are neglected in this comparison and may produce residual differential effects.

Causal isolation.

All cases in which weight is experienced (Cases 1 and 2) share one common condition: the presence of a sustained *net causal-potential gradient* (**MoA**).

All cases in which weight is absent (Cases 3 and 4) share one common condition: the absence of a sustained *net causal-potential gradient* (**MoD**).

Alternative causal hypotheses fail this test.

Gravitation alone cannot be the cause because, in free fall, the gravitational field remains present while weight vanishes (**MoD**), and acceleration produces weight in the absence of gravitation (**MoA**).

Coordinate acceleration alone cannot be the cause because free fall involves non-zero coordinate acceleration without weight (**MoD**), while a body held stationary on Earth has zero coordinate acceleration while experiencing weight (**MoA**).⁴

Contact forces also fail this test because they are consequences of the condition producing weight rather than its cause.

No other relevant candidate condition examined consistently coincides with the presence or absence of the sensation of weight across these cases. The only remaining condition is the presence or absence of a sustained net causal-potential gradient.

Inductive conclusion.

The causal origin of the sensation of weight is therefore isolated:

A sustained net causal-potential gradient is the cause of the sensation of weight, and its absence is the cause of weightlessness.

This establishes inductively that sustained gradients in causal potential generate the force associated with the sensation of weight in both gravitational and inertial contexts.

The causal-potential hypothesis, in which the negative gradient of causal potential generates a specific force, is therefore no longer a conjecture, but an inductively established causal law.

4.2.6 The Cause of Gravitation and Inertia

The previous subsection established that sustained net causal-potential gradients are the cause of weight. We now extend that result to gravitation and inertia more generally.

Because the force associated with weight was shown to arise from the negative gradient of causal potential, the corresponding force law may now be stated without qualification:

$$\vec{f}_{\nabla_0} = -\nabla_0 e_{U_*} = -\nabla_0 e_{I_*}. \quad (67)$$

⁴Proper acceleration is not treated here as an independent ontological cause; within the mechanistic paradigm it is an operational measure of the reaction associated with the weighted state and is related to coordinate acceleration as described in Appendix I.

Gravitation.

In gravitational settings, the nonuniform VDF establishes a spatial gradient in causal potential. When motion is constrained, the resulting internal imbalance manifests as weight. When motion is unconstrained, acceleration generates an opposing kinematic gradient that restores internal equilibrium and produces weightlessness.

Gravitation therefore arises from spatial gradients in causal potential, while the observed coordinate acceleration reflects both the energy-gradient and velocity-damping contributions identified previously.

Inertia.

In uniform VDF settings, acceleration itself generates a causal-potential gradient across the body through the Born-rigid gradient derived in Appendix I. The system responds by generating a specific reaction force equal and opposite to the imposed acceleration. This induced reaction force suggests a possible mechanistic basis for Newton’s Third Law. Though, given this foundation, it would not take much to show that reciprocal forces between two interacting bodies arise from this same mechanism.

This reaction force is inertia: an object’s resistance to a change in motion. It is the causal response of a system to a disturbance of its internal energetic equilibrium.

When the applied force ceases, the causal-potential gradient vanishes, internal stress relaxes, and the system returns to equilibrium. No force is then required to maintain uniform motion, providing a mechanistic explanation for Newton’s First Law.

Rotational motion provides an analogous extension of the same mechanism. Rotation induces radial causal-potential gradients throughout the rotating body, producing the internal stresses responsible for inertial reactions.

Having shown that gravitation and inertia share an identical causal origin—gradients in causal potential—we can now reformulate the equivalence principle as a law grounded in mechanism rather than as a postulate.

4.3 The Equivalence Law

The preceding analysis yields a single unifying result. In every case examined, measured weight arises from a sustained net causal-potential gradient. Whenever such a gradient is present, a specific force is produced; whenever it is absent, no force is sustained. This relation was not assumed, but isolated inductively through Mill’s Methods applied to the complete set of weighted and weightless conditions.

Traditionally, the equivalence principle states that the local effects experienced in a uniformly accelerating frame in empty space are indistinguishable from those experienced in a stationary frame within a gravitational field and that free fall is locally indistinguishable from inertial motion. These similarities are commonly attributed to spacetime geometry [34, 35].

Within the present framework, these similarities arise from a different cause: the presence or absence of a sustained net causal-potential gradient.

Standing on Earth and uniform acceleration both produce weight because both sustain a net causal-potential gradient. Conversely, free fall and floating in empty space are both weightless because no sustained net causal-potential gradient exists within the body.

What is equivalent is not the situations themselves, but the underlying cause responsible for the similarity between them. What lies beneath this understanding is a deeper causal law:

Equivalence Law:

Gravitational and inertial resistance forces, measured as weight, arise from spatial gradients in causal potential and are given by

$$\vec{f}_{\nabla_0} = -\nabla_0 e_{U_\star}.$$

The equivalence principle therefore emerges not as a fundamental postulate, but as the observational consequence of a more general causal law. The Equivalence Law identifies the common causal mechanism underlying the geometric description of that observational regularity.

4.4 Conclusion: *Ex Causa, Unitas Revelatur*

The analysis presented in this section establishes the causal potential, $e_{U_\star}$, as the unifying quantity underlying time dilation, gravitation, and inertia.

The relationships developed throughout this section may be summarized as

$$\begin{aligned} \frac{d\ell'}{dt'} &= c, \\ c_\star &\equiv \frac{d\ell'}{dt} = c \frac{dt'}{dt} = \frac{ds}{dt} = \sqrt{g_{\mu\nu} \frac{x^\mu}{dt} \frac{x^\nu}{dt}}, \\ e_{I_\star} &\equiv \frac{1}{2} c_\star^2 = \frac{1}{2} g_{\mu\nu} \frac{x^\mu}{dt} \frac{x^\nu}{dt}, \\ e_{I_\star} &\equiv e_{P_0} - e_{K_0}, \\ e_{I_\star} &= e_{I_\star}, \\ \mathcal{L}_0 &= -e_{I_\star} = -\frac{1}{2} g_{\mu\nu} \frac{x^\mu}{dt} \frac{x^\nu}{dt}, \\ e_{U_\star} &\equiv e_{I_\star} = \frac{1}{2} g_{\mu\nu} \frac{x^\mu}{dt} \frac{x^\nu}{dt}, \\ \vec{f}_0 &= -\nabla_0 e_{U_\star} = -\frac{1}{2} \nabla_0 \left(\frac{ds}{dt} \right)^2 = -\frac{1}{2} \nabla_0 \left(g_{\mu\nu} \frac{x^\mu}{dt} \frac{x^\nu}{dt} \right), \end{aligned}$$

and

$$\frac{dt'}{dt} = \sqrt{\frac{e_{U_\star}}{e_{\max}}} = \frac{ds}{dt c} = \frac{1}{c} \sqrt{g_{\mu\nu} \frac{x^\mu}{dt} \frac{x^\nu}{dt}},$$

where $e_{\max} = \frac{1}{I_0} c^2$. The notation used here distinguishes four measurement and representation bases. A prime denotes a direct, uncalibrated operational quantity measured by instruments co-located and comoving with the affected system. A subscript- $\star$ denotes a direct, uncalibrated operational quantity measured by instruments co-located and comoving with the affected system, but is calibrated for the preferred frame time coordinate. A subscript-0 denotes a direct, uncalibrated operational quantity measured by instruments co-located with the affected system, but stationary

with respect to the preferred frame, but is calibrated for the preferred frame time coordinate. Lastly, an unsubscripted and unprimed quantities denotes an operational measurement made by an instrument calibrated to the preferred frame coordinates, hence retaining the coordinate meaning assigned by the particular representation under consideration. Thus, in general,

$$q' : \text{ uncalibrated comoving measurement,} \quad (68)$$

$$q_\star : \text{ comoving measurement, calibrated time but not spatial measurements,} \quad (69)$$

$$q_0 : \text{ co-located measurement, calibrated time but not spatial measurements,} \quad (70)$$

$$q : \text{ measurement calibrated to preferred frame coordinates.} \quad (71)$$

These notations distinguish representations of a physical quantity. Accordingly, quantities expressed in different representations may be equal when they describe the same underlying physical condition, as in $e_{I_\star} = e_{I_\star}$.

These relationships have been established for whichever coordinate representation corresponds to the physically real present and reveal a common causal mechanism. Time dilation depends upon the local magnitude of the causal potential, whereas gravitation and inertia arise from its spatial gradient.

The same scalar quantity therefore governs both the rate at which internal processes unfold and the specific forces arising from their spatial imbalance. These phenomena are consequently not independent manifestations of nature, but complementary consequences of a common causal condition.

Thus, causal potential provides a unified causal mechanism for time dilation, gravitation, and inertia, revealing the deeper unity that awaited beneath the equivalence principle.

5 New Questions: Toward Quantum Gravity, a Unified Field Theory, and a Theory of Everything

The preceding sections established a unified causal mechanism for time dilation, gravitation, and inertia. Through axiomatic and causal induction, phenomena traditionally treated as distinct were shown to arise from variations in a common physical quantity: causal potential.

Historically, the identification of a common cause has rarely represented the end of scientific inquiry. Instead, it has often marked the beginning of new avenues of investigation.

Kepler's laws revealed a previously unrecognized unity governing planetary motion, motivating the question of what property of the Sun governed that motion [3]. Newton's law of gravitation answered that question while simultaneously unifying celestial and terrestrial motion, yet it left unanswered the question of the local mechanism responsible for gravitational interaction itself [26]. In each case, unification transformed previously independent phenomena into different manifestations of a common cause while simultaneously revealing questions that could not previously be formulated.

The present work follows the same historical progression. It identifies a common causal mechanism for time dilation, gravitation, and inertia, thereby resolving the question of the local mechanism responsible for gravitational interaction left unresolved by Newton.

Yet, as with Kepler and Newton before it, this unification simultaneously reveals new questions concerning the quantum origin of gravitation, the relationship between gravitation and electromagnetism, the possibility of a unified force, and the potential constitutional relationship between light and matter.

These questions should not be viewed as shortcomings of the present framework, but as natural consequences of the unification it provides. From this foundation, four major directions for future research emerge.

5.1 Toward Quantum Gravity

One of the most intriguing questions raised by the present framework concerns the relationship between gravitation and quantum theory.

The preceding analysis identified gravitation as a consequence of variations in causal potential, itself determined by variations in the speed of causality, c_* , which is represented through the local propagation of light. Because light is fundamentally quantum in nature [46–51], it is natural to ask whether the Velocity Damping Field responsible for these variations might also originate from quantum processes.

This possibility suggests a different perspective on the quantum-gravity problem. Rather than asking how to quantize spacetime itself, one may instead ask what quantum processes govern local variations in the speed of causality.

If quantum processes influence light propagation, and if light propagation determines the local speed of causality, then quantum fluctuations affecting light may also induce fluctuations in causal potential and, consequently, in gravitation itself.

Within this framework, the central challenge of quantum gravity becomes identifying the quantum origin of the Velocity Damping Field and determining how microscopic variations in causal propagation give rise to the macroscopic phenomena associated with gravitation.

Whether such a connection ultimately proves correct remains an open question. Nevertheless, the unification developed in the present work permits a question that could not previously be formulated: might gravitation itself emerge from the quantum behavior of light? The mechanistic relationship established between time dilation, gravitation, and causal propagation suggests that this possibility warrants further investigation.

5.2 Toward a Unified Field Theory

The present framework also raises the possibility of a deeper relationship between gravitation and electromagnetism.

Traditionally, these interactions have been treated as fundamentally distinct. Electromagnetic interactions are described by electromagnetic fields, whose radiative excitations propagate as light, whereas gravitation is typically described either as a

fundamental interaction or as a consequence of spacetime geometry. The mechanistic interpretation developed here suggests a different possibility.

The preceding analysis identified gravitation as a consequence of variations in causal potential, itself determined by variations in the speed of causality. Both are alternative representations of changes in the local propagation of light. If light participates in the mechanism governing gravitation, then a natural question emerges: might gravitation and electromagnetism represent different manifestations of the same underlying interaction?

An intriguing physical distinction motivates this possibility. Radiative electromagnetic fields possess transverse degrees of freedom relative to their direction of propagation, whereas the gravitational effects identified in the present framework arise through longitudinal variations in causal propagation. This distinction suggests that these phenomena may correspond to different physical aspects of a deeper causal process.

This possibility motivates the development of an electromagnetic–gravitational framework in which both interactions are treated within a common causal model. Unlike gravitoelectromagnetism (GEM), which establishes mathematical analogies between the two interactions [52], such an approach would seek a unified causal account.

Interestingly, one self-consistent extension of GEM known to the author employs complex-valued representations to preserve the orthogonality between gravitational and electromagnetic contributions while expressing both within a common mathematical framework [53]. A summary of this formulation is provided in Appendix K.

Einstein devoted considerable effort to the development of a unified field theory combining gravitation and electromagnetism [54–59]. Yet, without an identified causal mechanism connecting these phenomena, the path toward unification remained unclear.

Whether a mechanistic framework based on causal potential can ultimately achieve this goal remains an open question. Nevertheless, the present analysis identifies a previously unexplored causal relationship between gravitation and light, suggesting a new conceptual direction for future investigation.

5.3 Toward a Theory of Everything

If an electromagnetic–gravitational unification were ultimately confirmed, then an even broader question would naturally follow: could all known interactions arise from different manifestations of that same underlying mechanism?

Within the present framework, this possibility emerges from a simple observation. Within this speculative picture, radiative electromagnetic supply transverse degrees of freedom, whereas gravitation is associated with longitudinal variations in causal propagation. Together, these effects span all mutually orthogonal directions in three-dimensional space.

If these phenomena are indeed different manifestations of the same mechanism, then it is natural to ask whether this directional decomposition already exhausts the possible forms that a force can assume. If so, the traditional classification of the fundamental interactions may require reconsideration.

As a deliberately speculative example of the type of unification this possibility would invite, one might ask whether one possible interpretation is that the strong and weak nuclear interactions represent limiting cases of the same underlying mechanism. In these regimes, certain components may become dominant while others are suppressed.

For example, the strong interaction might correspond to a regime in which magnetic and gravitational effects dominate while electric effects become negligible. Conversely, the weak interaction might correspond to a regime in which electric and gravitational effects dominate while magnetic effects are suppressed. Conventional electromagnetism would then represent the complementary case in which gravitational effects become negligible.

Under such an interpretation, the apparent diversity of interactions would arise not because nature employs multiple fundamentally different force carriers, but because different physical regimes selectively emphasize different components of the same underlying mechanism.

Whether this possibility proves correct remains entirely speculative. Nevertheless, the mechanistic framework developed in the present work raises a question that could not previously be formulated: might all known interactions ultimately be different expressions of the same mechanism responsible for electromagnetism and gravity?

If so, the search for a theory of everything may shift away from identifying additional fundamental interactions and toward understanding how this single underlying mechanism gives rise to the diverse physical phenomena observed in nature.

5.4 Toward a Constitutional Theory of Everything

The preceding discussion considered the possibility that gravitation, electromagnetism, and perhaps even all known interactions arise from a common mechanism. A deeper question naturally follows.

If variations in the speed of causality govern both the propagation of light and the motion of material bodies, why should this be so? Why should changes in the local propagation of light influence the behavior of matter at all?

One possible answer is that light and matter are not fundamentally different categories of physical existence. Instead, they may represent different forms of the same constituent entity.

This possibility is motivated by the pervasive role of light throughout physics. Electromagnetic radiation participates in the formation, deconstruction, transformation, and interaction of matter across a wide range of physical phenomena. The behavior of charged particles is inseparable from electromagnetic interactions, particle creation and annihilation involve electromagnetic processes, and light participates in both the interactions and transformations of physical systems.

If light and matter share a common constitution, then the distinction between them may arise not from what they are, but from how the same constituent entity is organized.

Within such a picture, matter may correspond to stable, self-sustaining wave configurations, including standing waves, solitons, or other localized nonlinear excitations.

Particles would then be understood not as fundamental point-like objects, but as persistent configurations whose properties emerge from the same physical processes that govern the propagation of light.

This possibility has been explored in soliton-based models of elementary particles, where particles are treated as localized oscillatory solutions and their properties emerge from their internal dynamics [60]. More generally, nonlinear wave systems are known to support stable, particle-like solutions [61]. These results suggest that the distinction between particles and waves may reflect different organizations of the same physical substance rather than fundamentally different physical entities.

If this interpretation proves correct, then variations in the speed of causality would simultaneously influence both light and matter because both would arise from the same physical constitution.

Whether such a constitutional theory ultimately proves viable remains entirely speculative. Nevertheless, the mechanistic framework developed in the present work raises a question that could not previously be formulated: might light and matter be constituted from the same physical substance, differing only in form?

5.5 Conclusion: *Ex Unitate, Plures Inventiones Exspectant*

The purpose of this section has not been to establish additional physical laws, but to identify the questions that naturally emerge from the unified causal framework developed throughout this work.

Whether any of the possibilities explored here ultimately prove correct remains unknown. Their significance lies not in the specific hypotheses proposed, but in the recognition that causal unification expands what can be meaningfully investigated.

Scientific progress has repeatedly followed the same progression: observations reveal regularities, induction identifies their causes, causal discovery reveals unity, and unity exposes new questions.

The framework developed in the present work should therefore not be regarded as a final theory, but as a foundation from which these deeper questions may be investigated.

6 Conclusion: *Ultra Descriptionem, Inventiones Exspectant*

As the introduction made clear, relativistic physics has traditionally been developed through geometric descriptions that organize and predict observed regularities with extraordinary empirical success. The central question motivating the present work was whether those same regularities admit a deeper mechanistic explanation and whether pursuing such explanations contributes meaningfully to scientific progress.

To investigate this question, Section 2 established a mechanistic framework for interpreting relativistic phenomena by treating observations as manifestations of entities interacting according to their nature and by establishing induction as the epistemic foundation for causal discovery.

Applying Mill's Methods of Agreement and Difference, Section 3 isolated the physical cause of both kinetic and gravitational time dilation and identified it as

a metric-independent cause governing the interaction-cycle rate between physically separated entities.

Building upon this result, Section 4 developed a unified account in which time dilation depends upon the magnitude of causal potential while gravitational and inertial specific forces arise from its spatial gradients. Phenomena traditionally treated as distinct were thereby shown to be complementary manifestations of a common causal mechanism.

Section 5 demonstrated that the framework developed in the present work follows a broader historical pattern: observations reveal regularities, induction identifies their causes, causal discovery reveals unity, and unity exposes new questions and new directions for investigation. By moving beyond description toward causal explanation, the present work expands what can be meaningfully asked. In doing so, it identifies new avenues for investigating the quantum origin of gravitation, the relationship between gravitation and electromagnetism, the possibility of a unified force, and the constitutional relationship between light and matter.

In the end, the mechanistic framework developed throughout this work offers more than an alternative interpretation. It provides deeper insight into the entities involved, enables broader unification across apparently distinct phenomena, and creates opportunities for future discovery and, potentially, greater control over physical phenomena, fulfilling the promise outlined in Section 2.

From beginning to end, the work developed throughout this paper was guided by a single question concerning how scientific progress should proceed:

Descriptio an Explicatio
Description or Explanation

The progression developed throughout this work in answering that question may be summarized as follows:

Via ad Inventionem ab Inductione Incipit
The path to discovery begins with induction
Ex Inductione, Causa Invenitur
From induction, the cause is discovered
Ex Causa, Unitas Revelatur
From the cause, unity is revealed
Ex Unitate, Inventiones Expectant
From unity, discoveries await

The answer developed throughout this investigation may therefore be stated succinctly:

Ultra Descriptionem, Inventiones Expectant
Beyond description, discoveries await

7 Glossary

- **Axiomatic Induction:** An inductive method for identifying principles supported universally by observation and inescapable in experience and cognition. In this work, axiomatic induction is used to induce universal presentism from the observed singular present existence of physical entities, together with object permanence, the absence of observed exceptions, and the inescapable nature of the present in experience and cognition.
- **Calibrated Measurement:** A measurement corrected for changes imposed upon the measuring instrument itself, such as time dilation or length contraction, so that the result is expressed relative to the chosen physical reference standard.
- **Causal Capacity:** The capacity of a physical system for internal physical change, represented equivalently by its speed of causality, c_* , or its specific internal energy, $e_{I_*} = \frac{1}{2}c_*^2$. Causal capacity decreases as time dilation increases.
- **Causal Potential (e_{U_*}):** equivalent form of specific internal energy when recognizing that it quantifies the potential for internal action.

$$e_{U_*} \equiv e_{I_*} = \frac{1}{2}c_*^2. \quad (72)$$

It is the generalized state potential that may depend upon both spatial position and kinematic state. It's gradient-generated specific-force contribution is obtained from the spatial partial gradient while holding the generalized velocities fixed. Thus, returning to the same position and kinematic state returns the same value of causal potential; no assumption that e_{U_*} is conventional position-only conservative potential is required. The interpretation of specific internal energy as the potential for causal action:

$$\vec{f}_0 = -\nabla_0 e_{U_*}. \quad (73)$$

Within Universal Specificity, gravitational and inertial reaction forces are associated with such gradients.

- **Impressed Force ($\vec{f}_{\text{ext}}$):** An externally imposed specific force acting upon a system through interaction with another physical system, in the sense of Newton's impressed force. It is distinguished from reaction forces arising from gradients in causal potential and from additional velocity-dependent contributions.
- **Measurement Notation:** For relativistic measurement quantities in this work, a prime denotes an uncalibrated operational spatial measurement made by an instrument comoving with the affected system. A subscript-0 denotes an uncalibrated spatial measurement evaluated locally by a co-located instrument stationary with respect to the preferred frame. The instrument state and calibration state are conceptually distinct; neither notation should be interpreted as changing the physical location at which the measurement is evaluated.
- **Preferred Frame:** Given flat spacetime (Minkowski metric), the inertial frame whose simultaneity relation corresponds to the universal present and in which opposite one-way light propagation is isotropic. In gravitational settings, this

generalizes to a preferred foliation corresponding to the universal present together with the local preferred frames whose simultaneity agrees with that foliation. Where gravitational effects vanish, these local frames reduce to a single global preferred inertial frame.

- **Specific Force** ($\vec{f}_0$): The time rate of change of specific momentum, where specific momentum is identified with velocity:

$$\vec{f}_0 = \frac{d\vec{p}_s}{dt}, \quad \vec{p}_s = \vec{v}. \quad (74)$$

Specific force is therefore a mass-independent measure of acceleration. Gradient-generated specific forces satisfy

$$\vec{f}_0 = -\nabla_0 e_{U_\star}. \quad (75)$$

- **Specific Internal Energy** ($e_{I_\star}$): The scalar specific-energy representation of a system's capacity for internal physical change, determined by the rate at which causal interactions between spatially separated entities are completed and, consequently, by the rate at which physical change can occur within the system:

$$e_{I_\star} = \frac{1}{2}c_\star^2 = e_{P_0} - e_{K_0}. \quad (76)$$

A reduction in $e_{I_\star}$ corresponds to a reduction in the rate of internal physical processes and therefore to increased time dilation. In this framework, specific internal energy is constructed through specific momentum, specific force, and specific work rather than defined solely as total energy divided by mass.

- **Specific Lagrangian** ($\mathcal{L}_0$): The difference between specific kinetic and specific potential energy,

$$\mathcal{L}_0 = e_{K_0} - e_{P_0}, \quad (77)$$

and therefore an alternative representation of the same causal condition:

$$\mathcal{L}_0 = -e_{I_\star} = -e_{U_\star} = -\frac{1}{2}c_\star^2. \quad (78)$$

- **Speed of Causality** ($c_\star$): A speed-valued representation of the rate at which causal interactions between spatially separated entities are completed and, consequently, of the rate at which physical change can occur within a system. For timelike motion,

$$c_\star = \frac{ds}{dt} = c \frac{dt'}{dt}. \quad (79)$$

Reductions in $c_\star$ correspond directly to time dilation. Its numerical value is evaluated relative to the coordinate time of a chosen representation and therefore may

vary under a change of temporal coordiante. What remains metric-independent within the framework is its causal role and the relationship $c_*/c = dt'/dt$. The speed of causality is also an alternative representation of the same physical condition represented by specific internal energy:

$$e_{I_*} = \frac{1}{2}c_*^2. \quad (80)$$

- **Time:** The interval over which physical change occurs.
- **Time Dilation:** The lengthening of the interval over which physical change occurs as the rate of physical change is reduced through a reduction in the speed of causality, c_* . Because clocks depend upon the same internal physical processes as other systems, their measured rates change correspondingly.
- **Time Dilation Differential:** The ratio of an infinitesimal uncalibrated local clock interval to the corresponding calibrated reference interval:

$$\frac{dt'}{dt} = \frac{c_*}{c}. \quad (81)$$

It quantifies the relative rate of physical change and is less than or equal to unity.

- **Uncalibrated Measurement:** An operational measurement obtained directly from an instrument whose own physical rate or scale has been altered by relativistic conditions and has not been corrected for that alteration. Proper clock intervals and proper lengths measured by affected comoving instruments are treated as uncalibrated measurements in this work.
- **Universal Presentism:** The axiom that only the present physically exists and that all physical existence and interaction are confined to it. Universal presentism requires a single physical simultaneity relation and provides the foundational constraint from which the preferred frame is deduced.
- **Universal Specificity:** A mechanistic framework for explaining relativistic phenomena while recovering the observable relationships described geometrically by relativity. Grounded in universal presentism, it identifies the speed of causality, c_* , as the common physical condition governing kinetic and gravitational time dilation, represents that condition through specific internal energy and causal potential, and associates gradients in causal potential with gravitational and inertial specific forces.
- **Velocity Damping Field (VDF):** A hypothesized physical field proposed to account for the velocity-dependent contribution remaining after the causal-potential-gradient contribution is separated from the Schwarzschild coordinate acceleration. Its contribution is distinct from specific forces generated by gradients in causal potential. In the present work, the VDF is motivated by this residual and by the coordinate variation of light propagation; its independent existence is not established here.

8 Abbreviations

- **GEM**: Gravitoelectromagnetism
- **MoA**: Method of Agreement
- **MoD**: Method of Difference
- **VDF**: Velocity Damping Field

9 Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work the author used ChatGPT in order to improve the spelling, grammar, clarity and conciseness of the content within. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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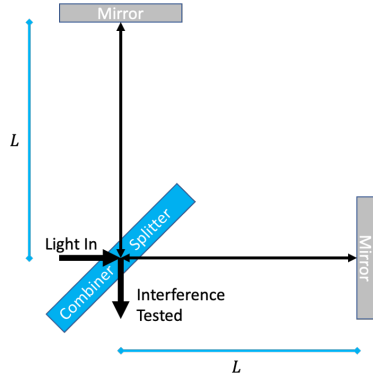
A Explaining the Apparent Isotropy of the Two-Way Speed of Light

This appendix examines how the measured two-way speed of light can remain isotropic in an inertial frame moving with respect to the preferred frame. The analysis begins with the physical propagation of light relative to the preferred frame and then considers how length contraction and time dilation affect measurements performed within the moving frame. The resulting relationship connects the apparent isotropy of the measured two-way speed of light directly to the speed of causality, c_* , developed in the main text.

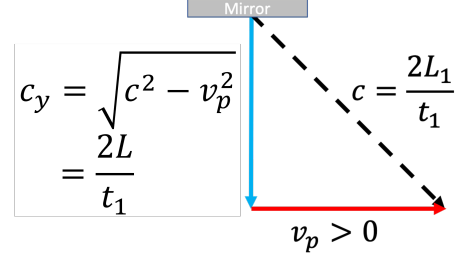
A.1 The Michelson–Morley Constraint

Consider an apparatus moving at any speed less than c with respect to the preferred frame. Because the coordinate axes of the preferred frame may be rotated without changing the physics, the x -axis can be chosen, without loss of generality, to lie parallel to the velocity of the apparatus, v_p . The orthogonal y -axis then lies transverse to that motion.

The Michelson–Morley experiment [62] provides the observational constraint that motivates the analysis. As illustrated in Figure 3a, an incoming light beam is divided into two orthogonal paths of equal measured length, reflected, and recombined. A difference in round-trip propagation time between the two paths would produce a detectable change in the resulting interference pattern. No such directional difference was observed.



(a) Michelson–Morley schematic.



(b) Trigonometric derivation of c_y .

Fig. 3: Apparatus setup and derivation of the transverse propagation speed.

If light propagates isotropically at c only in the preferred frame, an apparatus moving relative to that frame does not initially appear to satisfy this constraint. The purpose of the following analysis is therefore to determine the physical round-trip

propagation rates along the two axes and identify the conditions under which their measured values remain equal.

A.2 Two-Way Light Propagation in a Moving Frame

Consider first the transverse arm of the apparatus. From the preferred frame, the light must propagate diagonally so that it intercepts the transversely displaced mirror. The component of its propagation along the apparatus's y -axis is therefore

$$\begin{aligned} c_y &= \sqrt{c^2 - v_p^2} \\ &= c \sqrt{1 - \frac{v_p^2}{c^2}} \\ &= \gamma_K^{-1} c, \end{aligned} \tag{A1}$$

where

$$\gamma_K = \frac{1}{\sqrt{1 - \frac{v_p^2}{c^2}}}. \tag{A2}$$

Because the outbound and return portions of the transverse path are symmetric, c_y is also the average physical speed of propagation along the transverse arm as represented in the moving apparatus.

The longitudinal path is different. Light traveling toward the forward mirror must overtake a mirror moving in the same direction, whereas on the return path the light and beam splitter approach one another. For a physical longitudinal arm length L_x ,

$$\begin{aligned} t_2 &= \frac{L_x}{c - v_p}, \\ t_3 &= \frac{L_x}{c + v_p}. \end{aligned} \tag{A3}$$

The corresponding average two-way propagation speed along the longitudinal arm is therefore

$$\begin{aligned} c_x &= \frac{2L_x}{t_2 + t_3} \\ &= \frac{2L_x}{\frac{L_x}{c - v_p} + \frac{L_x}{c + v_p}} \\ &= c \left(1 - \frac{v_p^2}{c^2} \right) \end{aligned}$$

$$= \gamma_K^{-2} c. \quad (\text{A4})$$

Comparing the two directions gives

$$\begin{aligned} c_x &= \gamma_K^{-2} c, \\ c_y &= \gamma_K^{-1} c, \\ \therefore c_x &= \gamma_K^{-1} c_y. \end{aligned} \quad (\text{A5})$$

Thus, before accounting for the physical response of the apparatus to its motion relative to the preferred frame, the average two-way propagation speed along the longitudinal direction is smaller than that along the transverse direction.

A.3 Length Contraction and Equal Round-Trip Times

The Michelson–Morley null result requires the two light paths to return to the beam splitter simultaneously. Let L denote the transverse arm length and L_x the physical longitudinal arm length when the apparatus is moving relative to the preferred frame. Equality of the round-trip propagation times requires

$$\frac{2L}{c_y} = \frac{L_x}{c - v_p} + \frac{L_x}{c + v_p}. \quad (\text{A6})$$

Substituting $c_y = \gamma_K^{-1} c$ gives

$$\begin{aligned} \frac{2L}{\gamma_K^{-1} c} &= \frac{L_x(c + v_p) + L_x(c - v_p)}{c^2 - v_p^2} \\ \frac{2\gamma_K L}{c} &= \frac{2L_x c}{c^2 - v_p^2} \\ &= \frac{2L_x}{c} \frac{1}{1 - \frac{v_p^2}{c^2}} \\ &= \frac{2\gamma_K^2 L_x}{c}, \end{aligned} \quad (\text{A7})$$

and therefore

$$L_x = \gamma_K^{-1} L. \quad (\text{A8})$$

The required physical contraction is therefore exactly the Lorentz length-contraction factor. This is also the contraction obtained historically in Lorentz-type preferred-frame formulations [19, 20].

The directional difference in physical propagation speed is consequently accompanied by a corresponding directional difference in physical path length. For the transverse arm,

$$\begin{aligned}
t_y &= \frac{2L}{c_y} \\
&= \frac{2L}{\gamma_K^{-1}c} \\
&= \frac{2\gamma_K L}{c},
\end{aligned} \tag{A9}$$

while for the contracted longitudinal arm,

$$\begin{aligned}
t_x &= \frac{L_x}{c - v_p} + \frac{L_x}{c + v_p} \\
&= \frac{2L_x c}{c^2 - v_p^2} \\
&= \frac{2\gamma_K^{-1}L}{c} \gamma_K^2 \\
&= \frac{2\gamma_K L}{c}.
\end{aligned} \tag{A10}$$

Hence,

$$t_x = t_y. \tag{A11}$$

The Michelson–Morley null result therefore does not require the physical propagation of light to be directionally identical relative to a moving apparatus. It requires the complete round-trip propagation times to be identical, which follows when the directional propagation differences are accompanied by Lorentz contraction of the longitudinal path.

A.4 The Speed of Causality

The common rate associated with completion of the orthogonal round-trip propagation processes is

$$c_\star = \gamma_K^{-1}c. \tag{A12}$$

This is the same quantity identified in the main text as the *speed of causality*. It does not represent a new propagation law introduced specifically to explain the Michelson–Morley result. Rather, the Michelson–Morley geometry recovers the same

physical condition that governs the rate of interaction and, consequently, the rate of time dilation.

The same relationship follows directly from the flat-spacetime timelike interval:

$$\begin{aligned}
ds^2 &= c^2 dt^2 - dx^2 - dy^2 - dz^2, \\
\frac{ds}{dt} &= \sqrt{c^2 - v_p^2} \\
&= \gamma_K^{-1} c \\
&= c_\star.
\end{aligned} \tag{A13}$$

Thus, the quantity recovered from the Michelson–Morley analysis is identical to the speed-valued representation of the metric-independent physical condition developed from the general timelike interval in the main text.

A.5 Why the Measured Two-Way Speed Remains c

The remaining question is why observers moving relative to the preferred frame measure the two-way speed of light as c even though $c_\star < c$. The answer follows from the same reduction in the rate of physical change that produces time dilation.

For uniform motion relative to the preferred frame,

$$\frac{dt'}{dt} = \gamma_K^{-1} = \frac{c_\star}{c}, \tag{A14}$$

where dt is the calibrated preferred-frame-referenced interval and dt' is the corresponding uncalibrated time measurement made using physical processes within the moving frame. Because those processes proceed more slowly by the same factor that increases the physical light-propagation duration, the resulting time measurement is miscalibrated relative to the preferred-frame interval.

This relationship can also be obtained from the Lorentz transformation [19, 20]:

$$\begin{aligned}
t' &= \gamma_K \left(t - \frac{v_p}{c^2} x \right), \\
x' &= \gamma_K (x - v_p t), \\
y' &= y, \\
z' &= z.
\end{aligned} \tag{A15}$$

For an object comoving with the moving frame, $dx = v_p dt$, and therefore

$$dt' = \gamma_K \left(dt - \frac{v_p}{c^2} dx \right)$$

$$\begin{aligned}
&= \gamma_K \left(1 - \frac{v_p^2}{c^2} \right) dt \\
&= \gamma_K^{-1} dt,
\end{aligned} \tag{A16}$$

recovering Equation (A14).

A.5.1 Transverse Measurement

Consider first the transverse light path. Its physical round-trip duration is

$$t_y = \frac{2L_y}{c_y}. \tag{A17}$$

Because transverse lengths are unchanged,

$$L'_y = L_y, \tag{A18}$$

while the measured time interval satisfies

$$t'_y = \gamma_K^{-1} t_y. \tag{A19}$$

The measured two-way speed is therefore

$$\begin{aligned}
c'_y &= \frac{2L'_y}{t'_y} \\
&= \frac{2L_y}{\gamma_K^{-1} t_y} \\
&= \gamma_K c_y \\
&= \gamma_K (\gamma_K^{-1} c) \\
&= c.
\end{aligned} \tag{A20}$$

A.5.2 Longitudinal Measurement

For the longitudinal path, the physical arm length is contracted according to

$$L_x = \gamma_K^{-1} L'_x, \tag{A21}$$

where L'_x is the length assigned to the arm using measurements performed within the moving frame.

The physical outbound and return durations are

$$\begin{aligned}
t_2 &= \frac{L_x}{c - v_p}, \\
t_3 &= \frac{L_x}{c + v_p}.
\end{aligned} \tag{A22}$$

Applying the time-dilation relationship,

$$\begin{aligned}
t'_2 &= \gamma_K^{-1} t_2 = \frac{L'_x}{\gamma_K^2 (c - v_p)}, \\
t'_3 &= \gamma_K^{-1} t_3 = \frac{L'_x}{\gamma_K^2 (c + v_p)}.
\end{aligned} \tag{A23}$$

The measured two-way longitudinal speed is consequently

$$\begin{aligned}
c'_x &= \frac{2L'_x}{t'_2 + t'_3} \\
&= \frac{2L'_x}{\frac{L'_x}{\gamma_K^2 (c - v_p)} + \frac{L'_x}{\gamma_K^2 (c + v_p)}} \\
&= \gamma_K^2 \frac{2}{\frac{1}{c - v_p} + \frac{1}{c + v_p}} \\
&= \gamma_K^2 c \left(1 - \frac{v_p^2}{c^2} \right) \\
&= c.
\end{aligned} \tag{A24}$$

Therefore,

$$c'_x = c'_y = c. \tag{A25}$$

The measured two-way speed of light is isotropic even though the underlying physical propagation relative to the moving apparatus is not. The reason is not that the physical directional differences disappear, but that the same motion relative to the preferred frame also changes the physical lengths and rates of change used to measure them.

A.6 Conclusion

The apparent isotropy of the measured two-way speed of light is compatible with the existence of a preferred frame. Relative to an apparatus moving through that frame, the physical propagation of light differs between the longitudinal and transverse

directions. Lorentz contraction compensates for this difference by reducing the physical longitudinal path length so that orthogonal round-trip propagation times remain equal.

The common rate governing completion of these propagation processes is $c_\star = \gamma_K^{-1}c$, the same speed of causality identified independently from the timelike interval and shown in the main text to govern time dilation. Because all physical processes within the moving frame proceed at this reduced rate, time measurements made by those processes are miscalibrated by the corresponding factor. Together with the associated length contraction, this causes local measurements of the two-way speed of light to return c in every direction.

The Michelson–Morley null result therefore constrains what can be measured within a moving inertial frame without, by itself, distinguishing between a framework in which light is fundamentally isotropic in every inertial frame and one in which physical propagation is isotropic only in a preferred frame while the measuring processes themselves undergo compensating physical changes.

B Deriving Specific Work and Energy

This appendix derives the specific-work and specific-energy quantities used in Section 4.2. The objective is to establish these quantities from the corresponding definitions of momentum, force, and work, while expressing them in the preferred-frame-referenced quantities used throughout Universal Specificity. The resulting relations provide the kinetic and gravitational specific-energy terms that appear in the decomposition of specific internal energy.

B.1 First-Principles Definitions

In the kinetic context, energy and work are related through the work performed on a system; work is the integral of force over displacement; force is the time rate of change of momentum; and relativistic momentum is the product of relativistic mass and velocity. With all motion referenced to the preferred frame,

$$\Delta E = W, \tag{B1}$$

$$W = \int F dx, \tag{B2}$$

$$F = \frac{dP}{dt}, \tag{B3}$$

$$P = m_r v_p = \gamma m_0 v_p. \tag{B4}$$

The corresponding specific quantities may be constructed from specific momentum. Dividing relativistic momentum by the corresponding relativistic mass gives

$$p_s = \frac{P}{m_r} = v_p. \tag{B5}$$

Specific force is then defined as the time rate of change of specific momentum, specific work as the integral of specific force over displacement, and specific energy through the corresponding capacity for specific work:

$$p_s = v, \quad (\text{B6})$$

$$f = \frac{dp_s}{dt}, \quad (\text{B7})$$

$$w = \int f dx, \quad (\text{B8})$$

$$\Delta e = w. \quad (\text{B9})$$

These definitions remove mass from the hierarchy at the level of specific momentum. The resulting quantities therefore measure motion independently of the mass of the system:

$$p_s \longrightarrow f \longrightarrow w \longrightarrow \Delta e. \quad (\text{B10})$$

Under these definitions, specific quantities are not fundamentally defined by dividing their corresponding extensive quantities by mass. In general,

$$\begin{aligned} \Delta e &\neq \frac{\Delta E}{m_r}, & \Delta E &\neq m_r \Delta e, \\ w &\neq \frac{W}{m_r}, & W &\neq m_r w, \\ f &\neq \frac{F}{m_r}, & F &\neq m_r f. \end{aligned} \quad (\text{B11})$$

The familiar interpretation of specific energy as energy per unit mass is recovered when mass is constant. In that case,

$$\begin{aligned} W &= \int F dx = \int \frac{dP}{dt} dx = \int \frac{d(mv)}{dt} dx = m \int \frac{dv}{dt} dx = m \int f dx \\ &= mw, \end{aligned} \quad (\text{B12})$$

and therefore

$$\Delta E = W = mw = m \Delta e. \quad (\text{B13})$$

Thus, energy per unit mass and specific energy coincide in the constant-mass limit, while the specific quantities used here retain their definition directly through the hierarchy of motion.

B.2 Preferred-Frame-Referenced Specific Work

The specific-energy relations used in the main text require the corresponding work to be expressed using spatial intervals uncalibrated for relativistic length contraction. Let dx_0 denote such an interval evaluated locally by an uncalibrated instrument co-located with the system's position but stationary with respect to the preferred frame, while t denotes the preferred-frame-referenced time coordinate.

For an infinitesimal displacement, the corresponding specific work is

$$\begin{aligned} dw_0 &= f_0 dx_0 \\ &= \frac{d^2 x'}{dt^2} dx_0. \end{aligned} \tag{B14}$$

Integration along the path gives

$$\begin{aligned} w_0 &= \int f_0 dx_0 \\ &= \int \frac{d^2 x'}{dt^2} dx_0. \end{aligned} \tag{B15}$$

The subscript 0 therefore denotes the specific-work quantity obtained when motion is expressed using these preferred-frame-referenced, uncalibrated spatial intervals. This construction will now be applied separately to kinetic and gravitational motion.

B.3 Kinetic Specific Work without Gravitation

Consider first the purely kinetic case described by the Minkowski metric. In the absence of gravitation, no gravitational radial-length correction is present, so

$$\frac{dx_0}{dx} = \frac{dy_0}{dy} = \frac{dz_0}{dz} = 1. \tag{B16}$$

The kinetic specific work is therefore the sum of the work performed in the three orthogonal spatial directions:

$$\begin{aligned} w_{K_0} &= w_{x_0} + w_{y_0} + w_{z_0} \\ &= \int \frac{d^2 x'}{dt^2} dx_0 + \int \frac{d^2 y'}{dt^2} dy_0 + \int \frac{d^2 z'}{dt^2} dz_0 \\ &= \frac{1}{2} \left(\frac{dx_0}{dt} \right)^2 + \frac{1}{2} \left(\frac{dy_0}{dt} \right)^2 + \frac{1}{2} \left(\frac{dz_0}{dt} \right)^2 \\ &= \frac{1}{2} v_p^2. \end{aligned} \tag{B17}$$

Thus, in the absence of gravitation,

$$w_{K_0} = \frac{1}{2}v_p^2, \quad (\text{B18})$$

recovering the familiar kinetic specific-work form directly from the specific-force hierarchy.

B.4 Gravitational Specific Work

In a Schwarzschild gravitational field, radial distance and the corresponding radial acceleration must be expressed using the preferred-frame-referenced radial interval. The gravitational specific work is therefore

$$\begin{aligned} w_{g_0} &= \int g_0(r) dr_0 \\ &= \int \frac{d^2 r'}{dt^2} dr_0. \end{aligned} \quad (\text{B19})$$

The Schwarzschild radial coordinate acceleration derived in Appendix H is

$$\ddot{r} = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{3GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2 + r\dot{\Omega}^2\gamma_g^{-2}, \quad (\text{B20})$$

where

$$\gamma_g = \left(1 - \frac{2GM}{rc^2}\right)^{-1/2}. \quad (\text{B21})$$

Because gravitational work is performed through the radial specific force, the transverse kinematic contribution is excluded when determining gravitational specific work. The remaining radial coordinate acceleration is

$$\ddot{r} = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{3GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2. \quad (\text{B22})$$

As shown in Section 4.2, this radial expression contains both the specific-force contribution associated with the gradient of the causal potential and an additional velocity-dependent coordinate contribution. Isolating the gradient contribution gives

$$\ddot{r}_\nabla = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2. \quad (\text{B23})$$

The radial uncalibrated interval satisfies

$$\frac{dr_0}{dr} = \gamma_g. \quad (\text{B24})$$

Applying the chain rule,

$$\begin{aligned} \frac{dr_0}{dt} &= \frac{dr}{dt} \frac{dr_0}{dr}, \\ \frac{d^2 r'}{dt^2} &= \frac{d^2 r}{dt^2} \frac{dr_0}{dr} + \left(\frac{dr}{dt} \right)^2 \frac{d^2 r'}{dr^2}. \end{aligned} \quad (\text{B25})$$

Using Equation (B23) for the work-producing radial acceleration and Equation (B24) for the radial interval gives

$$\begin{aligned} \ddot{r}_0 &= \ddot{r}_\nabla \gamma_g + \dot{r}^2 \frac{d^2 r'}{dr^2} \\ &= -\frac{GM}{r^2} \gamma_g^{-1}. \end{aligned} \quad (\text{B26})$$

The corresponding gravitational specific work is therefore

$$\begin{aligned} w_{g_0} &= \int_{\infty}^r \ddot{r}_0 dr_0 \\ &= \int_{\infty}^r -\frac{GM}{r^2} \gamma_g^{-1} dr_0 \\ &= \int_{\infty}^r -\frac{GM}{r^2} dr \\ &= \frac{GM}{r}. \end{aligned} \quad (\text{B27})$$

Thus, the magnitude of the gravitational specific work accumulated from infinity to radius r is

$$\boxed{w_{g_0} = \frac{GM}{r}}. \quad (\text{B28})$$

B.5 Kinetic Specific Work within Gravitation

Within a Schwarzschild gravitational field, kinetic specific work contains both radial and transverse contributions. The radial contribution must account for the gravitational radial-length relation, whereas the transverse directions are unaffected by that radial contraction.

The total kinetic specific work is therefore

$$\begin{aligned}
 w_{K_0} &= w_{r_0} + w_{\Omega_0} \\
 &= \int a_0(r) dr' + \int a_0(\Omega) d\Omega' \\
 &= \frac{1}{2} \left(\frac{dr'}{dt} \right)^2 + \frac{1}{2} \left(r \frac{d\Omega'}{dt} \right)^2.
 \end{aligned} \tag{B29}$$

Using

$$\frac{dr'}{dr} = \gamma_g, \qquad \frac{d\Omega'}{d\Omega} = 1, \tag{B30}$$

gives

$$\begin{aligned}
 w_{K_0} &= \frac{1}{2} \left(\frac{dr}{dt} \frac{dr'}{dr} \right)^2 + \frac{1}{2} r^2 \left(\frac{d\Omega'}{dt} \right)^2 \\
 &= \frac{1}{2} \dot{r}^2 \gamma_g^2 + \frac{1}{2} r^2 \dot{\Omega}^2.
 \end{aligned} \tag{B31}$$

Equivalently, defining

$$\dot{r}_0 = \gamma_g \dot{r}, \tag{B32}$$

the same relation may be written

$$w_{K_0} = \frac{1}{2} \dot{r}_0^2 + \frac{1}{2} r^2 \dot{\Omega}^2. \tag{B33}$$

This is the kinetic specific-work quantity that appears in the Schwarzschild time-dilation relation.

B.6 Specific Kinetic and Potential Energy

Using the convention that kinetic specific energy is zero when the system is stationary with respect to the preferred frame, the kinetic specific energy is equal to the kinetic specific work required to reach its state of motion:

$$e_{K_0} = w_{K_0}. \quad (\text{B34})$$

Therefore,

$$\boxed{e_{K_0} = \frac{1}{2}\dot{r}_0^2 + \frac{1}{2}r^2\dot{\Omega}^2 = \frac{1}{2}\dot{r}^2\gamma_g^2 + \frac{1}{2}r^2\dot{\Omega}^2}. \quad (\text{B35})$$

The gravitational specific potential energy is defined using the convention

$$e_{P_0} = 0 \quad \text{as } r \rightarrow r_s, \quad (\text{B36})$$

and

$$e_{P_0} = e_{\max} \quad \text{as } r \rightarrow \infty, \quad (\text{B37})$$

where

$$r_s = \frac{2GM}{c^2} \quad (\text{B38})$$

is the Schwarzschild radius.

Because gravitational specific work reduces the available specific potential energy as the system moves inward,

$$\begin{aligned} e_{P_0} &= e_{\max} - w_{g_0} \\ &= e_{\max} - \frac{GM}{r}. \end{aligned} \quad (\text{B39})$$

Applying the boundary condition at the Schwarzschild radius,

$$\begin{aligned} 0 &= e_{\max} - \frac{GM}{r_s} \\ &= e_{\max} - \frac{GM}{2GM/c^2} \\ &= e_{\max} - \frac{1}{2}c^2, \end{aligned} \quad (\text{B40})$$

and therefore

$$e_{\max} = \frac{1}{2}c^2. \quad (\text{B41})$$

Substituting this result into Equation (B39) gives

$$e_{P_0} = \frac{1}{2}c^2 - \frac{GM}{r}. \quad (\text{B42})$$

The two specific-energy terms derived in this appendix are therefore

$$\begin{aligned} e_{K_0} &= \frac{1}{2}\dot{r}^2\gamma_g^2 + \frac{1}{2}r^2\dot{\Omega}^2, \\ e_{P_0} &= \frac{1}{2}c^2 - \frac{GM}{r}. \end{aligned} \quad (\text{B43})$$

These are precisely the quantities appearing in the specific internal-energy relation developed in the main text,

$$e_{I_*} = e_{P_0} - e_{K_0}. \quad (\text{B44})$$

B.7 Conclusion

Specific energy can be constructed directly from the hierarchy of specific momentum, specific force, and specific work rather than introduced only as a ratio of energy to mass. Beginning with $p_s = v$, integration of the corresponding specific force over preferred-frame-referenced spatial intervals yields the kinetic and gravitational specific-work quantities required by the relativistic time-dilation relations.

For Schwarzschild gravitation, the gravitational specific work accumulated from infinity to radius r is GM/r , while the kinetic specific work is determined by the radial and transverse motion according to

$$w_{K_0} = \frac{1}{2}\dot{r}^2\gamma_g^2 + \frac{1}{2}r^2\dot{\Omega}^2. \quad (\text{B45})$$

With the boundary conditions established above, these relations yield

$$\begin{aligned} e_{P_0} &= \frac{1}{2}c^2 - \frac{GM}{r}, \\ e_{K_0} &= \frac{1}{2}\dot{r}^2\gamma_g^2 + \frac{1}{2}r^2\dot{\Omega}^2. \end{aligned} \quad (\text{B46})$$

The specific internal energy introduced in the main text therefore follows from quantities independently constructed through specific work:

$$e_{I_\star} = e_{P_0} - e_{K_0}. \quad (\text{B47})$$

This establishes the specific-energy representation used in Universal Specificity directly from the corresponding measures of motion and work.

C Schwarzschild Coordinate Speed of Light

This appendix derives the radial and transverse coordinate speeds of light in Schwarzschild coordinates. These quantities are used in the main text to distinguish the direction-dependent coordinate propagation of light from the locally measured speed c and to establish the gravitational relationships associated with the speed of causality.

In Schwarzschild coordinates (t, r, θ, ϕ) , using the $(+, -, -, -)$ signature, the spacetime line element is

$$ds^2 = \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 - \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 - r^2 d\theta^2 - r^2 \sin^2 \theta d\phi^2. \quad (\text{C1})$$

Defining the gravitational time-dilation factor as

$$\gamma_g = \left(1 - \frac{2GM}{rc^2}\right)^{-1/2}, \quad (\text{C2})$$

the Schwarzschild metric may equivalently be written as

$$ds^2 = \gamma_g^{-2} c^2 dt^2 - \gamma_g^2 dr^2 - r^2 d\Omega^2, \quad (\text{C3})$$

where

$$d\Omega^2 \equiv d\theta^2 + \sin^2 \theta d\phi^2. \quad (\text{C4})$$

To determine the Schwarzschild coordinate speed of light, we consider null propagation, for which $ds^2 = 0$.

C.1 Radial Coordinate Speed of Light

For purely radial propagation,

$$d\theta = d\phi = 0, \quad (\text{C5})$$

and the Schwarzschild line element reduces to

$$0 = \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 - \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2. \quad (\text{C6})$$

Multiplying through by the Schwarzschild factor gives

$$\left(1 - \frac{2GM}{rc^2}\right)^2 c^2 dt^2 = dr^2. \quad (\text{C7})$$

Therefore,

$$\left(\frac{dr}{dt}\right)^2 = \left(1 - \frac{2GM}{rc^2}\right)^2 c^2, \quad (\text{C8})$$

and the radial coordinate speed of light is

$$\begin{aligned} c_r \equiv \frac{dr}{dt} &= \pm \left(1 - \frac{2GM}{rc^2}\right) c \\ &= \pm \gamma_g^{-2} c. \end{aligned} \quad (\text{C9})$$

Thus,

$$\boxed{c_r = \pm \gamma_g^{-2} c}. \quad (\text{C10})$$

This is the radial propagation speed expressed with respect to Schwarzschild radial coordinate r and Schwarzschild coordinate time t . It approaches c as $r \rightarrow \infty$ and approaches zero as $r \rightarrow r_s$, where

$$r_s = \frac{2GM}{c^2} \quad (\text{C11})$$

is the Schwarzschild radius.

C.2 Transverse Coordinate Speed of Light

For purely transverse propagation at fixed Schwarzschild radius,

$$dr = 0. \quad (\text{C12})$$

The null line element then becomes

$$0 = \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 - r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (\text{C13})$$

Using

$$d\Omega^2 \equiv d\theta^2 + \sin^2 \theta d\phi^2, \quad (\text{C14})$$

this becomes

$$0 = \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 - r^2 d\Omega^2. \quad (\text{C15})$$

Solving for the angular rate gives

$$\begin{aligned} r^2 d\Omega^2 &= \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2, \\ r^2 \left(\frac{d\Omega}{dt}\right)^2 &= c^2 \left(1 - \frac{2GM}{rc^2}\right). \end{aligned} \quad (\text{C16})$$

Defining the transverse coordinate speed as

$$c_{\perp} \equiv r \frac{d\Omega}{dt}, \quad (\text{C17})$$

we obtain

$$\begin{aligned} c_{\perp} &= \pm c \sqrt{1 - \frac{2GM}{rc^2}} \\ &= \pm \gamma_g^{-1} c. \end{aligned} \quad (\text{C18})$$

Thus,

$$\boxed{c_{\perp} = \pm \gamma_g^{-1} c}. \quad (\text{C19})$$

As in the radial case, the transverse coordinate speed approaches c as $r \rightarrow \infty$ and decreases as the Schwarzschild radius is approached.

C.3 Comparison of the Coordinate Speeds

The radial and transverse Schwarzschild coordinate speeds are therefore

$$c_r = \pm \left(1 - \frac{2GM}{rc^2}\right) c = \pm \gamma_g^{-2} c, \quad (\text{C20})$$

$$c_\perp = \pm c \sqrt{1 - \frac{2GM}{rc^2}} = \pm \gamma_g^{-1} c. \quad (\text{C21})$$

Consequently,

$$|c_r| = \gamma_g^{-1} |c_\perp|. \quad (\text{C22})$$

The Schwarzschild coordinate representation therefore assigns different propagation rates to radial and transverse light motion, with the radial coordinate speed decreasing more rapidly as the gravitational field strengthens.

These coordinate speeds should not be confused with the locally measured speed of light. A local freely falling observer measures light to propagate at c . The quantities derived here instead describe propagation relative to the Schwarzschild coordinates and their associated coordinate time.

C.4 Relation to Gravitational Propagation Delay

The reduction in Schwarzschild coordinate propagation speed is reflected observationally in the increased coordinate propagation time of light through a gravitational field. The Shapiro time-delay measurements provide an experimental example: light propagating through regions of stronger gravitational influence requires a greater round-trip propagation time than would be expected in the corresponding flat-spacetime case [42, 63].

The coordinate-speed expressions derived above provide the Schwarzschild-coordinate representation of this effect. In particular, radial propagation satisfies

$$\left| \frac{dr}{dt} \right| = \gamma_g^{-2} c, \quad (\text{C23})$$

while transverse propagation at fixed radius satisfies

$$r \left| \frac{d\Omega}{dt} \right| = \gamma_g^{-1} c. \quad (\text{C24})$$

These relationships are used in the main text to characterize how gravitational conditions alter propagation relative to Schwarzschild coordinate time while remaining consistent with the locally measured invariance of c .

C.5 Conclusion

For null propagation in Schwarzschild coordinates, the radial and transverse coordinate speeds of light are

$$\boxed{c_r = \pm \gamma_g^{-2} c}, \quad \boxed{c_\perp = \pm \gamma_g^{-1} c}. \quad (\text{C25})$$

Both approach c in the asymptotically flat limit, while both decrease as the Schwarzschild radius is approached, with the radial coordinate speed decreasing more rapidly than the transverse coordinate speed. These quantities describe light propagation relative to Schwarzschild coordinates rather than the result of a local measurement, for which the measured speed of light remains c .

D The River Model

This appendix examines the Schwarzschild solution in Painlevé–Gullstrand (PG) coordinates, commonly associated with the “river model.” Unlike Schwarzschild coordinates, in which radial light propagation is represented symmetrically in the inward and outward directions, PG coordinates represent the same spacetime using a radial flow velocity. This provides a useful alternative representation for testing whether the physical condition identified with time dilation depends on the particular coordinate description.

In PG coordinates, the Schwarzschild line element may be written as [35]

$$ds^2 = \left(1 - \frac{v_r^2}{c^2}\right) c^2 dt^2 + 2v_r dr dt - dr^2 - r^2 d\Omega^2, \quad (\text{D1})$$

where

$$d\Omega^2 \equiv d\theta^2 + \sin^2 \theta d\phi^2 \quad (\text{D2})$$

and the river velocity is

$$v_r = \pm \sqrt{\frac{2GM}{r}}. \quad (\text{D3})$$

Using the convention adopted here, $v_r < 0$ denotes inward flow and $v_r > 0$ denotes outward flow. For the usual black-hole PG coordinates, the inward-flow solution is therefore

$$v_r = -\sqrt{\frac{2GM}{r}}. \quad (\text{D4})$$

The transverse coordinate velocity is defined as

$$v_\perp = r \frac{d\Omega}{dt}. \quad (\text{D5})$$

D.1 Time Dilation in Painlevé–Gullstrand Coordinates

For a timelike trajectory,

$$ds^2 = c^2 dt'^2, \quad (\text{D6})$$

where dt' is the proper-time interval associated with the moving system. Dividing Equation (D1) by $c^2 dt^2$ gives

$$\left(\frac{dt'}{dt}\right)^2 = 1 - \frac{v_r^2}{c^2} + \frac{2v_r \dot{r}}{c^2} - \frac{\dot{r}^2}{c^2} - \frac{v_\perp^2}{c^2}, \quad (\text{D7})$$

where

$$\dot{r} = \frac{dr}{dt}. \quad (\text{D8})$$

Collecting the radial terms,

$$\left(\frac{dt'}{dt}\right)^2 = 1 - \frac{(\dot{r} - v_r)^2 + v_\perp^2}{c^2}. \quad (\text{D9})$$

Therefore,

$$\frac{dt'}{dt} = \sqrt{1 - \frac{(\dot{r} - v_r)^2 + v_\perp^2}{c^2}}. \quad (\text{D10})$$

The velocity of the system relative to the local river flow is

$$\vec{v}_p = \vec{v} - \vec{v}_r$$

$$\begin{aligned}
&= \dot{r} \hat{r} + v_{\perp} \hat{\Omega} - v_r \hat{r} \\
&= (\dot{r} - v_r) \hat{r} + v_{\perp} \hat{\Omega},
\end{aligned} \tag{D11}$$

so that

$$v_p^2 = (\dot{r} - v_r)^2 + v_{\perp}^2. \tag{D12}$$

Equation (D10) therefore reduces to

$$\boxed{\frac{dt'}{dt} = \sqrt{1 - \frac{v_p^2}{c^2}}}. \tag{D13}$$

Thus, when expressed relative to the river flow, the Schwarzschild gravitational time-dilation condition assumes the same mathematical form as the purely kinetic time-dilation relation.

D.2 Relation to the Speed of Causality

The speed of causality identified in the main text provides an alternative representation of this same condition. From

$$\frac{dt'}{dt} = \frac{c_{\star}}{c}, \tag{D14}$$

Equation (D13) gives

$$\frac{c_{\star}}{c} = \sqrt{1 - \frac{v_p^2}{c^2}}, \tag{D15}$$

and therefore

$$\begin{aligned}
c_{\star} &= \sqrt{c^2 - v_p^2} \\
&= \sqrt{c^2 - (\dot{r} - v_r)^2 - v_{\perp}^2}.
\end{aligned} \tag{D16}$$

Equivalently, the same relationship may be written in specific-energy form as

$$\frac{dt'}{dt} = \sqrt{\frac{\frac{1}{2}c^2 - \frac{1}{2}v_p^2}{\frac{1}{2}c^2}}. \tag{D17}$$

The PG representation therefore absorbs the gravitational contribution into the relative velocity between the system and the river flow. What appears in Schwarzschild coordinates as a combination of gravitational and kinetic terms appears here as a single relative-motion term. Despite this change in decomposition, the resulting time-dilation condition remains expressible through the same speed-valued quantity, c_* .

This provides a useful illustration of the distinction developed in the main text between a coordinate-dependent representation of a physical condition and the condition itself.

D.3 Asymmetric Radial Coordinate Speeds of Light

The difference between the Schwarzschild and PG coordinate representations is especially apparent for radial light propagation. Setting

$$ds^2 = 0, \quad d\Omega = 0, \quad (\text{D18})$$

in Equation (D1) gives

$$\begin{aligned} 0 &= \left(1 - \frac{v_r^2}{c^2}\right) c^2 dt^2 + 2v_r dr dt - dr^2 \\ &= c^2 dt^2 - v_r^2 dt^2 + 2v_r dr dt - dr^2 \\ &= c^2 dt^2 - (dr - v_r dt)^2. \end{aligned} \quad (\text{D19})$$

Therefore,

$$\left(\frac{dr}{dt} - v_r\right)^2 = c^2, \quad (\text{D20})$$

which gives the two radial coordinate velocities

$$\boxed{c_r = \frac{dr}{dt} = v_r \pm c}. \quad (\text{D21})$$

For the inward-flow solution, $v_r < 0$, the outgoing and ingoing coordinate velocities are

$$c_{r,\text{out}} = v_r + c, \quad (\text{D22})$$

$$c_{r,\text{in}} = v_r - c. \quad (\text{D23})$$

Their coordinate values are therefore asymmetric. Relative to the river flow itself, however,

$$\begin{aligned} c_{r,\text{out}} - v_r &= c, \\ c_{r,\text{in}} - v_r &= -c. \end{aligned} \tag{D24}$$

Thus,

$$\boxed{c_r - v_r = \pm c}. \tag{D25}$$

Light propagation is isotropic with magnitude c relative to the local river flow, while its radial coordinate velocity is shifted by the river velocity.

At the Schwarzschild radius,

$$r_s = \frac{2GM}{c^2}, \tag{D26}$$

the inward river velocity satisfies

$$v_r = -c. \tag{D27}$$

The outward radial coordinate speed consequently becomes

$$c_{r,\text{out}} = v_r + c = 0, \tag{D28}$$

while the inward solution is

$$c_{r,\text{in}} = -2c. \tag{D29}$$

These values are coordinate velocities and do not imply a violation of the locally measured speed of light, which remains c .

D.4 Comparison with Schwarzschild Coordinates

The same Schwarzschild spacetime therefore admits substantially different coordinate representations of radial light propagation. As derived in Appendix C, Schwarzschild coordinates give

$$c_r = \pm \gamma_g^{-2} c, \tag{D30}$$

which is symmetric between the inward and outward coordinate directions. Painlevé–Gullstrand coordinates instead give

$$c_r = v_r \pm c, \quad (\text{D31})$$

which is asymmetric with respect to the PG radial coordinate.

These different coordinate speeds describe the same underlying spacetime. Consequently, the directional coordinate speed assigned to light cannot by itself identify the physical mechanism responsible for gravitational time dilation.

The time-dilation relation provides a more useful comparison. In PG coordinates,

$$\frac{dt'}{dt} = \sqrt{1 - \frac{v_p^2}{c^2}} = \frac{c_\star}{c}. \quad (\text{D32})$$

The gravitational and kinetic contributions have therefore been reorganized by the change of coordinates, while the resulting condition governing the rate of time dilation remains representable by $c_\star$.

D.5 Implications for the Preferred Frame

The PG representation is particularly suggestive within a preferred-frame interpretation because light propagates isotropically at c relative to the local river flow:

$$c_r - v_r = \pm c. \quad (\text{D33})$$

Accordingly, if the local preferred frame is identified as the local frame in which opposite one-way light propagation is isotropic⁵, the PG river velocity identifies the motion of that local preferred frame relative to the PG radial coordinates.

For the inward-flow solution,

$$\vec{v}_r = -\sqrt{\frac{2GM}{r}} \hat{r}, \quad (\text{D34})$$

so the preferred-frame interpretation corresponds locally to a radially infalling frame whose speed increases as the central mass is approached.

This interpretation is not established merely by choosing PG coordinates. Rather, PG coordinates demonstrate that such a representation is mathematically compatible with the Schwarzschild solution. The physical identification of the isotropic-light frame with the preferred frame follows from the additional preferred-frame hypothesis developed in the main text.

⁵This is required for its local simultaneity relation to corresponds to the physically real present if this coordinate representation correctly describes the physical one-way propagation of light.

D.6 Conclusion

Painlevé–Gullstrand and Schwarzschild coordinates describe the same Schwarzschild spacetime while representing light propagation and gravitational time dilation differently. Schwarzschild coordinates assign symmetric but direction-dependent coordinate propagation rates,

$$c_r = \pm \gamma_g^{-2} c, \quad (\text{D35})$$

whereas PG coordinates assign asymmetric radial coordinate velocities,

$$c_r = v_r \pm c. \quad (\text{D36})$$

Relative to the PG river flow, however, light propagates isotropically:

$$c_r - v_r = \pm c. \quad (\text{D37})$$

The corresponding timelike relation reduces to

$$\frac{dt'}{dt} = \sqrt{1 - \frac{v_p^2}{c^2}} = \frac{c_\star}{c}. \quad (\text{D38})$$

Thus, changing the coordinate representation can redistribute what appears as gravitational and kinematic contributions and can transform symmetric coordinate light propagation into an asymmetric form. The condition governing time dilation, however, remains expressible through the same speed of causality, $c_\star$, for whichever metric corresponds to reality.

E Deriving the Speed of Causality

This appendix derives the speed of causality, $c_\star$, for purely kinetic, purely gravitational, and combined kinetic–gravitational cases. The speed of causality is an internally referenced rate constructed by retaining the uncalibrated comoving spatial path of an interaction cycle while correcting its temporal interval for time dilation. Thus,

$$\frac{d\ell'}{dt'} = c, \quad (\text{E1})$$

$$c_\star \equiv \frac{d\ell'}{dt} = c \frac{dt'}{dt}. \quad (\text{E2})$$

Thus, $c_\star$ is not introduced as an independent propagation law, but as a speed-valued representation of the same physical condition that determines the rate of time dilation.

E.1 Kinetic Case

Consider first a system moving with preferred-frame velocity v_p in otherwise flat space-time. Without loss of generality, choose the x -axis to lie along the direction of motion. As derived in Appendix A, the average two-way coordinate speeds of light parallel and transverse to the motion are

$$c_{\parallel} = \pm \gamma_K^{-2} c, \quad (\text{E3})$$

$$c_{\perp} = \pm \gamma_K^{-1} c, \quad (\text{E4})$$

where

$$\gamma_K = \frac{1}{\sqrt{1 - \frac{v_p^2}{c^2}}}. \quad (\text{E5})$$

The propagation rate may instead be evaluated independently along the longitudinal and transverse directions. Along the direction of motion, the uncalibrated spatial interval measured by the affected moving system satisfies

$$\frac{dx'}{dx} = \gamma_K. \quad (\text{E6})$$

Therefore,

$$\begin{aligned} c_{0,\parallel} &= |c_{\parallel}| \frac{dx'}{dx} \\ &= \gamma_K^{-2} c \gamma_K \\ &= \gamma_K^{-1} c. \end{aligned} \quad (\text{E7})$$

In a transverse direction, no length contraction occurs,

$$\frac{dy'}{dy} = 1, \quad (\text{E8})$$

and therefore

$$c_{0,\perp} = |c_{\perp}| = \gamma_K^{-1} c. \quad (\text{E9})$$

Thus, the longitudinal and transverse interaction cycles independently yield the same speed-valued rate,

$$\boxed{c_{\star} = \gamma_K^{-1} c = c \sqrt{1 - \frac{v_p^2}{c^2}}}. \quad (\text{E10})$$

The directional dependence of the coordinate propagation speeds therefore disappears when propagation is expressed using the corresponding uncalibrated spatial intervals. The resulting rate is independent of propagation direction and equals the kinetic speed of causality.

The flat-spacetime timelike interval gives the same result directly:

$$\begin{aligned} ds^2 &= c^2 dt^2 - dx^2 - dy^2 - dz^2 \\ &= (c^2 - v_p^2) dt^2. \end{aligned} \tag{E11}$$

Therefore,

$$\begin{aligned} \frac{ds}{dt} &= \sqrt{c^2 - v_p^2} \\ &= \gamma_K^{-1} c \\ &= c_\star. \end{aligned} \tag{E12}$$

E.2 Gravitational Case

Consider next a system stationary at Schwarzschild radius r . As derived in Appendix C, the radial and transverse Schwarzschild coordinate speeds of light are

$$c_r = \pm \gamma_g^{-2} c, \tag{E13}$$

$$c_\perp = \pm \gamma_g^{-1} c, \tag{E14}$$

where

$$\gamma_g = \left(1 - \frac{2GM}{rc^2} \right)^{-1/2}. \tag{E15}$$

The radial and transverse directions may be evaluated separately. For radial propagation, the preferred-frame-referenced uncalibrated radial interval satisfies

$$\frac{dr_0}{dr} = \gamma_g. \tag{E16}$$

Therefore,

$$\begin{aligned} c_{0,r} &= |c_r| \frac{dr_0}{dr} \\ &= \gamma_g^{-2} c \gamma_g \\ &= \gamma_g^{-1} c. \end{aligned} \tag{E17}$$

For transverse propagation, the corresponding spatial interval is unchanged, so

$$\begin{aligned} c_{0,\perp} &= |c_\perp| \\ &= \gamma_g^{-1} c. \end{aligned} \tag{E18}$$

Thus, the independently evaluated radial and transverse interaction cycles yield the same speed-valued rate,

$$c_\star = \gamma_g^{-1} c = c \sqrt{1 - \frac{2GM}{rc^2}}. \tag{E19}$$

As in the kinetic case, the direction-dependent coordinate propagation rates reduce to a single direction-independent rate when expressed using the corresponding uncalibrated spatial intervals.

For a stationary system, the Schwarzschild timelike interval independently gives

$$ds^2 = \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2, \tag{E20}$$

and therefore

$$\begin{aligned} \frac{ds}{dt} &= c \sqrt{1 - \frac{2GM}{rc^2}} \\ &= \gamma_g^{-1} c \\ &= c_\star. \end{aligned} \tag{E21}$$

E.3 Combined Kinetic and Gravitational Case

The general Schwarzschild case includes both gravitational influence and motion relative to the preferred frame. The gravitational condition first establishes the locally available propagation rate

$$c_{g_0} = \gamma_g^{-1} c. \tag{E22}$$

Motion relative to the preferred frame then reduces the resulting rate according to the same kinetic relationship derived above, but with c_{g_0} replacing the asymptotic value c .

Define

$$\gamma_{K_0}^{-1} = \sqrt{1 - \frac{v_{p_0}^2}{c_{g_0}^2}}, \tag{E23}$$

$$v_{p_0}^2 = v_{r_0}^2 + v_{\perp_0}^2. \quad (\text{E24})$$

The combined speed of causality is therefore

$$\begin{aligned} c_\star &= \gamma_{K_0}^{-1} c_{g_0} \\ &= c_{g_0} \sqrt{1 - \frac{v_{p_0}^2}{c_{g_0}^2}} \\ &= \sqrt{c_{g_0}^2 - v_{p_0}^2}. \end{aligned} \quad (\text{E25})$$

For Schwarzschild coordinates, the preferred-frame-referenced uncalibrated radial velocity is

$$v_{r_0} = \frac{dr_0}{dt} = \frac{dr}{dt} \frac{dr_0}{dr} = \gamma_g \dot{r}, \quad (\text{E26})$$

while the transverse component is

$$v_{\perp_0} = r \frac{d\Omega_0}{dt} = r \dot{\Omega}_0 = r \dot{\Omega}, \quad (\text{E27})$$

with the understood relation

$$\frac{d\Omega_0}{d\Omega} = 1 \quad (\text{E28})$$

Substituting Equations (E22), (E26), and (E27) into Equation (E25) gives

$$\begin{aligned} c_\star &= \sqrt{\gamma_g^{-2} c^2 - \gamma_g^2 \dot{r}^2 - r^2 \dot{\Omega}^2} \\ &= c \sqrt{1 - \frac{2GM}{rc^2} - \frac{\gamma_g^2 \dot{r}^2}{c^2} - \frac{r^2 \dot{\Omega}^2}{c^2}}. \end{aligned} \quad (\text{E29})$$

Equivalently,

$$\boxed{c_\star = c \sqrt{1 - \frac{2GM/r + \gamma_g^2 \dot{r}^2 + r^2 \dot{\Omega}^2}{c^2}}}. \quad (\text{E30})$$

E.4 Recovery from the Timelike Interval

The same general result follows directly from the Schwarzschild timelike interval. For arbitrary radial and transverse motion,

$$ds^2 = \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 - \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 - r^2 d\Omega^2. \quad (\text{E31})$$

Dividing by dt^2 gives

$$\left(\frac{ds}{dt}\right)^2 = \gamma_g^{-2} c^2 - \gamma_g^2 \dot{r}^2 - r^2 \dot{\Omega}^2. \quad (\text{E32})$$

Taking the positive timelike magnitude,

$$\frac{ds}{dt} = \sqrt{\gamma_g^{-2} c^2 - \gamma_g^2 \dot{r}^2 - r^2 \dot{\Omega}^2} = c_\star. \quad (\text{E33})$$

Since

$$ds = c dt', \quad (\text{E34})$$

it follows that

$$\boxed{\frac{c_\star}{c} = \frac{dt'}{dt}}. \quad (\text{E35})$$

Thus, the speed of causality is equivalently the speed-valued representation of the timelike interval accumulated per unit coordinate time.

E.5 Metric-Independent Relationship

The preceding derivations use Minkowski and Schwarzschild coordinates to make the physical meaning of $c_\star$ explicit. The final relationship, however, is not restricted to either metric.

For any timelike metric expressed in coordinates aligned with the physical reference under consideration,

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (\text{E36})$$

Parameterizing the trajectory by coordinate time t gives

$$\left(\frac{ds}{dt}\right)^2 = g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}. \quad (\text{E37})$$

The corresponding speed of causality is therefore

$$c_\star \equiv \frac{ds}{dt} = \sqrt{g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}}. \quad (\text{E38})$$

Because $ds = c dt'$ for a timelike trajectory,

$$c_\star = c \frac{dt'}{dt}. \quad (\text{E39})$$

The particular terms appearing inside this expression depend on the metric and coordinate representation, but the relationship between $c_\star$ and the rate of time dilation does not.

E.6 Conclusion

The speed of causality can be recovered independently from kinetic, gravitational, and combined kinetic–gravitational cases. In purely kinetic motion,

$$c_\star = \gamma_K^{-1} c, \quad (\text{E40})$$

while for a stationary system in Schwarzschild gravitation,

$$c_\star = \gamma_g^{-1} c. \quad (\text{E41})$$

For the general Schwarzschild case,

$$c_\star = \sqrt{c^2 - \frac{2GM}{r} - \gamma_g^2 \dot{r}^2 - r^2 \dot{\Omega}^2}. \quad (\text{E42})$$

Each result is a special case of the general timelike relation

$$c_\star = \frac{ds}{dt} = \sqrt{g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}} = c \frac{dt'}{dt}. \quad (\text{E43})$$

The directional coordinate speeds of light and the particular decomposition into kinetic and gravitational terms may vary with the metric and coordinate representation. Because $c_\star = ds/dt$ is evaluated relative to the coordinate time selected by that representation, its numerical value may change under a change of temporal coordinate. What remains metric-independent is the causal identification and functional relationship: within whichever admissible representation corresponds to the physically real present, $c_\star/c = dt'/dt$, and $c_\star$ is the quantity governing the rate of time dilation.

F Deriving the Specific Lagrangian

This appendix derives the specific Lagrangian used in the main text from the specific-energy relationships established in Appendix B. The derivation shows that the difference between specific kinetic and specific potential energy is equivalent to the negative of the specific internal energy and, therefore, to the negative of one-half the squared speed of causality. This provides a direct connection between the specific-energy representation developed in Universal Specificity and Lagrangian mechanics.

F.1 Specific Lagrangian

In classical mechanics, the Lagrangian is defined as the difference between kinetic and potential energy,

$$L = T - V. \quad (\text{F1})$$

The corresponding specific Lagrangian is defined analogously using specific kinetic and specific potential energy:

$$\mathcal{L}_0 \equiv e_{K_0} - e_{P_0}. \quad (\text{F2})$$

As derived in Appendix B, the Schwarzschild specific-energy terms are

$$\begin{aligned} e_{K_0} &= \frac{1}{2}\dot{r}_0^2 + \frac{1}{2}r^2\dot{\Omega}_0^2 \\ &= \frac{1}{2}\gamma_g^2\dot{r}^2 + \frac{1}{2}r^2\dot{\Omega}^2, \end{aligned} \quad (\text{F3})$$

$$e_{P_0} = \frac{1}{2}c^2 - \frac{GM}{r}. \quad (\text{F4})$$

Therefore,

$$\mathcal{L}_0 = e_{K_0} - e_{P_0}$$

$$= -\frac{1}{2}c^2 + \frac{GM}{r} + \frac{1}{2}\dot{r}_0^2 + \frac{1}{2}r^2\dot{\Omega}^2. \quad (\text{F5})$$

F.2 Relation to the Speed of Causality

From Appendix E, the speed of causality for the general Schwarzschild case is

$$\begin{aligned} c_\star &= \frac{ds}{dt} \\ &= \sqrt{c^2 - \frac{2GM}{r} - \gamma_g^2 \dot{r}^2 - r^2 \dot{\Omega}^2}. \end{aligned} \quad (\text{F6})$$

Squaring and dividing by two gives

$$\begin{aligned} \frac{1}{2}c_\star^2 &= \frac{1}{2} \left(\frac{ds}{dt} \right)^2 \\ &= \frac{1}{2}c^2 - \frac{GM}{r} - \frac{1}{2}\gamma_g^2 \dot{r}^2 - \frac{1}{2}r^2 \dot{\Omega}^2. \end{aligned} \quad (\text{F7})$$

Using Equations (F3) and (F4),

$$\frac{1}{2}c_\star^2 = e_{P_0} - e_{K_0}. \quad (\text{F8})$$

But the specific internal energy derived in the main text is

$$e_{I_\star} = e_{P_0} - e_{K_0}. \quad (\text{F9})$$

Consequently,

$$\boxed{e_{I_\star} = \frac{1}{2}c_\star^2}. \quad (\text{F10})$$

Comparing this with the definition of the specific Lagrangian gives

$$\boxed{\mathcal{L}_0 = -e_{I_\star} = -\frac{1}{2}c_\star^2 = -\frac{1}{2} \left(\frac{ds}{dt} \right)^2 = -\frac{1}{2}g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}}. \quad (\text{F11})$$

The specific Lagrangian is therefore not an independently introduced quantity. It is another representation of the same specific-energy condition already represented by $e_{I_\star}$ and $c_\star$.

F.3 Euler–Lagrange Equation

The specific Lagrangian may be used with the Euler–Lagrange equation

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}_0}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}_0}{\partial q} = 0, \quad (\text{F12})$$

where q is a generalized coordinate.

For the preferred-frame-referenced radial coordinate r_0 , evaluated locally by instruments stationary with respect to the preferred frame and not corrected for radial length change, let

$$q = r_0, \quad \dot{q} = \dot{r}_0 = \frac{dr_0}{dt}. \quad (\text{F13})$$

Equation (F5) may then be written as

$$\mathcal{L}_0 = -\frac{1}{2}c^2 + \frac{GM}{r} + \frac{1}{2}\dot{r}_0^2 + \frac{1}{2}r^2\dot{\Omega}^2. \quad (\text{F14})$$

The velocity derivative is

$$\frac{\partial \mathcal{L}_0}{\partial \dot{r}_0} = \dot{r}_0, \quad (\text{F15})$$

and therefore

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}_0}{\partial \dot{r}_0} \right) = \ddot{r}_0 = \frac{d^2 r_0}{dt^2}. \quad (\text{F16})$$

To evaluate the spatial derivative, use

$$\frac{dr_0}{dr} = \gamma_g, \quad (\text{F17})$$

so that

$$\frac{\partial r}{\partial r_0} = \gamma_g^{-1}. \quad (\text{F18})$$

Then

$$\begin{aligned}
\frac{\partial \mathcal{L}_0}{\partial r_0} &= \frac{\partial \mathcal{L}_0}{\partial r} \frac{\partial r}{\partial r_0} \\
&= \frac{\partial}{\partial r} \left(-\frac{1}{2}c^2 + \frac{GM}{r} + \frac{1}{2}\dot{r}_0^2 + \frac{1}{2}r^2\dot{\Omega}^2 \right) \gamma_g^{-1} \\
&= \left(-\frac{GM}{r^2} + r\dot{\Omega}^2 \right) \gamma_g^{-1}.
\end{aligned} \tag{F19}$$

Substituting Equations (F16) and (F19) into the Euler–Lagrange equation gives

$$\ddot{r}_0 - \left(-\frac{GM}{r^2} + r\dot{\Omega}^2 \right) \gamma_g^{-1} = 0. \tag{F20}$$

Therefore,

$$\boxed{\ddot{r}_0 = -\frac{GM}{r^2} \gamma_g^{-1} + r\dot{\Omega}^2 \gamma_g^{-1}}. \tag{F21}$$

This is the preferred-frame-referenced uncalibrated radial specific-force relation obtained from the specific Lagrangian.

F.4 Relation to Schwarzschild Coordinate Acceleration

The preferred-frame-referenced uncalibrated radial acceleration $\ddot{r}_0$ can be related to the Schwarzschild coordinate acceleration $\ddot{r}$ through the radial interval relation, dr_0 . Taking the first derivative with respect to coordinate time,

$$\begin{aligned}
\frac{dr_0}{dt} &= \dot{r}_0 = \frac{dr}{dt} \frac{dr_0}{dr} \\
&= \gamma_g \dot{r}.
\end{aligned} \tag{F22}$$

Differentiating again gives

$$\frac{d^2 r_0}{dt^2} = \ddot{r}_0 = \frac{d^2 r}{dt^2} \frac{dr_0}{dr} + \left(\frac{dr}{dt} \right)^2 \frac{d^2 r_0}{dr^2} = \ddot{r} \frac{dr_0}{dr} + \dot{r}^2 \frac{d \left(\frac{dr_0}{dr} \right)}{dr}. \tag{F23}$$

Since

$$\frac{dr_0}{dr} = \gamma_g, \tag{F24}$$

we obtain

$$\ddot{r}_0 = \gamma_g \ddot{r} + \dot{r}^2 \frac{d\gamma_g}{dr}. \quad (\text{F25})$$

For

$$\gamma_g = \left(1 - \frac{2GM}{rc^2}\right)^{-1/2}, \quad (\text{F26})$$

the radial derivative is

$$\frac{d\gamma_g}{dr} = -\frac{GM}{r^2 c^2} \gamma_g^3. \quad (\text{F27})$$

Therefore,

$$\ddot{r}_0 = \gamma_g \ddot{r} - \frac{GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^3. \quad (\text{F28})$$

Substituting Equation (F21) gives

$$-\frac{GM}{r^2} \gamma_g^{-1} + r \dot{\Omega}^2 \gamma_g^{-1} = \gamma_g \ddot{r} - \frac{GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^3. \quad (\text{F29})$$

Solving for $\ddot{r}$ yields

$$\boxed{\ddot{r}_\nabla = -\frac{GM}{r^2} \gamma_g^{-2} + \frac{GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^2 + r \dot{\Omega}^2 \gamma_g^{-2}}. \quad (\text{F30})$$

The subscript ∇ emphasizes that this is the Schwarzschild-coordinate representation of the specific-force contribution generated by the gradient of the causal potential. It is not, by itself, the complete Schwarzschild coordinate geodesic acceleration; the additional velocity-dependent coordinate contribution identified in the main text must also be included when reconstructing the full coordinate equation of motion.

F.5 Conclusion

The specific-energy decomposition developed in Universal Specificity naturally defines a corresponding specific Lagrangian,

$$\mathcal{L}_0 = e_{K_0} - e_{P_0}. \quad (\text{F31})$$

Because

$$e_{I_\star} = e_{P_0} - e_{K_0} = \frac{1}{2}c_\star^2, \quad (\text{F32})$$

the same quantity may be written as

$$\boxed{\mathcal{L}_0 = -e_{I_\star} = -\frac{1}{2}c_\star^2}. \quad (\text{F33})$$

Applying the Euler–Lagrange equation to this representation yields the preferred-frame-referenced radial specific acceleration

$$\ddot{r}_0 = -\frac{GM}{r^2}\gamma_g^{-1} + r\dot{\Omega}^2\gamma_g^{-1}, \quad (\text{F34})$$

which is equivalently the gradient of the causal potential (see Appendix G). The specific Lagrangian therefore emerges directly from the same physical condition already represented by $c_\star$, $e_{I_\star}$, and $e_{U_\star}$, providing a Lagrangian formulation of the specific-force relationship developed in the main text.

G Specific Force from Causal Potential

This appendix derives the radial specific force produced by the gradient of the causal potential, $e_{U_\star}$, introduced in Section 4.2. Because

$$e_{U_\star} \equiv e_{I_\star} = \frac{1}{2}c_\star^2, \quad (\text{G1})$$

the corresponding specific force is

$$\begin{aligned} \vec{f}_{g_0} &= -\nabla_0 e_{U_\star} \\ &= -\nabla_0 \left(\frac{1}{2}c_\star^2 \right). \end{aligned} \quad (\text{G2})$$

Because $e_{U_\star}$ may depend upon both position and generalized velocity, ∇_0 denotes here the spatial partial gradient on the system’s state space, evaluated at fixed generalized velocities. For the Schwarzschild case, the spatial variation of the potential is radial. The radial specific force is therefore

$$f_{g_0} = -\frac{\partial e_{U_\star}}{\partial r_0}$$

$$= -\frac{\partial}{\partial r_0} \left(\frac{1}{2} c_\star^2 \right), \quad (\text{G3})$$

where r_0 denotes the radial interval evaluated locally by an instrument co-located with the system's position but stationary with respect to the preferred frame, without correction for gravitationally induced radial-length contraction.

From the specific-energy relation,

$$\frac{1}{2} c_\star^2 = e_{I_\star} = e_{U_\star} = \frac{1}{2} c^2 - \frac{GM}{r} - \frac{1}{2} \dot{r}_0^2 - \frac{1}{2} r^2 \dot{\Omega}^2. \quad (\text{G4})$$

The transverse angular measure is unchanged in this Schwarzschild construction, $\dot{\Omega}_0 = \dot{\Omega}$, therefore:

$$e_{U_\star} = \frac{1}{2} c^2 - \frac{GM}{r} - \frac{1}{2} \dot{r}_0^2 - \frac{1}{2} r^2 \dot{\Omega}^2. \quad (\text{G5})$$

The preferred-frame uncalibrated radial coordinate is related to the Schwarzschild radial coordinate by

$$\frac{dr_0}{dr} = \gamma_g, \quad (\text{G6})$$

and therefore

$$\frac{dr}{dr_0} = \gamma_g^{-1}. \quad (\text{G7})$$

Applying the chain rule to Equation (G3) gives

$$f_{g_0} = -\frac{\partial \left(\frac{1}{2} c_\star^2 \right)}{\partial r} \frac{dr}{dr_0}. \quad (\text{G8})$$

Evaluating the partial derivative while holding the generalized velocities $\dot{r}_0$ and $\dot{\Omega}$ fixed,

$$\begin{aligned} f_{g_0} &= -\frac{\partial}{\partial r} \left(\frac{1}{2} c^2 - \frac{GM}{r} - \frac{1}{2} \dot{r}_0^2 - \frac{1}{2} r^2 \dot{\Omega}^2 \right) \gamma_g^{-1} \\ &= -\left(\frac{GM}{r^2} - r \dot{\Omega}^2 \right) \gamma_g^{-1}. \end{aligned} \quad (\text{G9})$$

Thus,

$$\boxed{f_{g_0} = -\frac{GM}{r^2}\gamma_g^{-1} + r\dot{\Omega}^2\gamma_g^{-1}}. \quad (\text{G10})$$

This is the same preferred-frame specific-force relation obtained independently from the specific Lagrangian in Appendix F. The agreement follows directly from

$$\mathcal{L}_0 = -e_{U_*}, \quad (\text{G11})$$

which gives

$$\frac{\partial \mathcal{L}_0}{\partial r_0} = -\frac{\partial e_{U_*}}{\partial r_0} = f_{g_0}. \quad (\text{G12})$$

G.1 Schwarzschild Coordinate Representation

The preferred-frame specific force may be expressed in Schwarzschild radial coordinates using the relation between r_0 and r . From

$$\dot{r}_0 = \frac{dr_0}{dt} = \gamma_g \dot{r}, \quad (\text{G13})$$

differentiating with respect to coordinate time gives

$$\ddot{r}_0 = \gamma_g \ddot{r} + \dot{r}^2 \frac{d\gamma_g}{dr}. \quad (\text{G14})$$

Using

$$\gamma_g = \left(1 - \frac{2GM}{rc^2}\right)^{-1/2}, \quad (\text{G15})$$

its radial derivative is

$$\frac{d\gamma_g}{dr} = -\frac{GM}{r^2 c^2} \gamma_g^3. \quad (\text{G16})$$

Therefore,

$$\ddot{r}_0 = \gamma_g \ddot{r} - \frac{GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^3. \quad (\text{G17})$$

Identifying the preferred-frame radial specific acceleration with the specific force,

$$\ddot{r}_0 = f_{g_0}, \quad (\text{G18})$$

and substituting Equation (G10) gives

$$-\frac{GM}{r^2}\gamma_g^{-1} + r\dot{\Omega}^2\gamma_g^{-1} = \gamma_g\ddot{r}_\nabla - \frac{GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^3. \quad (\text{G19})$$

Solving for the corresponding Schwarzschild coordinate contribution yields

$$f_g \equiv \ddot{r}_\nabla = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2 + r\dot{\Omega}^2\gamma_g^{-2}. \quad (\text{G20})$$

The subscript ∇ emphasizes that this is the Schwarzschild-coordinate representation of the acceleration produced by the gradient of the causal potential. As discussed in the main text, it is not by itself the complete Schwarzschild coordinate geodesic acceleration; additional velocity-dependent coordinate terms must be included when reconstructing the full geodesic equation.

G.2 Conclusion

The causal potential

$$e_{U_\star} = e_{I_\star} = \frac{1}{2}c_\star^2 \quad (\text{G21})$$

produces the preferred-frame uncalibrated radial specific force

$$f_{g_0} = -\frac{\partial e_{U_\star}}{\partial r_0} = -\frac{\partial}{\partial r_0} \left(\frac{1}{2}c_\star^2 \right) = -\frac{GM}{r^2}\gamma_g^{-1} + r\dot{\Omega}^2\gamma_g^{-1}. \quad (\text{G22})$$

This is the same relation obtained independently from the specific Lagrangian, demonstrating that both the energy-gradient and Lagrangian formulations produce the same preferred-frame specific force.

When expressed in Schwarzschild coordinates, the same gradient-produced contribution becomes

$$f_g = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2 + r\dot{\Omega}^2\gamma_g^{-2}. \quad (\text{G23})$$

The result therefore connects the scalar causal potential directly to the specific force governing motion through

$$\boxed{\vec{f}_{g_0} = -\nabla_0 e_{U_*} = -\nabla_0 \left(\frac{1}{2} c_*^2 \right)}. \quad (\text{G24})$$

H Deriving the Schwarzschild Radial Coordinate Acceleration

This appendix derives the Schwarzschild radial coordinate acceleration of a freely falling test particle directly from the geodesic equation. The derivation proceeds in two stages. First, the radial Schwarzschild coordinate is differentiated with respect to proper time. This result is then transformed to Schwarzschild coordinate time using the temporal geodesic equation and the chain rule.

The distinction between the two acceleration quantities is

$$\ddot{r}_{t'} \equiv \frac{d^2 r}{dt'^2}, \quad (\text{H1})$$

$$\ddot{r} \equiv \frac{d^2 r}{dt^2}, \quad (\text{H2})$$

where t' is proper time along the particle trajectory and t is Schwarzschild coordinate time. Both expressions involve the Schwarzschild radial coordinate r ; they differ only in the parameter with respect to which that coordinate is differentiated.

H.1 Schwarzschild Metric and Geodesic Equation

Using the $(+, -, -, -)$ signature, the Schwarzschild line element is [35]

$$c^2 dt'^2 = ds^2 = \gamma_g^{-2} c^2 dt^2 - \gamma_g^2 dr^2 - r^2 d\Omega^2, \quad (\text{H3})$$

where

$$\gamma_g^{-1} = \sqrt{1 - \frac{r_s}{r}} = \sqrt{1 - \frac{2GM}{rc^2}}, \quad (\text{H4})$$

$$r_s = \frac{2GM}{c^2}, \quad (\text{H5})$$

$$d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2. \quad (\text{H6})$$

Because the Schwarzschild spacetime is spherically symmetric, any geodesic may be taken without loss of generality to lie in the equatorial plane,

$$\theta = \frac{\pi}{2}, \quad d\theta = 0. \quad (\text{H7})$$

The angular coordinate may therefore be represented by $\Omega = \phi$, giving

$$d\Omega^2 = d\phi^2. \quad (\text{H8})$$

The metric then reduces to

$$ds^2 = \gamma_g^{-2} c^2 dt^2 - \gamma_g^2 dr^2 - r^2 d\Omega^2. \quad (\text{H9})$$

The geodesic equation for a freely falling test particle is [34, 35]

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0, \quad (\text{H10})$$

where λ is an affine parameter.

For a timelike geodesic, proper time t' may be used as the affine parameter, giving

$$\frac{d^2 x^\mu}{dt'^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{dt'} \frac{dx^\beta}{dt'} = 0. \quad (\text{H11})$$

The Christoffel symbols are defined by

$$\Gamma_{\alpha\beta}^\mu = \frac{1}{2} g^{\mu\nu} \left(\frac{\partial g_{\nu\beta}}{\partial x^\alpha} + \frac{\partial g_{\nu\alpha}}{\partial x^\beta} - \frac{\partial g_{\alpha\beta}}{\partial x^\nu} \right). \quad (\text{H12})$$

For coordinates (t, r, Ω) in the equatorial plane, the reduced metric tensor is

$$g_{\mu\nu} = \begin{bmatrix} \gamma_g^{-2} c^2 & 0 & 0 \\ 0 & -\gamma_g^2 & 0 \\ 0 & 0 & -r^2 \end{bmatrix}. \quad (\text{H13})$$

The Christoffel symbols required for the radial and temporal geodesic equations are

$$\Gamma_{tr}^t = \Gamma_{rt}^t = \frac{r_s}{2r(r - r_s)} = \frac{GM}{r^2} \frac{\gamma_g^2}{c^2}, \quad (\text{H14})$$

$$\Gamma_{tt}^r = \frac{c^2 r_s (r - r_s)}{2r^3} = \frac{GM}{r^2} \gamma_g^{-2}, \quad (\text{H15})$$

$$\Gamma_{rr}^r = -\frac{r_s}{2r(r - r_s)} = -\frac{GM}{r^2} \frac{\gamma_g^2}{c^2}, \quad (\text{H16})$$

$$\Gamma_{\Omega\Omega}^r = -(r - r_s) = -r\gamma_g^{-2}, \quad (\text{H17})$$

$$\Gamma_{r\Omega}^\Omega = \Gamma_{\Omega r}^\Omega = \frac{1}{r}. \quad (\text{H18})$$

H.2 Radial Coordinate Acceleration in Proper Time

Taking $\mu = r$ in Equation (H11) gives

$$\frac{d^2 r}{dt'^2} + \Gamma_{tt}^r \left(\frac{dt}{dt'} \right)^2 + \Gamma_{rr}^r \left(\frac{dr}{dt'} \right)^2 + \Gamma_{\Omega\Omega}^r \left(\frac{d\Omega}{dt'} \right)^2 = 0. \quad (\text{H19})$$

Therefore,

$$\ddot{r}_{t'} = -\Gamma_{tt}^r \left(\frac{dt}{dt'} \right)^2 - \Gamma_{rr}^r \left(\frac{dr}{dt'} \right)^2 - \Gamma_{\Omega\Omega}^r \left(\frac{d\Omega}{dt'} \right)^2. \quad (\text{H20})$$

Using

$$\frac{dr}{dt'} = \frac{dr}{dt} \frac{dt}{dt'} = \dot{r} \frac{dt}{dt'}, \quad (\text{H21})$$

$$\frac{d\Omega}{dt'} = \frac{d\Omega}{dt} \frac{dt}{dt'} = \dot{\Omega} \frac{dt}{dt'}, \quad (\text{H22})$$

we obtain

$$\ddot{r}_{t'} = \left(-\Gamma_{tt}^r - \Gamma_{rr}^r \dot{r}^2 - \Gamma_{\Omega\Omega}^r \dot{\Omega}^2 \right) \left(\frac{dt}{dt'} \right)^2. \quad (\text{H23})$$

Substituting the Christoffel symbols gives

$$\boxed{\ddot{r}_{t'} = \left(-\frac{GM}{r^2} \gamma_g^{-2} + \frac{GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^2 + r \dot{\Omega}^2 \gamma_g^{-2} \right) \left(\frac{dt}{dt'} \right)^2}. \quad (\text{H24})$$

This is the acceleration of the Schwarzschild radial coordinate when the particle trajectory is parameterized by proper time.

H.3 Transformation to Schwarzschild Coordinate Time

We now relate $\ddot{r}_{t'}$ to the radial coordinate acceleration

$$\ddot{r} = \frac{d^2 r}{dt^2}. \quad (\text{H25})$$

Beginning with

$$\frac{dr}{dt'} = \frac{dr}{dt} \frac{dt}{dt'}, \quad (\text{H26})$$

differentiate once more with respect to proper time:

$$\begin{aligned} \frac{d^2 r}{dt'^2} &= \frac{d}{dt'} \left(\frac{dr}{dt} \frac{dt}{dt'} \right) \\ &= \frac{d^2 r}{dt^2} \left(\frac{dt}{dt'} \right)^2 + \frac{dr}{dt} \frac{d^2 t}{dt'^2}. \end{aligned} \quad (\text{H27})$$

Thus,

$$\ddot{r}_{t'} = \ddot{r} \left(\frac{dt}{dt'} \right)^2 + \dot{r} \frac{d^2 t}{dt'^2}. \quad (\text{H28})$$

Solving Equation (H28) for $\ddot{r}$ gives

$$\ddot{r} = \ddot{r}_{t'} \left(\frac{dt'}{dt} \right)^2 - \dot{r} \frac{d^2 t}{dt'^2} \left(\frac{dt'}{dt} \right)^2. \quad (\text{H29})$$

The remaining term, $d^2 t / dt'^2$, follows from the temporal geodesic equation.

H.4 Temporal Geodesic Equation

Taking $\mu = t$ in Equation (H11) gives

$$\frac{d^2 t}{dt'^2} + \Gamma_{\alpha\beta}^t \frac{dx^\alpha}{dt'} \frac{dx^\beta}{dt'} = 0. \quad (\text{H30})$$

The only contributing Christoffel symbols are

$$\Gamma_{tr}^t = \Gamma_{rt}^t, \quad (\text{H31})$$

so

$$\begin{aligned}\frac{d^2 t}{dt'^2} &= -\Gamma_{tr}^t \frac{dt}{dt'} \frac{dr}{dt'} - \Gamma_{rt}^t \frac{dr}{dt'} \frac{dt}{dt'} \\ &= -2\Gamma_{tr}^t \frac{dr}{dt'} \frac{dt}{dt'}.\end{aligned}\tag{H32}$$

Using

$$\frac{dr}{dt'} = \dot{r} \frac{dt}{dt'},\tag{H33}$$

we find

$$\begin{aligned}\frac{d^2 t}{dt'^2} &= -2\Gamma_{tr}^t \dot{r} \left(\frac{dt}{dt'} \right)^2 \\ &= -2 \frac{GM}{r^2} \frac{\gamma_g^2}{c^2} \dot{r} \left(\frac{dt}{dt'} \right)^2.\end{aligned}\tag{H34}$$

H.5 Schwarzschild Radial Coordinate Acceleration

Substituting Equations (H24) and (H34) into Equation (H29) gives

$$\begin{aligned}\ddot{r} &= \left(-\frac{GM}{r^2} \gamma_g^{-2} + \frac{GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^2 + r \dot{\Omega}^2 \gamma_g^{-2} \right) \left(\frac{dt}{dt'} \right)^2 \left(\frac{dt'}{dt} \right)^2 \\ &\quad - \dot{r} \left[-2 \frac{GM}{r^2} \frac{\gamma_g^2}{c^2} \dot{r} \left(\frac{dt}{dt'} \right)^2 \right] \left(\frac{dt'}{dt} \right)^2.\end{aligned}\tag{H35}$$

Since

$$\left(\frac{dt}{dt'} \right)^2 \left(\frac{dt'}{dt} \right)^2 = 1,\tag{H36}$$

the expression reduces to

$$\begin{aligned}\ddot{r} &= -\frac{GM}{r^2} \gamma_g^{-2} + \frac{GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^2 + r \dot{\Omega}^2 \gamma_g^{-2} \\ &\quad + 2 \frac{GM}{r^2} \frac{\dot{r}^2}{c^2} \gamma_g^2.\end{aligned}\tag{H37}$$

Combining the radial velocity terms gives

$$\ddot{r} = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{3GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2 + r\dot{\Omega}^2\gamma_g^{-2}. \quad (\text{H38})$$

This is the complete Schwarzschild radial coordinate acceleration for a freely falling test particle expressed with respect to Schwarzschild coordinate time.

H.6 Decomposition of the Coordinate Acceleration

For comparison with the specific-force relation developed in the main text, Equation (H38) may be separated into two contributions:

$$\ddot{r} = \ddot{r}_\nabla + \ddot{r}_v, \quad (\text{H39})$$

where

$$\ddot{r}_\nabla = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2 + r\dot{\Omega}^2\gamma_g^{-2}, \quad (\text{H40})$$

and

$$\ddot{r}_v = 2\frac{GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2. \quad (\text{H41})$$

The first contribution is the Schwarzschild-coordinate form recovered from the gradient of the causal potential in Section 4.2, while the second arises from converting the proper-time-parameterized geodesic equation to Schwarzschild coordinate time.

Together,

$$\ddot{r}_\nabla + \ddot{r}_v = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{3GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2 + r\dot{\Omega}^2\gamma_g^{-2}, \quad (\text{H42})$$

recovering the complete relativistic coordinate equation of motion.

H.7 Conclusion

Beginning with the Schwarzschild geodesic equations parameterized by proper time, the radial coordinate acceleration is

$$\ddot{r}_{t'} = \left(-\frac{GM}{r^2}\gamma_g^{-2} + \frac{GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2 + r\dot{\Omega}^2\gamma_g^{-2} \right) \left(\frac{dt}{dt'} \right)^2. \quad (\text{H43})$$

Transforming from proper time to Schwarzschild coordinate time introduces an additional velocity-dependent term, yielding

$$\ddot{r} = -\frac{GM}{r^2}\gamma_g^{-2} + \frac{3GM}{r^2}\frac{\dot{r}^2}{c^2}\gamma_g^2 + r\dot{\Omega}^2\gamma_g^{-2}. \quad (\text{H44})$$

This provides the relativistic Schwarzschild coordinate equation of motion used in the main text as the comparison standard for the specific-force relation derived from the causal potential.

I Kinematic Acceleration and Specific Reaction Force

This appendix establishes from relativistic kinematics that acceleration through a spatially uniform VDF produces a spatial gradient in causal potential across an extended Born-rigid body, and that the negative gradient of this potential produces a specific reaction force opposing the imposed acceleration. In the main text, this result is considered together with gravitational cases to inductively identify the causal condition associated with weightedness.

I.1 The Clock-Rate Gradient of Born-Rigid Acceleration

Consider an extended system accelerating through otherwise empty space, where the VDF is spatially uniform. Although the surrounding condition is spatially uniform, the temporal rate throughout the accelerating system is not. For acceleration in the positive x -direction, clocks toward the front of a Born-rigid system advance more rapidly than clocks toward the rear [64].

This spatial variation in clock rate follows from the kinematics of Born-rigid acceleration. Maintaining a constant locally measured separation between different portions of an accelerating body requires different worldlines along the direction of acceleration and, consequently, different proper accelerations and rates of accumulated proper time. A derivation of this relation and its correspondence with the weak-field gravitational clock-rate gradient is provided in Appendix J.

For a Born-rigid accelerating system, take the rear of the system as the reference position $x' = 0$. The local clock interval at position x' relative to the rear clock interval is

$$\frac{dt'(x')}{dt'_r} = 1 + \frac{a'x'}{c^2}, \quad (\text{I1})$$

where a' is the proper acceleration at the rear reference point and x' is the proper distance measured from the rear in the direction of acceleration.

At the front of the system, $x' > 0$, this becomes

$$\frac{dt'_f}{dt'_r} = 1 + \frac{a'x'}{c^2}. \quad (\text{I2})$$

The proper acceleration is not uniform throughout a Born-rigid system. Rather, its value varies with position along the acceleration direction. Equation (I1) uses the rear acceleration a' as the reference acceleration and evaluates the resulting clock-rate gradient relative to that point.

This relationship follows from Born-rigid acceleration itself and does not require the equivalence principle.

I.2 Interpretation of Proper Measurements

The primed quantities in this appendix denote operational proper measurements made by instruments comoving with the accelerating system. In the mechanistic interpretation developed in this work, such quantities are directly measurable but are *not* taken to represent fundamental calibrated measures of the underlying physical intervals. Rather, the clocks and rulers used to make those measurements are themselves affected by the relativistic conditions being measured and lose calibration.

Accordingly, dt'_r and dt'_f denote the local clock intervals measured by clocks comoving with the rear and front of the system, respectively, while x' denotes the corresponding locally measured spatial separation. These quantities provide the natural operational calibration in which an accelerometer attached to the system reports its proper acceleration.

The objective here is therefore not to convert these quantities immediately into preferred-frame calibrated measures. Instead, the causal-potential gradient is first evaluated using the rear clock as the local normalization standard. This normalization is not a calibration to the preferred-frame standard; it amounts only to the setting the rear clock as the reference against which the other local clock rates are compared.

I.3 Rear-Normalized Causal Potential

Because the clock rate varies spatially across the accelerating system, the corresponding causal potential also varies spatially. To evaluate this variation using the rear clock as the reference standard, define the rear-normalized causal potential

$$e'_{U_0}(x') \equiv \frac{1}{2}c^2 \left(\frac{dt'(x')}{dt'_r} \right)^2. \quad (\text{I3})$$

At the rear reference point,

$$x' = 0, \quad (\text{I4})$$

and therefore

$$e'_{U_0}(0) = \frac{1}{2}c^2. \quad (\text{I5})$$

The quantity e'_{U_0} is not a distinct physical potential from the causal potential $e_{U_\star}$ developed in the main text. It represents the same underlying causal condition expressed using the rear clock and comoving ruler calibration of the accelerating system. Once the calibration between the proper and preferred-frame quantities is specified, a spatial variation in one representation corresponds to a spatial variation in the other.

For the present purpose, this explicit conversion is unnecessary. The rear-calibrated representation is sufficient to determine the specific reaction force in the same calibration in which that reaction is measured by a comoving accelerometer.

I.4 Specific Reaction Force

Within the mechanistic framework, specific force is produced by the negative spatial gradient of causal potential. Using the rear-calibrated quantities, define the causal-potential gradient at the rear as

$$\nabla'_0 e'_{U_0} \equiv \lim_{x' \rightarrow 0} \frac{e'_{U_0}(x') - e'_{U_0}(0)}{x'} \hat{x}. \quad (\text{I6})$$

The corresponding proper specific reaction force is

$$\vec{f}' \equiv -\nabla'_0 e'_{U_0}. \quad (\text{I7})$$

Substituting Equation (I1) into Equation (I3) gives

$$e'_{U_0}(x') = \frac{1}{2}c^2 \left(1 + \frac{a'x'}{c^2} \right)^2. \quad (\text{I8})$$

The specific reaction force is therefore

$$\begin{aligned} \vec{f}' &= - \lim_{x' \rightarrow 0} \frac{\frac{1}{2}c^2 \left[\left(1 + \frac{a'x'}{c^2} \right)^2 - 1 \right]}{x'} \hat{x} \\ &= - \lim_{x' \rightarrow 0} \frac{\frac{1}{2}c^2 \left[2\frac{a'x'}{c^2} + \frac{a'^2 x'^2}{c^4} \right]}{x'} \hat{x} \\ &= - \lim_{x' \rightarrow 0} \left(a' + \frac{a'^2 x'}{2c^2} \right) \hat{x} \end{aligned}$$

$$= -a' \hat{x}. \quad (\text{I9})$$

Thus,

$$\boxed{\vec{f}' = -a' \hat{x}}. \quad (\text{I10})$$

The causal-potential gradient generated across the Born-rigid accelerating system therefore produces a specific reaction force equal in magnitude and opposite in direction to the imposed proper acceleration.

Equivalently,

$$\boxed{\vec{f}' = -\nabla'_0 e'_{U_0} = -a' \hat{x}}. \quad (\text{I11})$$

This result is obtained from the kinematic clock-rate gradient alone.

I.5 Relation to Preferred-Frame Coordinate Acceleration

For one-dimensional motion parallel to the direction of acceleration, the proper acceleration a' and preferred-frame coordinate acceleration a are related by the standard special-relativistic expression

$$a' = \gamma_K^3 a, \quad (\text{I12})$$

or equivalently,

$$a = \frac{a'}{\gamma_K^3}, \quad (\text{I13})$$

where

$$\gamma_K = \frac{1}{\sqrt{1 - \frac{v_p^2}{c^2}}}. \quad (\text{I14})$$

The reaction derived in Equation (I10) is expressed in the rear-normalized operational representation because the clock intervals, spatial intervals, and acceleration used to construct the rear-normalized causal potential are all measured by instruments comoving with the accelerating system. Equation (I13) provides the corresponding relationship to preferred-frame coordinate acceleration. Within the mechanistic paradigm, the proper acceleration a' is used as the operational reading of the affected comoving accelerometer and as a convenient quantity related to preferred-frame coordinate acceleration; it is not assigned an independent ontological status.

I.6 Implication for Inertia and Weightedness

The derivation above begins with a system accelerating through a region in which the external VDF is spatially uniform. Consequently, the resulting reaction cannot be attributed to a pre-existing spatial gradient in the surrounding condition. Instead, Born-rigid acceleration itself produces a spatial variation in clock rate across the extended system:

$$\frac{dt'(x')}{dt'_r} = 1 + \frac{a'x'}{c^2}. \quad (\text{I15})$$

That temporal gradient corresponds to a spatial gradient in causal potential,

$$\nabla'_0 e'_{U_0} \neq 0, \quad (\text{I16})$$

which produces the specific reaction force

$$\vec{f}' = -\nabla'_0 e'_{U_0} = -a' \hat{x}. \quad (\text{I17})$$

Acceleration therefore supplies a purely kinematic case in which a causal-potential gradient and weighted reaction arise together. The gravitational cases considered in the main text supply the complementary comparison required to determine whether the same condition accompanies weightedness under gravitation.

I.7 Conclusion

Born-rigid acceleration necessarily produces a spatial clock-rate gradient across an extended accelerating system. Expressed relative to the rear clock,

$$\frac{dt'(x')}{dt'_r} = 1 + \frac{a'x'}{c^2}. \quad (\text{I18})$$

Representing this spatially varying temporal condition as a rear-normalized causal potential gives

$$e'_{U_0}(x') = \frac{1}{2}c^2 \left(\frac{dt'(x')}{dt'_r} \right)^2. \quad (\text{I19})$$

Its negative spatial gradient at the rear is

$$\boxed{-\nabla'_0 e'_{U_0} = -a' \hat{x}}. \quad (\text{I20})$$

Thus, acceleration through a spatially uniform external condition nevertheless creates a causal-potential gradient across an extended system, and that gradient produces a measurable specific reaction force opposing the acceleration. This provides the kinematic counterpart to the gravitational causal-potential gradients examined in the main text and supplies the inertial case used in the induction concerning the causal origin of weightedness.

J Equivalence of Time-Dilation Gradients: Weak-Field Gravity and Born-Rigid Acceleration

This appendix establishes the local equivalence between the time-dilation gradients produced by two distinct physical situations:

1. a static weak gravitational field evaluated across a small radial interval; and
2. a Born-rigid accelerating system in flat spacetime.

The comparison is made at the level of the measurable clock-rate gradient: the spatial variation in the ratio of clock intervals measured at neighboring locations. The mathematical equivalence can therefore be established independently of any particular interpretation of the physical mechanism responsible for that gradient.

Within Universal Specificity, the shared clock-rate gradient is interpreted as a shared spatial variation in the rate of causal processes and, consequently, in causal potential. This appendix establishes the mathematical equivalence of those gradients; their causal interpretation is developed in the main text.

Throughout this appendix, primed quantities denote uncalibrated operational measurements made by instruments comoving with the system under consideration. These clocks, rulers, and accelerometers are themselves physically affected by time dilation, length contraction, and acceleration. Their readings are therefore directly observable proper quantities, but they have not been corrected back to preferred-frame calibrated coordinates.

J.1 Born Rigidity and Position-Dependent Proper Acceleration

Consider an extended object accelerating along its length while maintaining constant proper length, meaning constant separation as measured by uncalibrated rulers comoving with the object. In special relativity, such Born-rigid acceleration cannot be maintained if every point along the object undergoes the same proper acceleration. Instead, the proper acceleration must vary systematically with position so that the locally measured separation between neighboring worldlines remains constant [35, 64].

For acceleration in the positive x -direction, let the rear of the system define the reference worldline at $x' = 0$, with proper acceleration a' . The proper acceleration of a comoving point located at position x' is

$$a'(x') = \frac{a'}{1 + \frac{a'x'}{c^2}}. \quad (\text{J1})$$

Thus, points farther toward the front of the system undergo smaller proper acceleration than points toward the rear.

This position-dependent acceleration is not an additional correction imposed upon otherwise uniform acceleration. It is the kinematic condition required to maintain Born rigidity. The corresponding spatial variation in accumulated clock time follows from the same construction.

J.2 Born-Rigid Time-Dilation Gradient

Begin with Minkowski spacetime in inertial coordinates (T, X, Y, Z) using the $(+, -, -, -)$ signature:

$$ds^2 = c^2 dT^2 - dX^2 - dY^2 - dZ^2. \quad (\text{J2})$$

A Rindler coordinate system adapted to the Born-rigid accelerating congruence may be defined by [64, 65]

$$cT = \left(x' + \frac{c^2}{a'}\right) \sinh\left(\frac{a't'_r}{c}\right), \quad (\text{J3})$$

$$X = \left(x' + \frac{c^2}{a'}\right) \cosh\left(\frac{a't'_r}{c}\right) - \frac{c^2}{a'}, \quad (\text{J4})$$

$$Y = y, \quad Z = z. \quad (\text{J5})$$

Here a' is the uncalibrated proper acceleration measured by an accelerometer comoving with the rear reference worldline, t'_r is the uncalibrated clock reading of the rear clock, and x' is the uncalibrated proper distance from that rear worldline in the direction of acceleration.

Substituting these transformations into the Minkowski line element gives the Rindler metric

$$ds^2 = \left(1 + \frac{a'x'}{c^2}\right)^2 c^2 dt'^2_r - dx'^2 - dy^2 - dz^2. \quad (\text{J6})$$

At the rear reference point, $x' = 0$, the Rindler coordinate interval dt'_r is equal to the proper clock interval measured there.

For a clock comoving with the accelerating system at fixed x' , $dx' = dy = dz = 0$, and therefore

$$c^2 dt'(x')^2 = \left(1 + \frac{a'x'}{c^2}\right)^2 c^2 dt_r'^2. \quad (\text{J7})$$

Taking the positive clock interval gives

$$dt'(x') = \left(1 + \frac{a'x'}{c^2}\right) dt_r'. \quad (\text{J8})$$

Hence,

$$\boxed{\frac{dt'(x')}{dt_r'} = 1 + \frac{a'x'}{c^2}}. \quad (\text{J9})$$

For a front clock located at $x' > 0$,

$$\frac{dt'_f}{dt'_r} = 1 + \frac{a'x'}{c^2}. \quad (\text{J10})$$

Thus, clocks farther forward in the direction of acceleration advance more rapidly relative to the rear clock.

Differentiating the clock-rate ratio with respect to the uncalibrated comoving distance gives

$$\boxed{\frac{d}{dx'} \left(\frac{dt'(x')}{dt_r'} \right) = \frac{a'}{c^2}}. \quad (\text{J11})$$

This is the local time-dilation gradient produced by Born-rigid acceleration.

J.3 Why No Separate Length-Contraction Correction Appears

Equation (J9) is not obtained by first calculating a time-dilation effect and then applying an independent length-contraction correction. The Rindler congruence itself is constructed so that neighboring worldlines maintain constant proper separation. The position-dependent accelerations required to accomplish this already incorporate the relativistic kinematic conditions necessary for Born rigidity [35, 64].

Consequently, the uncalibrated proper distance x' and the corresponding clock-rate gradient arise together from the same Born-rigid construction. No additional length-contraction term is required afterward.

J.4 Weak-Field Gravitational Time-Dilation Gradient

Consider now two clocks held stationary at nearby Schwarzschild radii. For a stationary clock at radius r ,

$$dt'(r) = \sqrt{1 - \frac{2GM}{rc^2}} dt. \quad (\text{J12})$$

Let the lower reference clock be located at radius R , and let a second clock be located a small radial distance z above it,

$$r = R + z, \quad |z| \ll R. \quad (\text{J13})$$

The ratio of the two uncalibrated clock intervals is

$$\frac{dt'(R+z)}{dt'(R)} = \sqrt{\frac{1 - \frac{2GM}{(R+z)c^2}}{1 - \frac{2GM}{Rc^2}}}. \quad (\text{J14})$$

In the weak-field limit,

$$\frac{GM}{Rc^2} \ll 1, \quad (\text{J15})$$

and over a sufficiently small height interval,

$$\frac{|z|}{R} \ll 1. \quad (\text{J16})$$

Introducing the Newtonian gravitational potential

$$\phi(r) = -\frac{GM}{r}, \quad (\text{J17})$$

the stationary-clock relation becomes, to first order,

$$dt'(r) \approx \left(1 + \frac{\phi(r)}{c^2}\right) dt. \quad (\text{J18})$$

Expanding the potential about R gives

$$\begin{aligned}
\phi(R+z) &\approx \phi(R) + \left. \frac{d\phi}{dr} \right|_R z \\
&= \phi(R) + gz,
\end{aligned} \tag{J19}$$

where

$$g \equiv \frac{GM}{R^2}. \tag{J20}$$

The ratio between the upper and lower clock intervals is therefore, to first order,

$$\begin{aligned}
\frac{dt'(R+z)}{dt'(R)} &\approx \frac{1 + \frac{\phi(R)+gz}{c^2}}{1 + \frac{\phi(R)}{c^2}} \\
&\approx 1 + \frac{gz}{c^2}.
\end{aligned} \tag{J21}$$

Defining

$$dt'_g \equiv dt'(R) \tag{J22}$$

as the uncalibrated clock interval measured by the lower reference clock,

$$\boxed{\frac{dt'(z)}{dt'_g} \approx 1 + \frac{gz}{c^2}}. \tag{J23}$$

The corresponding weak-field gravitational time-dilation gradient is

$$\boxed{\frac{d}{dz} \left(\frac{dt'(z)}{dt'_g} \right) \approx \frac{g}{c^2}}. \tag{J24}$$

J.5 Equivalence of the Time-Dilation Gradients

The Born-rigid accelerating system gives

$$\frac{dt'(x')}{dt'_r} = 1 + \frac{a'x'}{c^2}, \tag{J25}$$

while the static weak gravitational field gives

$$\frac{dt'(z)}{dt'_g} \approx 1 + \frac{gz}{c^2}. \quad (\text{J26})$$

These expressions have identical local form under the correspondence

$$a' \longleftrightarrow g, \quad (\text{J27})$$

$$x' \longleftrightarrow z. \quad (\text{J28})$$

Their spatial gradients are correspondingly

$$\frac{d}{dx'} \left(\frac{dt'(x')}{dt'_r} \right) = \frac{a'}{c^2}, \quad (\text{J29})$$

$$\frac{d}{dz} \left(\frac{dt'(z)}{dt'_g} \right) \approx \frac{g}{c^2}. \quad (\text{J30})$$

Therefore,

$$\boxed{a' = g \implies \frac{d}{dx'} \left(\frac{dt'(x')}{dt'_r} \right) = \frac{d}{dz} \left(\frac{dt'(z)}{dt'_g} \right) = \frac{g}{c^2}}. \quad (\text{J31})$$

The equivalence established here is local. The Rindler expression is exact within its coordinate domain, whereas the gravitational expression has been expanded in the weak-field and small-height limits. The result therefore does not imply the global equivalence of a Schwarzschild gravitational field and a uniformly accelerating system. It establishes that their local clock-rate gradients agree to leading order when $a' = g$.

This is the approximation invoked in the free-fall comparison in the main text: the body is assumed sufficiently small that the gravitational field may be treated as locally uniform across it, so tidal variations and high-order spatial gradients are neglected.

J.6 Relation to Causal Potential

The equivalence above is established entirely from the observable clock-rate relations. Within Universal Specificity, these rate gradients are interpreted as spatial variations in the rate of causal processes.

As shown in Appendix I, the uncalibrated clock-rate variation across the accelerating system may be represented by the rear-referenced causal potential

$$e'_{U_0}(x') = \frac{1}{2}c^2 \left(\frac{dt'(x')}{dt'_r} \right)^2. \quad (\text{J32})$$

Because

$$\frac{dt'(x')}{dt'_r} = 1 + \frac{a'x'}{c^2}, \quad (\text{J33})$$

a nonzero acceleration produces

$$\nabla'_0 e'_{U_0} \neq 0. \quad (\text{J34})$$

The gravitational clock-rate relation exhibits the same local functional dependence,

$$\frac{dt'(z)}{dt'_g} \approx 1 + \frac{gz}{c^2}, \quad (\text{J35})$$

so the gravitational case likewise corresponds locally to a spatial gradient in causal potential.

Thus, the mathematical equivalence established here may be summarized as

$$\text{Born-rigid acceleration} \longrightarrow \text{clock-rate gradient}, \quad (\text{J36})$$

$$\text{weak-field gravitation} \longrightarrow \text{clock-rate gradient}, \quad (\text{J37})$$

with the gradients taking the same local form when

$$a' = g. \quad (\text{J38})$$

The main text uses this shared condition, together with the associated weighted and weightless cases, in the induction concerning the causal origin of weightedness.

J.7 Calibration and Scope

The equivalence derived in this appendix is expressed directly in terms of uncalibrated measurements made by the affected clocks and rulers. No recalibration of those instruments is required to establish the result.

This appendix therefore does not require a unique transformation between the primed operational quantities and the preferred-frame calibrated quantities used elsewhere in the manuscript. Such transformations become necessary when the same physical condition is to be represented quantitatively in a different measurement basis, but they are not required to establish the equality of the locally observed clock-rate gradients.

Standard relativistic constructions, including Lorentz-transform stitching for Born-rigid motion and reconstruction from acceleration histories, provide the required mappings when needed [35, 64, 66].

J.8 Conclusion

Born-rigid acceleration produces the exact local clock-rate relation

$$\frac{dt'(x')}{dt'_r} = 1 + \frac{a'x'}{c^2}, \quad (\text{J39})$$

while a static weak gravitational field produces, over a sufficiently small height interval,

$$\frac{dt'(z)}{dt'_g} \approx 1 + \frac{gz}{c^2}. \quad (\text{J40})$$

Consequently, when $a' = g$,

$$\boxed{\frac{d}{dx'} \left(\frac{dt'(x')}{dt'_r} \right) = \frac{d}{dz} \left(\frac{dt'(z)}{dt'_g} \right) = \frac{g}{c^2}}. \quad (\text{J41})$$

The two physical situations therefore exhibit the same local time-dilation gradient in their uncalibrated operational measurements. Within Universal Specificity, this shared rate gradient corresponds to the same local form of causal-potential gradient, providing the mathematical connection between the accelerating and gravitational cases used in the analysis of weightedness.

K Complex Representation of Gravitoelectromagnetism

This appendix examines a formal complex-valued representation in which gravitational and electromagnetic field equations can be expressed within a common algebraic form. The construction begins by rescaling electric charge into a mass-dimensional quantity and then applying corresponding transformations to the electromagnetic source and field variables. The resulting electromagnetic equations take the same form as the gravitoelectromagnetic equations adopted here, allowing the gravitational and electromagnetic quantities to be represented as the real and imaginary components of generalized complex fields.

This construction establishes an algebraic correspondence between the two sets of field equations. It does not, by itself, establish that gravitation and electromagnetism are physically identical interactions or constitute a complete unified field theory.

K.1 Charge-to-Mass Scaling

Define the scaling constant

$$\kappa \equiv \sqrt{4\pi\epsilon_0 G}. \quad (\text{K1})$$

Electric charge may then be represented by the mass-dimensional quantity [53]

$$M_e \triangleq \frac{q}{\kappa} = \frac{q}{\sqrt{4\pi\epsilon_0 G}}. \quad (\text{K2})$$

For one coulomb,

$$\frac{1 \text{ C}}{\sqrt{4\pi\epsilon_0 G}} \approx 1.16042 \times 10^{10} \text{ kg}. \quad (\text{K3})$$

Using this definition, Coulomb's electrostatic force may be written

$$\begin{aligned} \vec{F}_e &= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r} \\ &= G \frac{M_{e1} M_{e2}}{r^2} \hat{r}. \end{aligned} \quad (\text{K4})$$

This has the same magnitude scaling as Newtonian gravitation but the opposite sign for like sources:

$$\vec{F}_g = -G \frac{M_{g1} M_{g2}}{r^2} \hat{r}. \quad (\text{K5})$$

The sign difference can be encoded algebraically by representing the gravitational and electric source quantities along orthogonal axes of the complex plane. Define

$$\begin{aligned} \mathbb{M} &= M_g + iM_e \\ &= m + i \frac{q}{\sqrt{4\pi\epsilon_0 G}}. \end{aligned} \quad (\text{K6})$$

For two purely electric source components, application of the gravitational force sign gives

$$\begin{aligned} -G \frac{(iM_{e1})(iM_{e2})}{r^2} \hat{r} &= -G \frac{i^2 M_{e1} M_{e2}}{r^2} \hat{r} \\ &= +G \frac{M_{e1} M_{e2}}{r^2} \hat{r}, \end{aligned} \quad (\text{K7})$$

because

$$i^2 = -1. \quad (\text{K8})$$

The complex factor therefore provides an algebraic representation of the opposite signs of gravitational attraction and electrostatic interaction between like charges.

The imaginary component in Equation (K6) should not be interpreted as a literal imaginary inertial mass. Rather, it provides a mathematical means of representing the gravitational and electric source quantities as independent components of a generalized source.

K.2 Gravitational and Electromagnetic Field Equations

Using the gravitoelectromagnetic normalization adopted in this work [53], the gravitational field equations are written as

$$\nabla \cdot \vec{G}_g = -4\pi G \rho_g, \quad (\text{K9})$$

$$\nabla \cdot \vec{B}_g = 0, \quad (\text{K10})$$

$$\nabla \times \vec{G}_g = -\frac{\partial \vec{B}_g}{\partial t}, \quad (\text{K11})$$

$$\nabla \times \vec{B}_g = -\frac{4\pi G}{c^2} \vec{J}_g + \frac{1}{c^2} \frac{\partial \vec{G}_g}{\partial t}. \quad (\text{K12})$$

The corresponding Maxwell equations are

$$\nabla \cdot \vec{E} = \frac{\rho'}{\varepsilon_0}, \quad (\text{K13})$$

$$\nabla \cdot \vec{B} = 0, \quad (\text{K14})$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad (\text{K15})$$

$$\nabla \times \vec{B} = \frac{1}{\varepsilon_0 c^2} \vec{J} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}. \quad (\text{K16})$$

The positive current term in Equation (K16) is important. The corresponding negative sign in the rescaled electromagnetic equation below arises from the field transformation rather than from Maxwell's equation itself.

K.3 Rescaling the Electromagnetic Variables

Using the same charge-to-mass conversion introduced in Equation (K2), define the mass-dimensional electromagnetic source variables

$$\rho_e \triangleq \frac{\rho'}{\sqrt{4\pi\varepsilon_0 G}}, \quad \vec{J}_e \triangleq \frac{\vec{J}'}{\sqrt{4\pi\varepsilon_0 G}}, \quad (\text{K17})$$

and the corresponding rescaled field variables

$$\vec{G}_e \triangleq -\sqrt{4\pi\varepsilon_0 G} \vec{E}, \quad \vec{B}_e \triangleq -\sqrt{4\pi\varepsilon_0 G} \vec{B}. \quad (\text{K18})$$

Applying these transformations to Gauss's law gives

$$\begin{aligned} \nabla \cdot \vec{G}_e &= -\sqrt{4\pi\varepsilon_0 G} \nabla \cdot \vec{E} \\ &= -\sqrt{4\pi\varepsilon_0 G} \frac{\rho'}{\varepsilon_0} \\ &= -4\pi G \rho_e. \end{aligned} \quad (\text{K19})$$

Similarly,

$$\nabla \cdot \vec{B}_e = 0, \quad (\text{K20})$$

and Faraday's law becomes

$$\nabla \times \vec{G}_e = -\frac{\partial \vec{B}_e}{\partial t}. \quad (\text{K21})$$

Finally, applying the transformation to the Maxwell–Ampère equation gives

$$\begin{aligned} \nabla \times \vec{B}_e &= -\sqrt{4\pi\varepsilon_0 G} \left[\frac{1}{\varepsilon_0 c^2} \vec{J}' + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} \right] \\ &= -\frac{4\pi G}{c^2} \vec{J}_e + \frac{1}{c^2} \frac{\partial \vec{G}_e}{\partial t}. \end{aligned} \quad (\text{K22})$$

The gravitational and rescaled electromagnetic equations therefore take the same algebraic form:

$$\begin{aligned} \nabla \cdot \vec{G}_g &= -4\pi G \rho_g, & \nabla \cdot \vec{G}_e &= -4\pi G \rho_e, \\ \nabla \cdot \vec{B}_g &= 0, & \nabla \cdot \vec{B}_e &= 0, \\ \nabla \times \vec{G}_g &= -\frac{\partial \vec{B}_g}{\partial t}, & \nabla \times \vec{G}_e &= -\frac{\partial \vec{B}_e}{\partial t}, \end{aligned}$$

$$\nabla \times \vec{B}_g = -\frac{4\pi G}{c^2} \vec{J}_g + \frac{1}{c^2} \frac{\partial \vec{G}_g}{\partial t}, \quad \nabla \times \vec{B}_e = -\frac{4\pi G}{c^2} \vec{J}_e + \frac{1}{c^2} \frac{\partial \vec{G}_e}{\partial t}. \quad (\text{K23})$$

Thus, after the charge-to-mass and field transformations, the two sets of equations differ only in the physical quantities represented by their respective variables.

K.4 Complex Generalized Fields

Because the two transformed systems are linear and have identical algebraic form, they may be represented as orthogonal components in the complex plane. Define the generalized source and field quantities

$$\begin{aligned} \varrho &= \rho_g + i\rho_e, & \vec{\mathbb{J}} &= \vec{J}_g + i\vec{J}_e, \\ \vec{\mathbb{G}} &= \vec{G}_g + i\vec{G}_e, & \vec{\mathbb{B}} &= \vec{B}_g + i\vec{B}_e. \end{aligned} \quad (\text{K24})$$

The gravitational quantities occupy the real component of this representation, while the rescaled electromagnetic quantities occupy the imaginary component:

$$\begin{aligned} \Re(\varrho) &= \rho_g, & \Im(\rho) &= \rho_e, \\ \Re(\vec{\mathbb{G}}) &= \vec{G}_g, & \Im(\vec{\mathbb{G}}) &= \vec{G}_e. \end{aligned} \quad (\text{K25})$$

Because divergence, curl, and temporal differentiation are linear operators,

$$\nabla \cdot (\vec{G}_g + i\vec{G}_e) = \nabla \cdot \vec{G}_g + i\nabla \cdot \vec{G}_e, \quad (\text{K26})$$

with analogous relations for the remaining field operators. The two systems can consequently be combined into

$$\begin{aligned} &\boxed{\nabla \cdot \vec{\mathbb{G}} = -4\pi G \varrho}, \\ &\boxed{\nabla \cdot \vec{\mathbb{B}} = 0}, \\ &\boxed{\nabla \times \vec{\mathbb{G}} = -\frac{\partial \vec{\mathbb{B}}}{\partial t}}, \\ &\boxed{\nabla \times \vec{\mathbb{B}} = -\frac{4\pi G}{c^2} \vec{\mathbb{J}} + \frac{1}{c^2} \frac{\partial \vec{\mathbb{G}}}{\partial t}}. \end{aligned} \quad (\text{K27})$$

The real components of Equation (K27) recover the gravitational equations, while the imaginary components recover the rescaled Maxwell equations. The complex representation therefore packages both field systems into a single algebraic expression without eliminating their independent physical components.

K.5 Interpretation and Scope

Multiplication by i represents a 90° rotation in the complex plane and therefore provides a natural algebraic means of representing the gravitational and electromagnetic quantities as orthogonal components. This mathematical orthogonality should be distinguished from a claim that the corresponding physical vector fields must always be spatially perpendicular.

Within the mechanistic interpretation explored in this work, gravitation and electromagnetism may act differently upon the propagation and motion of physical systems. The complex representation developed here provides one possible mathematical language for representing such distinct contributions while retaining their common field-equation form. The existence of this representation alone, however, does not establish the physical mechanism connecting the two interactions.

The result should also be distinguished from the statement that the full non-linear gravitational field equations have been unified with electromagnetism. The construction applies to the gravitoelectromagnetic field equations adopted above and demonstrates a formal correspondence with Maxwell's equations after the stated rescaling.

K.6 Conclusion

Electric charge can be expressed in mass-dimensional units through

$$M_e = \frac{q}{\sqrt{4\pi\epsilon_0 G}}, \quad (\text{K28})$$

and the opposite interaction sign between Newtonian gravitation and electrostatics can be represented algebraically by assigning the electric component to the imaginary axis,

$$\mathbb{M} = M_g + iM_e. \quad (\text{K29})$$

Applying the corresponding transformations to the electromagnetic source and field variables causes the Maxwell equations to assume the same algebraic form as the gravitoelectromagnetic equations used here. The two sets may consequently be represented by the generalized complex quantities

$$\begin{aligned} \varrho &= \rho_g + i\rho_e, & \vec{\mathbb{J}} &= \vec{J}_g + i\vec{J}_e, \\ \vec{\mathbb{G}} &= \vec{G}_g + i\vec{G}_e, & \vec{\mathbb{B}} &= \vec{B}_g + i\vec{B}_e, \end{aligned} \quad (\text{K30})$$

which satisfy

$$\boxed{\nabla \cdot \vec{\mathbb{G}} = -4\pi G \varrho}, \quad \boxed{\nabla \cdot \vec{\mathbb{B}} = 0},$$

$$\boxed{\nabla \times \vec{\mathbb{G}} = -\frac{\partial \vec{\mathbb{B}}}{\partial t}}, \quad \boxed{\nabla \times \vec{\mathbb{B}} = -\frac{4\pi G}{c^2} \vec{\mathbb{J}} + \frac{1}{c^2} \frac{\partial \vec{\mathbb{G}}}{\partial t}}. \quad (\text{K31})$$

This establishes a compact complex representation in which gravitoelectromagnetic and electromagnetic fields occupy independent real and imaginary components of a common mathematical system. Whether this formal correspondence reflects a deeper physical relationship between gravitation and electromagnetism, and whether it can be extended into a predictive theory beyond the field equations considered here, remain questions for further investigation.

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